

Extensive true-vacuum quantum blocking in the full spatial Wilson box

Exact path-product maps, associative memory, and physical spectral channels

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1 Original objects and the finite interacting operator

Let $L \geq 2$ be an integer. The vertex set is $\mathbf{V}_L = \{-L, -L+1, \dots, L\}^3$. For each $i \in \{1, 2, 3\}$, the physical edge $e = (n, i)$ goes from n to $n + \mathbf{e}_i$ when both vertices are present. Its variable is $U_i(n) \in \mathrm{SU}(2)$. Inverse traversal uses $U_i(n)^{-1}$. Every such edge belongs to $\mathbf{E}_L$. Every elementary face $p = (n; i, j)$, $i < j$, with four vertices in the box belongs to $\mathbf{P}_L$; its ordered holonomy and trace are

$$U_p = U_i(n)U_j(n + \mathbf{e}_i)U_i(n + \mathbf{e}_j)^{-1}U_j(n)^{-1}, \quad W_p = \mathrm{tr} U_p. \quad (1)$$

The counts are $N = 3(2L)(2L+1)^2$ and $M = 3(2L)^2(2L+1)$: fix a direction or a face plane and count its initial and transverse coordinates. The configuration space is $\mathcal{Q} = \mathrm{SU}(2)^{\mathbf{E}_L}$ with product Haar probability measure λ . Put

$$T_a = -i\sigma_a/2, \quad X_{e,a}f = \left. \frac{d}{dt} f(\dots, e^{tT_a} U_e, \dots) \right|_{t=0}, \quad E_e = -\sum_{a=1}^3 X_{e,a}^2, \quad (2)$$

$$H = \kappa \sum_{e \in \mathbf{E}_L} E_e + b \sum_{p \in \mathbf{P}_L} (2 - W_p), \quad \kappa = \frac{2g^2}{a}, \quad b = \frac{1}{2g^2a}, \quad \xi = \frac{b}{\kappa} = \frac{1}{4g^4}, \quad a, g > 0. \quad (3)$$

Here a in $a, g > 0$ denotes spatial spacing; the color index in (2) is summed only when indicated. Haar integration by parts makes $X_{e,a}$ skew-adjoint. The Pauli identities give $-2\mathrm{tr}(T_a T_b) = \delta_{ab}$ and $-\sum_a T_a^2 = 3I/4$. The original metric $c(A, B) = -\mathrm{tr}(AB)/2$ has orthonormal basis $2T_a$; hence $X^c = 2X$, $\Delta^c = 4\Delta^T$, and the same kinetic term is $-(\kappa/4) \sum_e \Delta_e^c$. All faces, including boundary faces and faces in every plane, remain in (3). In particular the exact scalar term $2bM = M/(g^2a)$ is part of H .

Gauge transformations are $(h \cdot U)_e = h_{s(e)}^{-1} U_e h_{t(e)}$. The form of H on $H^1(\mathcal{Q})$ is

$$q_H(u, v) = \kappa \sum_{e,a} \int \overline{X_{e,a}u} X_{e,a}v \, d\lambda + b \sum_p \int (2 - W_p) \bar{u}v \, d\lambda.$$

It is closed, since the smooth potential is bounded between 0 and $4bM$. Its associated self-adjoint operator has domain H^2 by elliptic regularity; smooth functions are operator and form cores. Compact Sobolev embedding gives compact resolvent. Taking the modulus of a minimizing vector does not increase its form: apply the chain rule first to $(|u|^2 + \epsilon^2)^{1/2}$ and then let ϵ decrease to zero.

The resulting nonnegative lowest eigenfunction is smooth by elliptic regularity and strictly positive by the elliptic strong maximum principle. Write it as $\psi > 0$, with $H\psi = E_0\psi$ and $\int \psi^2 d\lambda = 1$. This fixes the state norm; none of the physical coefficients is changed.

Set $\rho = \psi^2$, $\mu = \rho\lambda$, and $\mathcal{H} = L^2(\mu)$. Multiplication $\mathcal{U}u = \psi u$ is a unitary $\mathcal{H} \rightarrow L^2(\lambda)$, with inverse division by ψ . The product rule and $H\psi = E_0\psi$ give

$$\mathcal{L} = \mathcal{U}^{-1}(H - E_0)\mathcal{U} = -\kappa \sum_{e,a} [X_{e,a}^2 + (X_{e,a} \log \rho) X_{e,a}], \quad (4)$$

$$q(u, v) = \kappa \sum_{e,a} \int \overline{X_{e,a} u} X_{e,a} v d\mu, \quad D(q) = \mathcal{V} = H^1(\mathcal{Q}). \quad (5)$$

Integration by parts proves the form identity. Multiplication and division by the smooth positive ψ preserve every integer Sobolev space, so the form and operator domains are respectively H^1, H^2 . If $q(u, u) = 0$, all its derivatives vanish and connectedness makes u constant. This proves uniqueness of ψ , and its gauge invariance follows because gauge transformations preserve H , positivity and norm. It also proves that the finite first nonzero eigenvalue δ of $\mathcal{L}$ is strictly positive. No uniform lower bound for δ is used in the uniform estimates below. Inner products are conjugate-linear in their first arguments.

2 A uniform single-link estimate for the actual vacuum

This section supplies estimates independent of L , with every $b > 0$ allowed. Define the constants

$$D = 16b, \quad d = D/\kappa = 16\xi = 4/g^4, \quad t_0 = \frac{2 \log 12}{\kappa}, \quad R = 3e^{Dt_0} = 3 \cdot 12^{2d}, \quad \alpha = R^{-2}, \quad (6)$$

and

$$G = R \exp\left(\frac{d}{1+d}\right) \left(1 + \frac{2d}{1+d}\right) \sqrt{\frac{1+d}{2}}, \quad C_* = \frac{3}{2} + 2G. \quad (7)$$

Here D has units of energy and t_0 of inverse energy. The constants d, R, α, G, C_* are dimensionless and depend on the original coupling g , but not on the number of links, the blocking length, or a separately.

Lemma 2.1. *For the single-link heat operator $P_t = e^{-t\kappa E}$, its kernel at $t = t_0$ lies between $1/2$ and $3/2$. For every real smooth f ,*

$$2\kappa t \sum_a |X_a P_t f|^2 \leq P_t(f^2) - (P_t f)^2, \quad \|\nabla P_t f\|_\infty \leq \frac{\|f\|_\infty}{\sqrt{2\kappa t}}. \quad (8)$$

Proof. The spin- j representation has dimension $2j + 1$ and Casimir $j(j + 1)$ in (2). One can verify the latter by the eigenvalues $m = -j, \dots, j$ of iT_3 and the raising/lowering coefficients $\sqrt{(j \mp m)(j \pm m + 1)}$; their quadratic sum is $j(j + 1)$. Peter-Weyl completeness therefore gives the heat kernel $p_t(U) = \sum_{j \geq 0} (2j + 1) e^{-\kappa t j(j+1)} \chi_j(U)$. This series is absolutely uniformly convergent for $t > 0$, since $|\chi_j| \leq 2j + 1$. Writing $k = 2j$, using $j(j + 1) = k(k + 2)/4 \geq k/2$ and $(k + 1)^2 \leq 4^k$ for $k \geq 1$, gives

$$|p_{t_0} - 1| \leq \sum_{k \geq 1} 4^k e^{-\kappa t_0 k/2} = \sum_{k \geq 1} 3^{-k} = \frac{1}{2}.$$

For the gradient assertion, the bi-invariant Casimir commutes with each X_a : the commutator of its sum of squares with X_b contracts antisymmetric Lie structure constants with $X_a X_c + X_c X_a$ and is

zero. Thus $X_a P_s = P_s X_a$ on smooth functions, and Jensen's inequality gives $\Gamma(P_s f) \leq P_s \Gamma(f)$ for $\Gamma(f) = \sum_a |X_a f|^2$. Differentiate $P_s[(P_{t-s} f)^2]$ and use the product rule:

$$P_t(f^2) - (P_t f)^2 = 2\kappa \int_0^t P_s \Gamma(P_{t-s} f) ds \geq 2\kappa t \Gamma(P_t f).$$

The last inequality applies the preceding gradient contraction with input $P_{t-s} f$. The supremum estimate follows because $P_t(f^2) \leq \|f\|_\infty^2$. This is the exact single-group version of the reverse Poincare inequality [2, Eq. (1.7)], with the time and generator coefficient displayed explicitly. $\square$

Theorem 2.2. *Fix one physical link e and write all other variables as y . For all $u, u' \in \text{SU}(2)$ and y ,*

$$R^{-1} \leq \frac{\psi(u, y)}{\psi(u', y)} \leq R, \quad \alpha \leq \frac{\rho(u, y)}{\int \rho(v, y) dv} \leq \alpha^{-1}. \quad (9)$$

Moreover,

$$\left(\sum_a |X_{e,a} \log \psi|^2 \right)^{1/2} \leq G, \quad |\nabla_e \log \rho| \leq 2G. \quad (10)$$

All bounds are uniform in e and L at the fixed positive a, g .

Proof. Let $V_e = b \sum_{p \ni e} (2 - W_p)$ and let H_{-e} contain all other kinetic terms and all faces not incident to e . There are at most four incident faces in the full three-dimensional box, including the boundary cases with fewer faces. Therefore $0 \leq V_e \leq D$, and the exact operator is

$$H = A_e + V_e, \quad A_e = \kappa E_e + H_{-e}, \quad T_t = e^{-tA_e} = P_t^{(e)} \otimes e^{-tH_{-e}}.$$

The two factors of T_t commute because H_{-e} is independent of u . They have positive kernels. The parabolic comparison principle with the bounded multiplication $0 \leq V_e \leq D$ gives, for $f \geq 0$,

$$e^{-Dt} T_t f \leq e^{-tH} f \leq T_t f. \quad (11)$$

For example $T_t f$ is a supersolution for $\partial_t + A_e + V_e$, while $e^{-Dt} T_t f$ is a subsolution; their initial data agree. The maximum principle gives (11). This proves the comparison for this compact group product directly; no Euclidean configuration-space identification is needed.

Apply it to ψ . At t_0 , positivity of the other-link kernel and Lemma 2.1 imply

$$\frac{(T_{t_0} \psi)(u, y)}{(T_{t_0} \psi)(u', y)} \leq 3.$$

Since $e^{-tH} \psi = e^{-tE_0} \psi$, comparison at both u, u' gives $\psi(u, y)/\psi(u', y) \leq 3e^{Dt_0} = R$. Interchanging u, u' proves both sides. Squaring and averaging the denominator over u' proves the conditional density bounds. The extensive energy E_0 cancels from this ratio exactly.

Here is a derivative proof that does not differentiate an unknown vacuum estimate. If a positive function $F(u, y)$ satisfies the first ratio bound, so does $f_t(u) = (e^{-tH_{-e}} F(u, \cdot))(y)$ at each fixed y , by positivity. Hence $\|f_t\|_\infty \leq R \min f_t \leq R P_t f_t(u)$. Lemma 2.1 yields

$$|\nabla_e T_t \psi| \leq \frac{R}{\sqrt{2\kappa t}} T_t \psi.$$

The function $(e^{-tH_{-e}} (V_e \psi))(u, y)$ is nonnegative and bounded above by $D(e^{-tH_{-e}} \psi)(u, y)$. Its supremum in u is consequently at most $DR \min_u (e^{-tH_{-e}} \psi)(u, y)$. The same heat estimate gives

$$|\nabla_e T_t (V_e \psi)| \leq \frac{DR}{\sqrt{2\kappa t}} T_t \psi.$$

The exact Duhamel identity on the smooth eigenvector is

$$\psi = e^{E_0 t} T_t \psi - \int_0^t e^{E_0 \tau} T_\tau (V_e \psi) d\tau. \quad (12)$$

It follows either by differentiating $e^{E_0 t} T_t \psi$ and using $A_e \psi = (E_0 - V_e) \psi$, or by the bounded-potential Duhamel formula. Comparison gives $e^{E_0 \tau} T_\tau \psi \leq e^{D\tau} \psi$. The two gradient estimates, whose $\tau^{-1/2}$ singularity is integrable, justify differentiation of (12) and give

$$\frac{|\nabla_e \psi|}{\psi} \leq R \left[\frac{e^{Dt}}{\sqrt{2\kappa t}} + D \int_0^t \frac{e^{D\tau}}{\sqrt{2\kappa \tau}} d\tau \right] \leq \frac{Re^{Dt}(1 + 2Dt)}{\sqrt{2\kappa t}}.$$

Choose the actual time $t = (\kappa + D)^{-1}$. This last expression is exactly G in (7). Finally $X \log \rho = 2X \log \psi$. No gap bound, perturbation disk, or correlation factorization was used. $\square$

3 A specified extensive spatial blocking sequence

Let $m \geq 1$ divide $2L$, and put $q_m = 2L/m$. Retain as skeleton links those direction- i fine edges whose two transverse coordinates lie in $-L + m\{0, 1, \dots, q_m\}$. Every longitudinal fine edge on each such line is included. Denote this subset by R_m and its complement by S_m . Its exact count is

$$|R_m| = 3(2L)(q_m + 1)^2, \quad |S_m| = 3(2L)[(2L + 1)^2 - (q_m + 1)^2]. \quad (13)$$

Thus a fixed $m > 1$ eliminates an extensive number of links as the box increases. If $m \mid n \mid 2L$, then $R_n \subset R_m$. In particular $m = 1, 2, 4, \dots, 2^k$ gives a repeated blocking sequence whenever $2^k \mid 2L$. The Hamiltonian still has all N link and M face terms.

The coarse vertices are the points $x \in [-L, L]^3$ whose coordinates all lie in $-L + m\mathbb{Z}$. A coarse direction- i edge $\gamma = (x, i)$ is the ordered path of m fine edges from x to $x + m\mathbf{e}_i$. The number of coarse edges is

$$N_m = 3q_m(q_m + 1)^2, \quad |R_m| = mN_m.$$

The chains have disjoint physical-edge supports and meet only at coarse vertices. Write their variables as $u_{\gamma,1}, \dots, u_{\gamma,m}$ in positive order, and define

$$z_\gamma = u_{\gamma,1} \cdots u_{\gamma,m}, \quad \pi_m : \mathcal{Q} \longrightarrow \mathcal{Q}_m := \text{SU}(2)^{N_m}. \quad (14)$$

No original coordinate was replaced by a unit coarse spacing: the physical coarse spacing is ma , while the original coordinates remain x .

For every chain put $v_{\gamma,k} = u_{\gamma,k}$ for $k < m$. Let y comprise all v and all untouched variables on S_m . The full typed coordinate map is

$$\Phi_m : \mathcal{Q} \longrightarrow \mathcal{Y}_m \times \mathcal{Q}_m, \quad U \longmapsto (y, z), \quad \mathcal{Y}_m = \text{SU}(2)^{S_m} \times \prod_{\gamma=1}^{N_m} \text{SU}(2)^{m-1}.$$

It is a global smooth bijection. Its inverse $\Phi_m^{-1} : \mathcal{Y}_m \times \mathcal{Q}_m \rightarrow \mathcal{Q}$ uses

$$u_{\gamma,m} = (v_{\gamma,1} \cdots v_{\gamma,m-1})^{-1} z_\gamma,$$

and keeps every other variable. Substituting the original ordered product cancels its prefix on the left and recovers $u_{\gamma,m}$. Conversely multiplying the retained prefix by the displayed last link

recovers z_γ . Every v and every S_m coordinate is fixed, so both compositions are the identity on their respective full spaces. There are $|S_m| + (m-1)N_m = N - N_m$ factors in $\mathcal{Y}_m$; none of those coordinates is omitted from Φ_m . Integrating $u_{\gamma,m}$ last, left invariance of Haar proves the exact measure identity

$$d\lambda(U) = d\lambda_y(y) d\lambda_m(z). \quad (15)$$

This proves the map and its measure factor, rather than postulating independent coarse links in the interacting vacuum.

The complete kinetic operator in these coordinates is as follows. Define the prefix and the real orthogonal matrix $O_{\gamma,k}$ by

$$p_{\gamma,k} = v_{\gamma,1} \cdots v_{\gamma,k-1}, \quad p_{\gamma,k} T_a p_{\gamma,k}^{-1} = \sum_b O_{\gamma,k;ba} T_b.$$

Orthogonality follows by applying $-2 \operatorname{tr}(AB)$ to both conjugated factors; its determinant is 1 by connectedness. Let $D_{\gamma,b}$ be left multiplication differentiation on z_γ , with the same T_b . For $k < m$ the transformed fine vector field is

$$X_{\gamma,k,a} = X_{v_{\gamma,k},a} + \sum_b O_{\gamma,k;ba} D_{\gamma,b};$$

for $k = m$ it is $\sum_b O_{\gamma,m;ba} D_{\gamma,b}$. The prefix $p_{\gamma,k}$ does not involve $v_{\gamma,k}$, and no O depends on z . The full sum of squares is consequently

$$\begin{aligned} \sum_{e,a} X_{e,a}^2 &= \sum_{e \in S_m, a} X_{e,a}^2 + \sum_{\gamma} \left[\sum_{k < m, a} X_{v_{\gamma,k},a}^2 + m \sum_b D_{\gamma,b}^2 \right] \\ &\quad + 2 \sum_{\gamma, k < m, a, b} O_{\gamma,k;ba} X_{v_{\gamma,k},a} D_{\gamma,b}. \end{aligned} \quad (16)$$

Each square before expansion is the square of the displayed original field, so this formula includes all mixed derivatives with their signs and coefficients. Every original W_p in (3) is evaluated at the explicit inverse coordinate map. It is not replaced by a coarse face trace or truncated to selected words.

Gauge transformations on a chain cancel at its internal vertices and give $z_\gamma \mapsto h_x^{-1} z_\gamma h_{x+me_i}$. The marginal density on R_m is invariant under the internal gauge transformations, by gauge invariance of ρ and Haar substitution on S_m . A non-coarse skeleton vertex is bivalent in the skeleton. For a fixed chain with endpoints fixed, set successively $h_{x+ke_i} = (u_1 \cdots u_k)^{-1}$, $k < m$. This sends its first $m-1$ links to the identity and its last to z_γ . Different chain interiors are disjoint, so these choices are simultaneous. It follows that this skeleton marginal is a function of z alone. This statement concerns the marginal; the full conditional distribution of y given z need not factorize.

4 The actual coarse density, domains, and conditional coupling

Let $\tilde{\rho}_m(y, z) = \rho(U(y, z))$ and define

$$r_m(z) = \int \tilde{\rho}_m(y, z) d\lambda_y, \quad \mu_m = r_m \lambda_m, \quad p_m(y | z) = \tilde{\rho}_m(y, z) / r_m(z), \quad (17)$$

$$\begin{aligned} \mathcal{K}_m &= L^2(\mu_m), \quad J_m f = f \circ \pi_m, \quad P_m = J_m J_m^*, \quad Q_m = I - P_m, \\ (J_m^* u)(z) &= \int p_m(y | z) u(U(y, z)) d\lambda_y. \end{aligned} \quad (18)$$

In particular the types are $J_m : \mathcal{K}_m \rightarrow \mathcal{H}$ and $J_m^* : \mathcal{H} \rightarrow \mathcal{K}_m$. Their full coordinate expressions are $J_m f(U) = f(z(U))$ and the fibre integral (18); the operator $P_m : \mathcal{H} \rightarrow J_m \mathcal{K}_m \subset \mathcal{H}$ is their composition. Smoothness and strict positivity follow from compact integration. Fubini proves the adjoint formula and $J_m^* J_m = I$; Jensen proves that J_m^* is a contraction. Thus P_m is the true-vacuum orthogonal projection. All these maps intertwine the stated fine and coarse gauge actions. The restriction of the conditional projection onto skeleton functions, on the fine invariant Hilbert space, equals this path projection: its output is internally gauge invariant, hence depends only on z , as proved in the preceding section. The exact invariant Hilbert-space unitary, its almost-everywhere inverse and the equality of the full complementary operator domains are proved in Proposition 10.1.

For each chain and color set, in the original coordinates,

$$Y_{\gamma,b} = \frac{1}{m} \sum_{k=1}^m \sum_a O_{\gamma,k;ba} X_{\gamma,k,a}. \quad (19)$$

Then $Y_{\gamma,b} J_m f = J_m D_{\gamma,b} f$. Indeed each summand differentiates z_γ in the direction $\text{Ad}_{p_{\gamma,k}} T_a$, and orthogonality of O gives one copy of $D_{\gamma,b}$ per k . Also $Y_{\gamma,b}$ has zero Haar divergence: every coefficient $O_{\gamma,k;ba}$ is independent of its own differentiated fine edge. Consequently integration by parts against a coarse smooth test function gives the exact derivative identities

$$D_{\gamma,b} \log r_m = J_m^*(Y_{\gamma,b} \log \rho), \quad (20)$$

$$D_{\gamma,b} J_m^* u = J_m^*(Y_{\gamma,b} u) + J_m^*(\beta_{\gamma,b} u), \quad (21)$$

$$\beta_{\gamma,b} = Y_{\gamma,b} \log \rho - J_m D_{\gamma,b} \log r_m, \quad J_m^* \beta_{\gamma,b} = 0. \quad (22)$$

For example integrate $Y_{\gamma,b}(J_m \varphi \rho)$ over the full configuration space; its integral is zero. The resulting identity is $\int (D_{\gamma,b} \varphi) r_m = - \int \varphi J_m^*(Y_{\gamma,b} \log \rho) r_m$, which proves (20). Apply the same calculation to $u\rho$ to prove (21). This also avoids differentiating fibre coordinates while holding an incompatible fine coordinate fixed.

Theorem 2.2, the average in (19), and conditional Jensen give

$$|Y_\gamma \log \rho| \leq 2G, \quad |D_\gamma \log r_m| \leq 2G, \quad |\beta_\gamma| \leq 4G. \quad (23)$$

Here each absolute value is the Euclidean norm of its three color components. The first inequality uses the average of m rotated vectors, each with norm at most $2G$; it does not assume independence.

Proposition 4.1. *The exact lifted form and its generator are*

$$a_m(f, g) = q(J_m f, J_m g) = \kappa m \sum_{\gamma,b} \int \overline{D_{\gamma,b} f} D_{\gamma,b} g \, d\mu_m, \quad D(a_m) = \mathcal{V}_m = H^1(\mathcal{Q}_m), \quad (24)$$

$$A_m = -\kappa m r_m^{-1} \sum_{\gamma,b} D_{\gamma,b}(r_m D_{\gamma,b}), \quad D(A_m) = H^2(\mathcal{Q}_m). \quad (25)$$

The maps J_m, J_m^* preserve all integer Sobolev spaces, and P_m, Q_m preserve $\mathcal{V}$. On $\mathcal{V}_m$ the full eliminated coupling is

$$B_m f = -\kappa m \sum_{\gamma,b} \beta_{\gamma,b} J_m D_{\gamma,b} f, \quad J_m^* B_m f = 0, \quad \mathcal{L} J_m f = J_m A_m f + B_m f \quad (f \in H^2). \quad (26)$$

In particular the retained kinetic coefficient is exactly κm .

Proof. The fine derivative of $J_m f$ on edge (γ, k) is the orthogonal rotation $O_{\gamma, k}^T D_\gamma f$; on S_m it is zero. Summing the squared derivatives over the m edges proves (24). The smooth positive weight gives closedness and the H^2 elliptic domain of (25). The compact smooth coordinate bijection (15) preserves each integer Sobolev class with finite norm bounds, since its derivatives and inverse derivatives are bounded. The fibre integral with smooth positive kernel p_m has bounded derivatives of every finite order. This proves the Sobolev mapping claims.

There is also a useful explicit first-order estimate. Define

$$\eta_m = \kappa m \left\| \sum_{\gamma, b} \beta_{\gamma, b}^2 \right\|_\infty \leq 16 \kappa m N_m G^2.$$

Orthogonality of the rotations and Cauchy–Schwarz in the k sum give the pointwise inequality

$$m \sum_{\gamma, b} |Y_{\gamma, b} u|^2 \leq \sum_{e \in R_m, a} |X_{e, a} u|^2.$$

Conditional Jensen and (21) consequently give

$$a_m(J_m^* u, J_m^* u) \leq 2q(u, u) + 2\eta_m \|u\|_\mu^2. \quad (27)$$

This extends by smooth approximation. The same derivative identities show that $J_m f \in H^1$ implies $f \in H^1$, so the retained form space has exactly the claimed domain. On a lifted smooth f , (16) gives $\sum_{e, a} X_{e, a}^2 J_m f = m J_m \sum_{\gamma, b} D_{\gamma, b}^2 f$. The drift in (4) is $-\kappa m \sum_{\gamma, b} (Y_{\gamma, b} \log \rho) J_m D_{\gamma, b} f$. Subtract (25), and use (22); this proves (26) for smooth f . Density gives the stated H^2 identity and the bounded map $B_m : \mathcal{V}_m \rightarrow Q_m \mathcal{H}$, with

$$\|B_m f\|_\mu^2 \leq \eta_m a_m(f, f). \quad (28)$$

□

The full correlation tensor of the induced first-order terms is

$$I_{\gamma b, \eta c}(z) = J_m^*(\beta_{\gamma, b} \beta_{\eta, c})(z), \quad \|B_m f\|^2 = (\kappa m)^2 \int \sum_{\gamma, b, \eta, c} I_{\gamma b, \eta c} \overline{D_{\gamma, b} f} D_{\eta, c} f \, d\mu_m. \quad (29)$$

Every off-diagonal index is retained. This is the conditional covariance of the vectors $Y_{\gamma, b} \log \rho$. It is positive semidefinite because its quadratic form is a conditional expectation of a squared absolute value. Positivity does not assert that its different entries vanish or that it is uniformly diagonal-dominant.

There is an exact Haar-space description of the instantaneous marginal operator. Multiplication $f \mapsto \sqrt{r_m} f$ is a unitary $\mathcal{H}_m \rightarrow L^2(\lambda_m)$, with inverse $v \mapsto v/\sqrt{r_m}$. Both maps preserve H^1 and H^2 , since r_m is smooth and strictly positive on the compact group product. Direct product differentiation gives

$$\sqrt{r_m} A_m r_m^{-1/2} = -\kappa m \sum_{\gamma, b} D_{\gamma, b}^2 + \kappa m \frac{\sum_{\gamma, b} D_{\gamma, b}^2 \sqrt{r_m}}{\sqrt{r_m}}. \quad (30)$$

The last scalar function equals $\kappa m \sum_{\gamma, b} [D_{\gamma, b}^2 \log r_m / 2 + (D_{\gamma, b} \log r_m)^2 / 4]$. It depends on the entire induced density. No equality with a sum of one-plaquette Wilson terms is asserted. Moreover (30) is only the instantaneous compression; the next section gives the rest of the physical operator relation.

5 Exact memory, resolvent reconstruction, and physical norms

Set $\mathcal{H}_{Q_m} = Q_m \mathcal{H}$ and $\mathcal{V}_{Q_m} = \mathcal{V} \cap \mathcal{H}_{Q_m}$. The restriction $c_m(h, k) = q(h, k)$ on $\mathcal{V}_{Q_m}$ is closed and densely defined. Closedness uses the L^2 -closed subspace, and density follows by applying Q_m to smooth approximants, using (27). Define its self-adjoint operator C_m by

$$D(C_m) = \{h \in \mathcal{V}_{Q_m} : \exists v \in \mathcal{H}_{Q_m} \text{ such that } c_m(k, h) = \langle k, v \rangle \ \forall k \in \mathcal{V}_{Q_m}\}, \quad C_m h = v. \quad (31)$$

For smooth $h \in \mathcal{H}_{Q_m}$ this is $Q_m \mathcal{L} h$. Every such h is orthogonal to $1 = J_m 1$, so $C_m \geq \delta I > 0$. For form-domain arguments no formal compression domain is substituted for (31).

For $f, g \in \mathcal{V}_m$ and $h, k \in \mathcal{V}_{Q_m}$, integration by parts on a smooth core, followed by continuity, gives

$$q(J_m f + h, J_m g + k) = a_m(f, g) + \langle B_m f, k \rangle + \langle h, B_m g \rangle + c_m(h, k). \quad (32)$$

For smooth $h \in \mathcal{H}_{Q_m}$ the adjoint coupling is

$$B_m^* h = \kappa m r_m^{-1} \sum_{\gamma, b} D_{\gamma, b} (r_m J_m^* (\beta_{\gamma, b} h)). \quad (33)$$

To prove the sign, integrate the negative derivative in (26) against h using Haar integration by parts in z . Differentiating the conditional integral by (21) expands each derivative in (33) into terms with Yh , $Y\beta$, $\beta^2 h$ and $(D \log r_m) \beta h$. All coefficients are smooth and bounded in a fixed box, so $B_m^* : \mathcal{V}_{Q_m} \rightarrow \mathcal{H}_m$ is bounded. Without that regularity the adjoint is the bounded dual map $\mathcal{H}_{Q_m} \rightarrow \mathcal{V}_m^*$.

For $z > 0$ define on $\mathcal{V}_m$ the sesquilinear form

$$s_{m,z}(f, g) = a_m(f, g) + z \langle f, g \rangle - \langle B_m f, (C_m + z)^{-1} B_m g \rangle. \quad (34)$$

The use of z as a positive spectral shift here is distinct from the tuple z_γ of coarse link variables. The argument of a resolvent always denotes the scalar shift.

Theorem 5.1. *The form $s_{m,z}$ is closed with domain $\mathcal{V}_m$, and its associated operator $S_{m,z}$ satisfies $S_{m,z} \geq zI$. Its exact full resolvent and reconstruction maps are*

$$J_m^* (\mathcal{L} + z)^{-1} J_m = S_{m,z}^{-1}, \quad (35)$$

$$(\mathcal{L} + z)^{-1} J_m g = [J_m - (C_m + z)^{-1} B_m] S_{m,z}^{-1} g. \quad (36)$$

Every product in these formulas is defined by the displayed form domains.

Proof. Completing the square in (32) gives

$$q(J_m f + h, J_m f + h) + z(\|f\|^2 + \|h\|^2) = s_{m,z}(f, f) + \|(C_m + z)^{1/2}[h + (C_m + z)^{-1} B_m f]\|^2. \quad (37)$$

Its unique minimizing h is $-(C_m + z)^{-1} B_m f$, which lies in $D(C_m) \subset \mathcal{V}_{Q_m}$. Nonnegativity of q proves $s_{m,z}(f, f) \geq z\|f\|^2$. Apply (27) to $u = J_m f + h$. It gives

$$a_m(f, f) + \kappa \|f\|^2 \leq 2q(u, u) + (2\eta_m + \kappa) \|u\|^2 \leq K_{m,z} [q(u, u) + z\|u\|^2],$$

where $K_{m,z} = \max\{2, (2\eta_m + \kappa)/z\}$ in the form norm with reference energy κ . This constant is dimensionless. Taking the minimizing h proves a lower form-norm bound. The upper bound $s_{m,z}(f, f) \leq a_m(f, f) + z\|f\|^2$ follows by choosing $h = 0$. Their equivalence proves closedness. For $v = (\mathcal{L} + z)^{-1} J_m g$, its form-domain decomposition $v = J_m f + h$ is legitimate. Testing the weak resolvent equation against $k \in \mathcal{V}_{Q_m}$ gives $h = -(C_m + z)^{-1} B_m f$; testing against $J_m f'$ gives $s_{m,z}(f', f) = \langle f', g \rangle$. This proves both identities. $\square$

The notation in (34) is a closed-form implementation of the Feshbach–Schur map. The general method and reconstruction are established in [1, Theorem 1.2, Eqs. (1.10)–(1.16)]. That theorem’s bounded effective operator assumption is not silently applied to our infinite-rank projection; the coercivity and domain proof above supplies the needed extension for this particular operator.

For smooth initial f_0 , define $K_m(t) = J_m^* e^{-t\mathcal{L}} J_m$, $f(t) = K_m(t)f_0$. The full evolution and (32) give

$$f'(t) = -A_m f(t) + \int_0^t B_m^* e^{-(t-s)C_m} B_m f(s) ds, \quad f(0) = f_0, \quad (38)$$

$$K_m(t) = J_m^* e^{-t\mathcal{L}} J_m, \quad \int_0^\infty e^{-zt} K_m(t) dt = S_{m,z}^{-1}. \quad (39)$$

Here is the domain justification and the sign calculation. Smooth initial data lie in all powers of the compact elliptic $\mathcal{L}$. Spectral calculus and elliptic regularity keep $v(t) = e^{-t\mathcal{L}} J_m f_0$, $f(t) = J_m^* v(t)$ and $h(t) = Q_m v(t)$ smooth, continuously in the required Sobolev norms. They satisfy $f' = -A_m f - B_m^* h$, $h' = -B_m f - C_m h$, $h(0) = 0$. Variation of constants gives $h(t) = -\int_0^t e^{-(t-s)C_m} B_m f(s) ds$. Since $\sup_{x \geq 0} \sqrt{x} e^{-\tau x} = (2e\tau)^{-1/2}$, $e^{-\tau C_m}$ maps $\mathcal{H}_{Q_m}$ to its form space with norm at most $\sqrt{\kappa} + (2e\tau)^{-1/2}$ when the squared form norm is $c_m(h, h) + \kappa \|h\|^2$. The singularity is integrable; the bounded form-space map (33) makes (38) a Bochner integral in $\mathcal{H}_m$. The positive sign follows from the two negative cross terms. The Laplace identity follows by the spectral theorem and Theorem 5.1. This explicitly realizes projection memory of the kind introduced in [4], with this Euclidean generator.

For every smooth f , orthogonality of the two components of $\mathcal{L} J_m f$ gives the full second moment

$$\|\mathcal{L} J_m f\|^2 = \|A_m f\|^2 + \|B_m f\|^2. \quad (40)$$

Thus $K_m(t)$ is replaced by e^{-tA_m} for all t only when B_m vanishes: compare their scalar second derivatives at $t = 0$ on a smooth core. Conversely $B_m = 0$ makes (32) a direct-sum form and proves that replacement exactly. We do not make that replacement.

The dependence on energy also records the actual reconstructed norm. For fixed $0 \neq f \in \mathcal{V}_m$ with $\mu_m(f) = 0$ and scalar $z \geq 0$, put

$$w_z = (C_m + z)^{-1} B_m f, \quad u_z = \psi(J_m f - w_z), \quad d_f = \|f\|^2. \quad (41)$$

The $z = 0$ case exists because $C_m \geq \delta > 0$ in every fixed box. The vector is in the original form domain, is orthogonal to ψ , and is gauge invariant when f is. Its exact norm and energy are

$$\|u_z\|_\lambda^2 = d_f + \|w_z\|_\mu^2 = \partial_z s_{m,z}(f, f), \quad (42)$$

$$q_{H-E_0}[u_z] = a_m(f, f) - \langle B_m f, w_z \rangle - z \|w_z\|^2 = s_{m,z}(f, f) - z \partial_z s_{m,z}(f, f). \quad (43)$$

To verify these, orthogonality gives the norm; differentiation of the bounded resolvent gives $\partial_z (C_m + z)^{-1} = -(C_m + z)^{-2}$. Testing $(C_m + z)w_z = B_m f$ against w_z gives $c_m(w_z, w_z) = \langle B_m f, w_z \rangle - z \|w_z\|^2$; insert it in (32). Nonnegativity of q proves the numerator is nonnegative. For $B_m f \neq 0$ the quotient in (43) divided by (42) is strictly below $a_m(f, f)/d_f$. These are energies of actual full-system vectors; no artificial effective vacuum norm has replaced their eliminated components.

6 Associativity of repeated blocking with all prior memory

Let $m \mid n \mid 2L$. A coarse n -edge is the ordered concatenation of n/m coarse m -edges along its line. Let $\pi_{m,n}$ take these products and discard the other m -coarse edges. Literal ordered multiplication

gives

$$\pi_n = \pi_{m,n} \circ \pi_m, \quad J_n = J_m I_{m,n}, \quad \mu_n = (\pi_{m,n})_* \mu_m, \quad J_n^* = I_{m,n}^* J_m^*. \quad (44)$$

Here $I_{m,n}f = f \circ \pi_{m,n}$ is an isometry $\mathcal{K}_n \rightarrow \mathcal{K}_m$; its adjoint is conditional integration using r_m , not Haar integration in place of that weight. Fubini proves each equality, including the adjoint identity. In the full Hilbert space $P_n P_m = P_m P_n = P_n$. All maps preserve the relevant Sobolev spaces by the same compact path-product coordinate construction. Coefficient composition in the instantaneous forms is exactly $\kappa m(n/m) = \kappa n$.

Let $\hat{Q} = I - I_{m,n} I_{m,n}^*$ on $\mathcal{K}_m$. On $\hat{Q}\mathcal{K}_m \cap \mathcal{V}_m$, let $t_{22,z}$ be the restriction of the already completed form $s_{m,z}$. Define its other block forms by

$$\begin{aligned} t_{11,z}(f, g) &= s_{m,z}(I_{m,n}f, I_{m,n}g), \\ t_{21,z}(h, f) &= s_{m,z}(h, I_{m,n}f), \\ t_{12,z}(f, h) &= s_{m,z}(I_{m,n}f, h). \end{aligned}$$

The operator $T_{22,z}$ associated to $t_{22,z}$ is positive with lower bound z . Its inverse also acts from its form dual to its form space, by the coercive weak equation. In that sense the next eliminated component is $h_f = -T_{22,z}^{-1} t_{21,z}(\cdot, f)$, and the next form is

$$s_{n,z}(f, g) = t_{11,z}(f, g) - t_{12,z}(f, T_{22,z}^{-1} t_{21,z}(\cdot, g)). \quad (45)$$

This notation means substitution of the unique weak solution, so does not require multiplying unspecified unbounded compressions.

Theorem 6.1. *Equation (45) equals the direct fine-to- n form (34). In particular*

$$S_{n,z}^{-1} = I_{m,n}^* S_{m,z}^{-1} I_{m,n} = J_n^* (\mathcal{L} + z)^{-1} J_n. \quad (46)$$

No memory generated at an earlier level is lost in the next level.

Proof. For prescribed $f \in \mathcal{V}_n$, every $u \in \mathcal{V}$ with $J_n^* u = f$ has a unique orthogonal decomposition $u = J_n f + J_m h + k$, where $h \in \hat{Q}\mathcal{K}_m \cap \mathcal{V}_m$ and $k \in \mathcal{H}_{Q_m} \cap \mathcal{V}$; obtain it by the nested projections in (44). Put $v_h = J_m(I_{m,n}f + h)$ to keep both arguments of the sesquilinear form explicit. The exact variational identity (37) gives

$$\begin{aligned} s_{n,z}(f, f) &= \inf_{J_n^* u = f} [q(u, u) + z\|u\|^2] \\ &= \inf_h \inf_k [q(v_h + k, v_h + k) + z\|v_h + k\|^2] \\ &= \inf_h s_{m,z}(I_{m,n}f + h, I_{m,n}f + h). \end{aligned}$$

The infima have unique minimizers by coercivity and the closed form-space constraints. The weak equation for the last minimizer is precisely the $T_{22,z}$ equation above. Expanding the last form and polarizing proves (45). Alternatively compress (35) by $I_{m,n}$ and use (44); this proves (46) directly and confirms the same unique closed form. Both arguments include all intermediate eliminated components. $\square$

To display the generated terms explicitly, write $\mathcal{M}_{m,z}(f, g) = \langle B_m f, (C_m + z)^{-1} B_m g \rangle$. Then each of $t_{11,z}, t_{12,z}, t_{21,z}, t_{22,z}$ is the corresponding block of $a_m + z\langle \cdot, \cdot \rangle - \mathcal{M}_{m,z}$. In particular the two off-diagonal blocks contain the corresponding off-diagonal terms of $-\mathcal{M}_{m,z}$, and $T_{22,z}$ contains its entire diagonal term. Setting these terms to zero would change (45). The exact recursion is independent of the order of the specified nested eliminations; it does not reinitialize the dynamics at each marginal A_m .

7 A large-block Wilson channel with uniform norm and spectral bounds

Let C be a closed coarse edge word using each coarse physical edge at most once. Its fine path then uses each of its fine physical edges once; let r be its number of coarse edges and $\ell = mr$ its number of fine edges. For example, every elementary coarse face has $r = 4$, fine perimeter $\ell = 4m$, and physical side length ma . Keep its original ordered word, with inverses on the negative traversals, and set

$$W_C(z) = \text{tr} \prod_{(\gamma, \epsilon) \in C} z_\gamma^\epsilon, \quad \overline{W}_C = \mu_m(W_C), \quad r_C = \mu_m(W_C^2),$$

$$f_C = W_C - \overline{W}_C, \quad u_C = \psi J_m f_C, \quad d_C = \|u_C\|_\lambda^2 = r_C - (\overline{W}_C)^2. \quad (47)$$

SU(2) traces are real. The fine and coarse gauge actions both fix the closed trace, so u_C is a physical invariant vector, orthogonal to ψ . The subtraction is the actual interacting mean.

Lemma 7.1. *For every such loop, at arbitrary positive coupling,*

$$\alpha \leq d_C \leq 4, \quad \alpha \leq r_C \leq 4 - 3\alpha. \quad (48)$$

The exact unnormalized excitation energy and bounds are

$$\mathfrak{q}_{H-E_0}[u_C] = \kappa\ell \left(1 - \frac{r_C}{4}\right), \quad \frac{3}{4}\alpha\kappa\ell \leq \mathfrak{q}_{H-E_0}[u_C] \leq \kappa\ell. \quad (49)$$

Consequently

$$\frac{3\alpha}{16}\kappa\ell \leq \frac{\mathfrak{q}_{H-E_0}[u_C]}{d_C} = \kappa\ell \frac{1 - r_C/4}{r_C - (\overline{W}_C)^2} \leq \frac{\kappa\ell}{\alpha}. \quad (50)$$

Proof. Fix any one fine edge used by the loop and condition on all other fine variables. The ordered holonomy is $Au^{\pm 1}B$ for fixed group matrices A, B , so its Haar distribution is Haar by invariance and inversion. Writing $U = u_0I + i\sum_a u_a\sigma_a$ identifies Haar measure with the uniform measure on the unit sphere in $\mathbb{R}^4$. Its coordinate second moments are $1/4$ by sign and permutation symmetry and $\sum_{a=0}^3 u_a^2 = 1$. Thus $\int W_C du = 0$, $\int W_C^2 du = 1$, and $\int (1 - W_C^2/4) du = 3/4$. The conditional density is bounded below by α from Theorem 2.2. Integrating the two nonnegative functions W_C^2 and $4 - W_C^2$ proves the two bounds on r_C . For every real constant c , conditional integration gives $\int (W_C - c)^2 p(u | y) du \geq \alpha \int (W_C - c)^2 du = \alpha(1 + c^2) \geq \alpha$. Taking c to be that conditional mean proves conditional variance at least α . The law of total variance, obtained by expanding around the conditional mean and integrating the cross term to zero, proves $d_C \geq \alpha$. Since $|W_C| \leq 2$, $d_C \leq 4$.

For a positive occurrence, a based form of the derivative is $\text{tr}(T_a A_C)$ with $A_C \in \text{SU}(2)$ having trace W_C ; for a negative occurrence it is $-\text{tr}(T_a A'_C)$. This follows from $X_a u^{-1} = -u^{-1}T_a$ and cyclic trace invariance. The Pauli expression above gives $\sum_a |\text{tr}(T_a A_C)|^2 = 1 - W_C^2/4$. There are exactly ℓ used fine edges. Insert this identity in (5); centering leaves derivatives unchanged. The exact energy and its lower and upper bounds follow. Divide using $\alpha \leq d_C \leq 4$ to obtain (50). $\square$

These bounds are for the same density used by the block projection. For an additional consistency check, that coarse density itself has a single-coarse-link conditional lower bound α , independent of m . Indeed change only the last fine edge of a specified chain, holding all y and all other coarse links fixed in (15). Theorem 2.2 bounds the ratio of the two full densities by R^2 . Integrating the same y on both sides bounds the ratio $r_m(z_\gamma, z_{\neq \gamma})/r_m(z'_\gamma, z_{\neq \gamma})$ by R^2 . Averaging over z'_γ proves the assertion. This exact argument does not equate single-link conditional bounds with independence or a global Poincare inequality.

Theorem 7.2. *Define the unnormalized actual full-Hamiltonian spectral measure*

$$\mathbf{M}_C(B) = \langle u_C, \mathbf{1}_B(H - E_0)u_C \rangle, \quad \mathbf{M}_C([0, \infty)) = d_C. \quad (51)$$

No division by d_C is made in this measure. It has no atom at zero. Its full first and second moments satisfy

$$\int \omega d\mathbf{M}_C = \kappa\ell(1 - r_C/4),$$

$$\int \omega^2 d\mathbf{M}_C = \|A_m f_C\|^2 + \|B_m f_C\|^2 \leq C_*^2 \kappa^2 \ell^2. \quad (52)$$

$$\Omega_C = \frac{3\alpha}{32} \kappa\ell, \quad c_H = \frac{9\alpha^3}{1024C_*^2} > 0, \quad \boxed{\mathbf{M}_C([\Omega_C, \infty)) \geq c_H d_C}. \quad (53)$$

In particular c_H is independent of the increasing blocking length m and of total box volume. The full compressed resolvent, including its memory, obeys at the scalar shift $z = \kappa\ell$

$$z \langle f_C, S_{m,z}^{-1} f_C \rangle \leq d_C \left[1 - c_H \frac{3\alpha/32}{1 + 3\alpha/32} \right]. \quad (54)$$

Proof. The state is smooth and orthogonal to the unique vacuum. Spectral calculus and (49) prove the first moment and the absence of a zero atom. Every used edge has fundamental Casimir $3/4$, including inverse occurrences, so $-\sum_{e,a} X_{e,a}^2 W_C = 3\ell W_C/4$. Combining (4) with (10) and $|\nabla_e W_C| \leq 1$ gives the pointwise bound

$$|\mathcal{L} J_m f_C| = |\mathcal{L} J_m W_C| \leq \kappa\ell \left(\frac{3}{2} + 2G \right) = \kappa\ell C_*.$$

The second-moment bound follows by integration and (40). In particular the term $\|B_m f_C\|^2$ has not been deleted from this spectral measure.

Put $A = 3\alpha/16$. The first moment is at least $A\kappa\ell d_C$ by (50). With $\Omega_C = A\kappa\ell/2$, its part on $[\Omega_C, \infty)$ is at least $A\kappa\ell d_C/2$, since the integral over $[0, \Omega_C)$ is at most $\Omega_C d_C$. Cauchy-Schwarz against the unnormalized $\mathbf{M}_C$ gives

$$(A\kappa\ell d_C/2)^2 \leq \left(\int \omega^2 d\mathbf{M}_C \right) \mathbf{M}_C([\Omega_C, \infty)).$$

Using (52) gives the stronger intermediate bound $\mathbf{M}_C([\Omega_C, \infty)) \geq A^2 d_C^2 / (4C_*^2)$. Since $d_C \geq \alpha$, this is at least $A^2 \alpha d_C / (4C_*^2) = c_H d_C$, with the state mass retained. Finally (35) gives $z \langle f_C, S_{m,z}^{-1} f_C \rangle = \int z/(\omega + z) d\mathbf{M}_C$. On $[\Omega_C, \infty)$ its integrand is at most $z/(\Omega_C + z)$, and elsewhere it is at most 1. Its upper bound is therefore $d_C - \Omega_C \mathbf{M}_C([\Omega_C, \infty)) / (\Omega_C + z)$. Insert (53) and $z = \kappa\ell$ to obtain (54). $\square$

The same channel can be dressed by the complete eliminated resolvent with a quantitative bound that continues to hold for growing m . Define

$$\tau_* = \frac{8G}{\sqrt{\alpha c_H}}, \quad z_C = \tau \kappa\ell, \quad \tau \geq \tau_*. \quad (55)$$

For $f = f_C$ use the full u_{z_C} of (41), with all fine eliminated degrees of freedom in C_m .

Corollary 7.3. *This is a nonzero physical vector with exact norm and energy (42)–(43). Quantitatively,*

$$\begin{aligned} d_C &\leq \|u_{z_C}\|^2 \leq d_C + \alpha c_H/4, \\ \|\mathbf{1}_{[\Omega_C, \infty)}(H - E_0)u_{z_C}\|^2 &\geq \frac{c_H}{4 + c_H} \|u_{z_C}\|^2, \end{aligned} \tag{56}$$

$$\frac{\Omega_C c_H}{4 + c_H} \leq \frac{\mathfrak{q}_{H-E_0}[u_{z_C}]}{\|u_{z_C}\|^2} \leq \kappa \ell \frac{1 - r_C/4}{d_C}. \tag{57}$$

Proof. By (23), each used coarse edge contributes at most $4G\kappa m$ to the pointwise absolute value of $B_m f_C$; there are r such edges and $|D_\gamma W_C| \leq 1$. Thus $\|B_m f_C\| \leq 4G\kappa \ell$. Since $C_m \geq 0$, its shifted inverse has norm at most $1/z_C$. Consequently

$$\|w_{z_C}\| \leq 4G/\tau \leq \frac{1}{2}\sqrt{\alpha c_H} \leq \frac{1}{2}\sqrt{d_C c_H}.$$

Orthogonality gives the norm bound. Multiplication by ψ is unitary, and an orthogonal spectral projection is contractive. Hence the triangle inequality and (53) give

$$\|\mathbf{1}_{[\Omega_C, \infty)}(H - E_0)u_{z_C}\| \geq \sqrt{d_C c_H} - \|w_{z_C}\| \geq \frac{1}{2}\sqrt{d_C c_H}.$$

The retained norm satisfies $\|u_{z_C}\|^2 = d_C + \|w_{z_C}\|^2 \leq d_C(1 + c_H/4)$, so the squared lower bound $d_C c_H/4$ is at least $c_H \|u_{z_C}\|^2/(4 + c_H)$. Integrating $\omega \geq \Omega_C$ over this weight proves the lower energy bound; (43) proves the upper bound. The large positive shift here is explicitly fixed in (55); this argument does not assert the same lower bound for the separately defined vector minimizing the energy numerator at $z = 0$. Its full Rayleigh variational problem is calculated in Section 8. $\square$

8 Zero-shift reconstruction and affine physical states

We now calculate the zero-shift channel without transferring the positive-shift estimate by assertion. Gauge invariance is imposed throughout this section. Let $\mathcal{H}^G$ be the closed subspace of gauge-invariant functions in $\mathcal{H}$, and let $\mathcal{K}_m^G$ be the corresponding coarse subspace. The gauge group is the compact product $\text{SU}(2)^{V_L}$; its Haar average is an orthogonal projection. The intertwining of the path products and the gauge descent of the conditional density show that J_m, J_m^*, P_m, Q_m commute with the corresponding gauge actions. Consequently B_m, C_m and their resolvents preserve these subspaces. All physical vectors below are obtained by the same unitary multiplication by ψ .

Take $m \geq 2$ dividing $2L$. In this section write C for the restriction of C_m to $\mathcal{K}_{Q_m}^G$, retaining precisely its form domain $\mathcal{V}_{Q_m} \cap \mathcal{H}^G$. This is a reducing restriction of the operator, not a change of interaction. Put

$$c = \inf \sigma(C), \quad \delta_G = \inf_{\substack{0 \neq u \in \mathcal{V} \cap \mathcal{H}^G \\ \mu u = 0}} \frac{q(u, u)}{\|u\|^2}.$$

This complementary physical space is nonzero. For example, the fine face at $n = (-L, -L, -L)$ in the first two directions contains the direction-2 edge with first coordinate $-L + 1$, which is outside R_m . Varying that edge with every other fine edge fixed leaves every coarse product unchanged and changes the fine face trace. Positivity of the conditional density implies $Q_m W_p \neq 0$, and this function is gauge invariant. Compact embedding of the restricted form domain into L^2 gives compact resolvent for C : a bounded sequence in its form norm is bounded in the original H^1 norm. Thus c is an eigenvalue.

Proposition 8.1. *With the original number of fine edges $N = 3(2L)(2L + 1)^2$, define*

$$\Delta_L = \frac{3\kappa}{4}\alpha^N = \frac{3g^2}{2a}\alpha^{3(2L)(2L+1)^2}. \quad (58)$$

Then $\delta_G \geq \delta \geq \Delta_L > 0$ and $c \geq \delta_G$. For mean-zero $f \in \mathcal{V}_m \cap \mathcal{K}_m^G$, define the linear map

$$\mathcal{R}_m f = J_m f - C^{-1} B_m f, \quad s_{m,0}(f, g) = a_m(f, g) - \langle B_m f, C^{-1} B_m g \rangle. \quad (59)$$

The form $s_{m,0}$ is closed on the indicated retained form space, with

$$\Delta_L \|f\|^2 \leq s_{m,0}(f, f) \leq a_m(f, f), \quad a_m(f, f) \leq 2(1 + \eta_m/\Delta_L) s_{m,0}(f, f). \quad (60)$$

The map $\mathcal{R}_m$ takes this space into the original physical mean-zero form domain and satisfies $J_m^ \mathcal{R}_m f = f$. For every $h \in \mathcal{V}_{Q_m} \cap \mathcal{H}^G$,*

$$q(\mathcal{R}_m f, h) = 0, \quad q(\mathcal{R}_m f, \mathcal{R}_m g) = s_{m,0}(f, g). \quad (61)$$

For its associated mean-zero operator $S_{m,0}$, the exact domain map is

$$f \in D(S_{m,0}) \iff \mathcal{R}_m f \in D(\mathcal{L}), \quad \mathcal{L} \mathcal{R}_m f = J_m S_{m,0} f. \quad (62)$$

The equivalence in (62) is for f already in the retained mean-zero form domain.

Proof. Change fine variables one at a time between any two configurations. Theorem 2.2 gives $\rho_{\max}/\rho_{\min} \leq R^{2N} = \alpha^{-N}$. The product Haar Laplacian has first positive eigenvalue $3/4$: tensor products of the complete matrix-coefficient bases in Lemma 2.1 have eigenvalues $\sum_e j_e(j_e + 1)$, whose least positive value is $3/4$. Expanding a smooth function in that basis, and then using H^1 density, gives the Haar Poincare inequality. Hence

$$\text{Var}_\mu u \leq \rho_{\max} \text{Var}_\lambda u \leq \frac{4\rho_{\max}}{3} \sum_{e,a} \int |X_{e,a} u|^2 d\lambda \leq \frac{4\rho_{\max}}{3\kappa\rho_{\min}} q(u, u).$$

The first inequality evaluates the infimum over constants defining Var_μ at the Haar mean. This proves (58) for all mean-zero functions, and restricting to gauge-invariant functions preserves the lower bound. Every complementary physical function has zero mean, so $c \geq \delta_G$.

The bounded resolvent C^{-1} maps its Hilbert space to $D(C)$ and therefore to the form domain. Completing the square at zero gives

$$q(J_m f + h, J_m f + h) = s_{m,0}(f, f) + c_m(h + C^{-1} B_m f, h + C^{-1} B_m f).$$

The unique minimizer is $\mathcal{R}_m f$, proving (61) and the upper bound in (60). Its mean is zero and its squared norm is $\|f\|^2 + \|C^{-1} B_m f\|^2$. The already proved full Poincare inequality gives the lower bound and $\|\mathcal{R}_m f\|^2 \leq s_{m,0}(f, f)/\Delta_L$. Apply (27) to $\mathcal{R}_m f$ for the second inequality in (60). These two-sided form-norm bounds prove closedness with exactly the retained H^1 domain.

If $f \in D(S_{m,0})$, decompose any physical form test vector as $J_m g + h$. Equations (61) give

$$q(J_m g + h, \mathcal{R}_m f) = s_{m,0}(g, f) = \langle J_m g + h, J_m S_{m,0} f \rangle.$$

Constants give zero on both sides. This is the defining weak operator identity. It also holds against arbitrary, noninvariant test vectors: average that test vector over the gauge group, since the operator and the vector being tested are invariant. Thus $\mathcal{R}_m f$ belongs to the full $D(\mathcal{L}) = H^2$ and has the displayed image. Conversely if $\mathcal{R}_m f \in D(\mathcal{L})$, tests in the complementary space show $Q_m \mathcal{L} \mathcal{R}_m f = 0$; retained tests then prove $f \in D(S_{m,0})$ and the claimed image. $\square$

The exponential dependence on N in (58) is explicit. It provides a proved finite-box inverse bound, not a volume-uniform gap.

Fix a nonzero, mean-zero, gauge-invariant $f \in \mathcal{V}_m$. Keep its norm unchanged and set

$$d_f = \|f\|^2, \quad a_f = a_m(f, f), \quad b_f = B_m f, \quad \sigma_f(I) = \|\mathbf{1}_I(C)b_f\|^2.$$

This σ_f is a finite positive measure on $[c, \infty)$ of total mass $\|b_f\|^2$, with no division by that mass. For every real $z > -c$, put

$$M_j(z) = \int_{[c, \infty)} (t+z)^{-j} d\sigma_f(t), \quad j = 1, 2, 3, \quad v_z = J_m f - (C+z)^{-1} b_f. \quad (63)$$

All these moments are finite by $t+z \geq c+z > 0$.

Theorem 8.2. *The actual physical state ψv_z has exact squared norm, excitation energy and Rayleigh quotient*

$$N_f(z) = d_f + M_2(z), \quad e_f(z) = a_f - M_1(z) - zM_2(z), \quad Q_f(z) = \frac{e_f(z)}{N_f(z)}. \quad (64)$$

For all $z > -c$,

$$Q'_f(z) = \frac{2M_3(z)}{N_f(z)}[z + Q_f(z)]. \quad (65)$$

In particular Q_f is strictly increasing on $[0, \infty)$ whenever $b_f \neq 0$. Its zero-shift value is strictly smaller than the lifted quotient a_f/d_f , and some negative shifts give a quotient strictly smaller still. Zero shift minimizes the energy numerator at fixed retained data, but it does not minimize the physical quotient when $b_f \neq 0$.

The full affine variational problem, with all physical eliminated states allowed, is determined as follows:

$$\begin{aligned} \chi_f &= \inf_{h \in \mathcal{V}_{Q_m} \cap \mathcal{H}^G} \frac{q(J_m f + h, J_m f + h)}{d_f + \|h\|^2}, \\ F_f(E) &= a_f - E d_f - M_1(-E), \quad 0 \leq E < c, \quad F'_f(E) = -d_f - M_2(-E) < 0, \\ F_f(0) &> 0, \quad \mathfrak{l}_f = \lim_{E \uparrow c} F_f(E) \in [-\infty, \infty). \end{aligned} \quad (66)$$

If $\mathfrak{l}_f < 0$, there is exactly one root $E_f \in (0, c)$ and

$$\begin{aligned} \chi_f &= E_f, \quad h_f = -(C - E_f)^{-1} b_f, \quad \|\psi(J_m f + h_f)\|^2 = -F'_f(E_f), \\ \mathfrak{q}_{H-E_0}[\psi(J_m f + h_f)] &= -E_f F'_f(E_f). \end{aligned} \quad (67)$$

This is the unique affine minimizer. If $\mathfrak{l}_f \geq 0$, then $\chi_f = c$. In this case b_f is orthogonal to $\ker(C - c)$. The value c is attained exactly when $\mathfrak{l}_f = 0$, and all minimizers are

$$h = -(C - c)^\dagger b_f + k, \quad k \in \ker(C - c). \quad (68)$$

Here the dagger is the inverse on $\ker(C - c)^\perp$ and zero on the kernel. When $\mathfrak{l}_f > 0$, no finite affine vector attains c ; the infimum is approached by $h = tk$, $t \rightarrow \infty$, for a nonzero $k \in \ker(C - c)$. These alternatives are exhaustive for the actual interacting C and b_f , including $b_f = 0$.

Proof. The spectral theorem identifies M_1 with $\langle b_f, (C+z)^{-1}b_f \rangle$ and M_2 with $\|(C+z)^{-1}b_f\|^2$. The orthogonal decomposition of v_z and the block form (32) prove (64) for negative as well as positive z . Differentiating these bounded resolvents gives $M'_1 = -M_2$, $M'_2 = -2M_3$, $N'_f = -2M_3$, and $e'_f = 2zM_3$. The quotient rule proves (65). Each v_z is nonzero and mean-zero, so $Q_f(z) \geq \delta_G > 0$. For $b_f \neq 0$, $M_3 > 0$ and the derivative is positive at zero and at every positive shift. Also

$$\frac{a_f}{d_f} - Q_f(0) = \frac{M_1(0) + (a_f/d_f)M_2(0)}{d_f + M_2(0)} > 0.$$

Continuity of the derivative at zero proves strict improvement for sufficiently small negative shifts. The statement about the numerator follows independently from (61); increasing the norm is part of the exact quotient calculation.

For $0 \leq E < c$, complete the shifted square for arbitrary h :

$$q(J_m f + h, J_m f + h) - E(d_f + \|h\|^2) = F_f(E) + \|(C - E)^{1/2}[h + (C - E)^{-1}b_f]\|^2. \quad (69)$$

No discarded term or approximation enters this identity. The sign and derivative statements in (66) follow from Proposition 8.1 and the spectral derivative. Monotonicity bounds the limit above by $F_f(0)$ and allows the value $-\infty$ below. The intermediate value theorem proves the unique root when the limit is negative. At this root (69) bounds every quotient below by E_f , and equality holds precisely for the stated h_f . The norm and energy in (67) follow from (64) and $F_f(E_f) = 0$.

If the limit is nonnegative, $F_f(E) > 0$ for every $E < c$. Equation (69), followed by $E \uparrow c$, bounds every affine quotient below by c . A complementary eigenvector k of eigenvalue c yields

$$\frac{a_f + 2t \operatorname{Re}\langle b_f, k \rangle + t^2 c \|k\|^2}{d_f + t^2 \|k\|^2} \longrightarrow c,$$

proving equality of the infimum. If b_f had a nonzero projection on that eigenspace, its contribution to $M_1(-E)$ would diverge as $(c - E)^{-1}$, contradicting $\mathfrak{l}_f \geq 0$. Compact resolvent isolates c from the rest of $\sigma(C)$, so $(C - c)^\dagger$ is bounded on the complementary eigenspaces and maps into $D(C)$. Completing the square at $E = c$ now gives

$$q(J_m f + h, J_m f + h) - c(d_f + \|h\|^2) = \mathfrak{l}_f + \|(C - c)^{1/2}[h + (C - c)^\dagger b_f]\|^2.$$

It proves both the attainment criterion and every vector in (68). $\square$

The scalar root in (66) solves the affine problem for this fixed retained direction. It is not, in general, an eigenvalue of the full operator: a full eigenvector also requires the weak equation $s_{m,-E}(g, f) = 0$ for every retained test direction g . Here $s_{m,-E}$ denotes the sesquilinear expression (34) at $z = -E$, restricted to physical retained functions; it is well defined on $\mathcal{V}_m \cap \mathcal{K}_m^G$ because $E < c$. Indeed complementary testing gives $h = -(C - E)^{-1}B_m f$, and retained testing gives exactly that additional equation. Conversely these equations imply the full weak eigenvector equation and its H^2 operator domain. This proves the exact map in both directions, and explains precisely which test directions the affine minimization uses. The energy-dependent eigenvalue and reconstruction principle agrees with [1, Theorem 1.2 and Corollary 1.3]; the affine classification and the physical domains above were proved directly.

Corollary 8.3. *For the loop $f = f_C$, the zero-shift state $u_0 = \psi \mathcal{R}_m f_C$ obeys the exact identities*

$$\begin{aligned} \|u_0\|^2 &= d_C + M_2(0), \\ \mathfrak{q}_{H-E_0}[u_0] &= \kappa \ell(1 - r_C/4) - M_1(0). \end{aligned} \quad (70)$$

$$\alpha \leq d_C \leq \|u_0\|^2 \leq d_C + \min \left\{ \frac{\kappa\ell(1-r_C/4)}{\Delta_L}, \frac{16G^2\kappa^2\ell^2}{\Delta_L^2} \right\}, \quad (71)$$

$$\begin{aligned} \|u_0\|^2 &\leq 4 + \frac{4\ell}{3\alpha^N}, \\ \Delta_L &\leq \frac{\mathfrak{q}_{H-E_0}[u_0]}{\|u_0\|^2} \leq \frac{\kappa\ell(1-r_C/4)}{d_C}. \end{aligned} \quad (72)$$

For an elementary coarse face, the last norm bound is $4 + 16m/(3\alpha^N)$, and its energy numerator is exactly $8g^2m(1-r_C/4)/a - M_1(0)$. The same $\Delta_L = 3g^2\alpha^N/(2a)$ bounds the quotient below. These bounds are finite and explicit; their volume dependence is retained.

Proof. The identities are (64) at zero. Positivity gives $M_1(0) \leq a_f$, and the spectral inequality $t^{-2} \leq c^{-1}t^{-1}$ gives $M_2(0) \leq a_f/c \leq a_f/\Delta_L$. The other estimate uses $M_2(0) \leq \|B_m f_C\|^2/\Delta_L^2$ and the already proved $\|B_m f_C\| \leq 4G\kappa\ell$. Insert $a_f = \kappa\ell(1-r_C/4) \leq \kappa\ell$ and $d_C \leq 4$. The lower quotient bound is the full Poincare inequality, and the upper one follows from the exact numerator and norm. Finally use $\ell = 4m$ and $\kappa = 2g^2/a$. $\square$

9 Exact comparison of all physical low-energy states

The previous section treated a fixed retained observable. We now give a quantitative relation encompassing all physical states, with no assumed complementary gap uniform in volume. Define the actual zero-shift reconstructed infimum

$$\lambda_m^R = \inf_{\substack{0 \neq f \in \mathcal{V}_m \cap \mathcal{H}_m^G \\ \mu_m f = 0}} \frac{s_{m,0}(f, f)}{\|f\|^2 + \|C^{-1}B_m f\|^2}. \quad (73)$$

This denominator is the full physical norm. In every fixed box it is positive, and $\lambda_m^R \geq \delta_G \geq \Delta_L$.

Theorem 9.1. *For every allowed L, m, a, g with $m \geq 2$, the following comparison holds with numerical constants independent of all four parameters:*

$$\frac{\lambda_m^R c}{\lambda_m^R + c} \leq \delta_G \leq \min\{\lambda_m^R, c\}, \quad \frac{1}{2} \min\{\lambda_m^R, c\} \leq \delta_G. \quad (74)$$

It has the following explicit state map. For every mean-zero physical $u \in \mathcal{V}$, put

$$f = J_m^* u, \quad v = \mathcal{R}_m f, \quad k = u - v = Q_m u + C^{-1}B_m f. \quad (75)$$

Then $k \in \mathcal{V}_{Q_m} \cap \mathcal{H}^G$ and

$$q(u, u) = s_{m,0}(f, f) + c_m(k, k), \quad \|u\|^2 = \|v\|^2 + \|k\|^2 + 2\operatorname{Re}\langle v, k \rangle. \quad (76)$$

At least one of the nonzero components v, k has physical Rayleigh quotient at most $2q(u, u)/\|u\|^2$. Conversely each such component is an actual admissible physical state under multiplication by ψ .

For every sequence of allowed boxes, block lengths, positive spacings and couplings, δ_G tends to zero if and only if $\min\{\lambda_m^R, c\}$ tends to zero. Uniform positive lower bounds for δ_G are equivalent to simultaneous uniform positive lower bounds for these two actual sector infima. These are equivalences proved by (74), not assumed estimates for either sector.

Proof. Sobolev preservation of J_m^* and Proposition 8.1 make all components in (75) well defined in their stated form domains. The identity $J_m^* v = f$ puts k in the complement. Equation (61) gives $q(v, k) = 0$, which proves the energy identity; the norm identity retains its cross term. Cauchy–Schwarz in two scalar components gives

$$\|u\| \leq \|v\| + \|k\| \leq \sqrt{\frac{q(v, v)}{\lambda_m^R}} + \sqrt{\frac{c_m(k, k)}{c}} \leq \sqrt{\left(\frac{1}{\lambda_m^R} + \frac{1}{c}\right) q(u, u)}.$$

Zero components contribute zero. Taking the infimum over u proves the first lower bound. The upper bound follows because the full variational set contains both the reconstructed vectors and all complementary physical vectors. For positive x, y , $xy/(x + y) \geq \min(x, y)/2$, proving the last bound.

For the explicit factor-two state assertion, $\|u\|^2 \leq 2(\|v\|^2 + \|k\|^2)$ and the energy identity show

$$\min_{\substack{w \in \{v, k\} \\ w \neq 0}} \frac{q(w, w)}{\|w\|^2} \leq \frac{q(v, v) + q(k, k)}{\|v\|^2 + \|k\|^2} \leq 2 \frac{q(u, u)}{\|u\|^2}.$$

This construction changes no physical coefficient or state norm. All assertions about sequences and uniform bounds follow directly from the two-sided inequalities for each member of the sequence. $\square$

This comparison calculates how a putative collective low-energy sector must appear under the exact block map: it appears either among the full zero-shift reconstructed states, with their eliminated norm retained, or among physical states in the complementary sector. The decomposition (75) exhibits the vectors and controls their energy quotients by the numerical factor two. It does not assert which alternative occurs in the interacting large-volume limit, or that either one actually becomes soft. The fixed-loop high-energy fraction in Theorem 7.2 continues to hold in its full scope. Equations (70)–(72) and Theorem 9.1 now supply the proved zero-shift and all-state relations without extending that fraction to an unsupported family of vectors. For $m = 1$, J_1 is the identity, $Q_1 = B_1 = 0$, and reconstruction is the identity; the physical gap relation is then exactly $\lambda_1^R = \delta_G$ with no complementary sector.

10 Physical skeleton morphism and uniform complementary inverse

This section uses the complete fixed-point and physical-resolvent proof retained in **references/LANE_A_ELIMINATION_INPUTS.md**, Sections 1, 10, 11, 12 and 15. The input concerns the identical open box, every original link and face, and the actual positive unit vacuum. Here is its exact dictionary. Its g_{YM} is our g , its $\mathcal{E}$ is our E_0 , and its coordinate action $\alpha_h(U)_e = h_{s(e)} U_e h_{t(e)}^{-1}$ obeys

$$(h \cdot U)_e = h_{s(e)}^{-1} U_e h_{t(e)} = (\alpha_{h^{-1}} U)_e.$$

Thus its Hilbert-space representation $\mathsf{T}_h u = u \circ \alpha_{h^{-1}}$ is exactly $\mathsf{T}_h u(U) = u(h \cdot U)$ here. The identity map on functions intertwines these representations, on L^2 , H^1 and H^2 ; no assertion that parameter inversion is a group homomorphism is needed. The two Hamiltonians and their domains coincide. Uniqueness of the positive unit vacuum then proves equality of the two ψ , of $\rho = \psi^2$, and of the ground energies. The source scalar in its (12.7) will be written $c_{\text{vac}} = \langle 1, \psi \rangle$ here; it is not the density constant α in (58).

For precision, the proved input is

$$\delta_G \geq \Delta_{\text{sc}} := 3\kappa \left(1 - \frac{512}{3}\xi\right) = \kappa(3 - 512\xi) \geq \frac{287\kappa}{96}, \quad 0 < \xi \leq \frac{1}{49152}. \quad (77)$$

Its entire proof is retained, not a proposed additional hypothesis. The exact source construction maps a collection of nonconstant Haar sectors z_I to $\phi = e^{\mathfrak{C}}1$, where $\mathfrak{C} = \sum_{I \neq \emptyset} |z_I\rangle\langle 1_I| \otimes I_{I^c}$ is bounded and nilpotent in each finite box. The inverse is $e^{-\mathfrak{C}}$, with both exponentials finite polynomials preserving $D(H_0) = H^2$. Its full fixed-point proof gives

$$\psi = c_{\text{vac}}\phi, \quad c_{\text{vac}} > 0, \quad c_{\text{vac}}^2 \|\phi\|^2 = 1, \quad e^{-\mathfrak{C}}(H - E_0)e^{\mathfrak{C}} = H_0 + \mathcal{K}, \quad H_0 = \kappa \sum_e E_e.$$

These maps preserve the vacuum scalar and all of $2bM$ in the eigenvalue equation. The source proves that every iterate, hence both similarities, commutes with every vertex gauge transformation. In the orthogonal sector decomposition a physical nonempty support has no degree-one vertex: averaging at such a vertex would give both the same invariant vector and its zero one-link Haar average. Every nonempty surviving support therefore contains at least four edges of this open bipartite cubic graph. With the original E_e Casimir, $H_0 \geq 3\kappa$ on the physical nonconstant sectors. The complete interaction estimate proved there is $\|\mathcal{K}f\|_{\oplus,1} \leq (512\xi/3)\|H_0f\|_{\oplus,1}$, where $\|f\|_{\oplus,1} = \sum_I \|\Pi_I f\|$ and $\|f\| \leq \|f\|_{\oplus,1} \leq 2^{N/2}\|f\|$. The retained proof supplies every commutator and counting estimate and the contraction on $\|z\|_* \leq 1/16$, with constant at most $11/36$ in the displayed range. The sector norm equivalence is explicitly volume dependent. The physical Neumann resolvent excludes every $0 < t < 3\kappa(1 - 512\xi/3)$, transfers under the displayed domain-preserving similarity, and proves (77) for the original self-adjoint operator. No form positivity of a nonunitary similarity or deletion of its norm factors is used.

The retained proof derives numerical constants for the creation method of Yarotsky [7], Section 2, source equations `mmo`, `c1`, `vsum`, `th1`, `sumj`. Its physical-space reference is Baez [8], the section “Gauge Theory on a Graph,” Lemmas 1, 2 and the source lemma `lem2.5`. Baez’s link convention $A_e \mapsto h_{t(e)}A_e h_{s(e)}^{-1}$ is related to the input’s $U_e \mapsto h_{s(e)}U_e h_{t(e)}^{-1}$ by the involution $j(U)_e = U_e^{-1}$: direct multiplication gives $j(\alpha_h U)_e = h_{t(e)}j(U)_e h_{s(e)}^{-1}$. The pullback is a Haar-unitary map with itself as inverse, preserves H^1, H^2 because inversion is a bi-invariant metric isometry, and intertwines each E_e . The raw Wilson products in our Hamiltonian are unchanged. Both original TeX sources are retained.

The source conditional complement retains individual fine links. Our complement retains their ordered products. The following proves the exact relation, including the Hilbert and operator domains.

Proposition 10.1. *Let $2 \leq m \mid 2L$, $s = (U_e)_{e \in S_m}$, $r = (U_e)_{e \in R_m}$, and define*

$$\mathcal{A}_m = \text{SU}(2)^{R_m}, \quad \bar{\rho}_R(r) = \int \rho(s, r) d\lambda_{S_m}(s), \quad \mathcal{K}_R = L^2(\mathcal{A}_m, \bar{\rho}_R \lambda_R).$$

Let $J_S : \mathcal{K}_R \rightarrow \mathcal{H}$ be $J_SF(s, r) = F(r)$, with

$$(J_S^* u)(r) = \frac{1}{\bar{\rho}_R(r)} \int \rho(s, r) u(s, r) d\lambda_{S_m}(s), \quad P_S = J_S J_S^*.$$

Write $\mathcal{K}_R^G$ for invariance under the endpoint action of all original vertices. There is a unitary map of physical spaces

$$\iota_m : \mathcal{K}_m^G \longrightarrow \mathcal{K}_R^G, \quad (\iota_m f)(r) = f((u_{\gamma,1} \cdots u_{\gamma,m})_{\gamma}). \quad (78)$$

Its inverse on arbitrary L^2 classes is

$$(\iota_m^{-1}F)(z) = \int F(\Theta_m^{-1}(v, z)) d\lambda_v(v), \quad \Theta_m : \mathcal{A}_m \longrightarrow \left(\prod_{\gamma} \mathrm{SU}(2)^{m-1} \right) \times \mathcal{Q}_m, \quad r \longmapsto (v, z). \quad (79)$$

The inverse coordinates are $u_{\gamma,k} = v_{\gamma,k}$ for $k < m$ and $u_{\gamma,m} = (v_{\gamma,1} \cdots v_{\gamma,m-1})^{-1} z_{\gamma}$. Both maps identify the invariant H^1 and H^2 spaces, and

$$J_S \iota_m = J_m, \quad J_S^*|_{\mathcal{H}^G} = \iota_m J_m^*|_{\mathcal{H}^G}, \quad P_S|_{\mathcal{H}^G} = P_m|_{\mathcal{H}^G}. \quad (80)$$

Consequently the identity on functions is an isometry from the source physical conditional complement for $S = S_m$ onto $\mathcal{H}_{Q_m}^G$. It identifies their full form domains, their complementary operator domains, their operators and their resolvents. In the range (77),

$$C \geq \Delta_{\mathrm{sc}} I, \quad \|C^{-1}\| \leq \Delta_{\mathrm{sc}}^{-1} \leq \frac{96}{287\kappa}. \quad (81)$$

Proof. The two inverse coordinate compositions and the Haar identity for Θ_m follow by the exact prefix cancellations already proved for Φ_m , with all S_m factors omitted from both sides of this particular map. Integrating those factors in the density gives

$$\bar{\rho}_R(\Theta_m^{-1}(v, z)) = r_m(z), \quad \bar{\rho}_R d\lambda_R = r_m(z) d\lambda_v d\lambda_m.$$

To prove the first equality, perform simultaneously the internal vertex transformations with fixed coarse endpoints as in Section 3; they send (v, z) to $(1, z)$, preserve the marginal by substitution on S_m , and leave z unchanged. Integrating in v then identifies the remaining function with the definition of r_m .

There is also an L^2 proof of the descent, without evaluation on a measure-zero section. For a chain with endpoints fixed, the internal gauge action is $v'_1 = u_1 h_1$ and $v'_k = h_{k-1}^{-1} u_k h_k$ for $1 < k < m$. For fixed input r , Haar integration successively over $h_1, \dots, h_{m-1}$ makes the variables $v'_1, \dots, v'_{m-1}$ product Haar; the chain product remains z . Applying this to every disjoint interior shows that the compact internal gauge average on $L^2(\bar{\rho}_R \lambda_R)$ is precisely fibre integration in v followed by constant lift. Hence every invariant F equals that lift almost everywhere. The remaining coarse endpoint action is $z_{\gamma} \mapsto h_x^{-1} z_{\gamma} h_{x+m\mathbf{e}_i}$; this proves the full coarse invariance of (79) and the fine invariance of (78). Substitution proves both inverse identities. Moreover

$$\|\iota_m f\|_{\mathcal{K}_R}^2 = \int |f(z)|^2 r_m(z) d\lambda_v d\lambda_m = \|f\|_{\mathcal{H}_m}^2.$$

No vector is divided by its mass. Smooth coordinate pullback on these compact products preserves each Sobolev order. Differentiating under the compact fibre integral and applying Cauchy–Schwarz bounds the inverse in H^1 and H^2 . Smooth positive weights give the same Sobolev sets in each finite box. Approximation, followed by compact gauge averaging, proves the claimed domain identifications.

The first identity in (80) follows from the coordinate product. For physical u , substitution in the conditional integral shows that $J_S^* u$ is invariant under every skeleton endpoint action. In (v, z) coordinates it is therefore independent of v almost everywhere. Its v integral equals $r_m(z)^{-1} \int \rho(s, \Theta_m^{-1}(v, z)) u(s, \Theta_m^{-1}(v, z)) d\lambda_{S_m} d\lambda_v$, which is exactly $J_m^* u$. This proves the second identity and then the third. It does not assert equality of the projections on all noninvariant vectors.

Thus the two physical complements are the same closed subspace of $\mathcal{H}$, with the same form q , including derivatives in every fine link. Both have form domain H^1 intersected with that space. The weak operator-domain rule is, in both presentations,

$$\begin{aligned} D(C) &= \{u \in H^1 \cap \mathcal{H}_{Q_m}^G : \text{there exists } w \in \mathcal{H}_{Q_m}^G \text{ such that} \\ &\quad q(v, u) = \langle v, w \rangle \text{ for every } v \in H^1 \cap \mathcal{H}_{Q_m}^G\}, \\ Cu &= w. \end{aligned}$$

Equality of the tests, inner products and forms proves equality of the operators and their resolvents, not merely of their formal differential expressions. Finally every vector in the complement has $\mu u = 0$, and $\mathcal{U}u = \psi u$ is physical and perpendicular to ψ . The already-proved (77) applied under this unchanged unitary gives (81). $\square$

Theorem 10.2. *In the range (77), for every mean-zero $f \in \mathcal{V}_m \cap \mathcal{H}_m^G$, put $b_f = B_m f$, $d_f = \|f\|^2$, $a_f = a_m(f, f)$ and retain the raw measure $\sigma_f(I) = \|1_I(C)b_f\|^2$. Then*

$$s_{m,0}(f, f) \geq \Delta_{\text{sc}}(d_f + \|C^{-1}b_f\|^2), \quad \lambda_m^R \geq \Delta_{\text{sc}}, \quad c \geq \Delta_{\text{sc}}. \quad (82)$$

For the unchanged mean-centered simple loop f_C , let ℓ be its fine perimeter and d_C, r_C retain their earlier definitions. The physical zero-shift state $u_0 = \psi \mathcal{R}_m f_C$ has the exact mass and energy

$$\|u_0\|^2 = d_C + \int t^{-2} d\sigma_{f_C}(t), \quad (83)$$

$$\mathfrak{q}_{H-E_0}[u_0] = \kappa\ell(1 - r_C/4) - \int t^{-1} d\sigma_{f_C}(t) \geq \Delta_{\text{sc}}\|u_0\|^2. \quad (84)$$

Its quantitative upper mass bound is

$$\|u_0\|^2 \leq d_C + \min \left\{ \frac{\kappa\ell(1 - r_C/4)}{\Delta_{\text{sc}}}, \frac{16G^2\kappa^2\ell^2}{\Delta_{\text{sc}}^2} \right\} \leq 4 + \frac{96\ell}{287}. \quad (85)$$

For every $z \geq 0$, define $w_z = (C + z)^{-1}b_f$, $N_f(z) = d_f + \|w_z\|^2$ and $e_f(z) = a_f - \int (t + z)^{-1} d\sigma_f - z \int (t + z)^{-2} d\sigma_f$. These are the full physical mass and excitation energy of $\psi(J_m f - w_z)$. The exact differences are

$$w_0 - w_z = zC^{-1}(C + z)^{-1}b_f, \quad (86)$$

$$e_f(z) - e_f(0) = \int \frac{z^2}{t(t + z)^2} d\sigma_f(t), \quad (87)$$

$$N_f(0) - N_f(z) = \int \left(\frac{1}{t^2} - \frac{1}{(t + z)^2} \right) d\sigma_f(t), \quad (88)$$

$$s_{m,z}(f, f) - s_{m,0}(f, f) - zd_f = \int \frac{z}{t(t + z)} d\sigma_f(t). \quad (89)$$

Writing $\theta = z/(\Delta_{\text{sc}} + z)$, their estimates are

$$\begin{aligned} \|w_0 - w_z\| &\leq \theta \sqrt{a_f/\Delta_{\text{sc}}}, \\ 0 &\leq e_f(z) - e_f(0) \leq \theta^2 a_f, \\ 0 &\leq N_f(0) - N_f(z) \leq \left[1 - \left(\frac{\Delta_{\text{sc}}}{\Delta_{\text{sc}} + z} \right)^2 \right] \frac{a_f}{\Delta_{\text{sc}}}, \\ 0 &\leq s_{m,z}(f, f) - s_{m,0}(f, f) - zd_f \leq \theta a_f. \end{aligned} \quad (90)$$

The constants controlling the inverse do not depend on box or block size. The displayed a_f , ℓ , G , κ , raw norms and spectral measures keep their full dependence.

Proof. The map $\mathcal{R}_m$ and its exact domain and energy identities were proved in Proposition 8.1. Its image under $\mathcal{U}$ is physical and perpendicular to ψ . Applying (77) proves (82), including the infimum inequality. The previous exact inverse-moment formulas give (83)–(84). Positivity of $s_{m,0}$ gives $\int t^{-1} d\sigma_f \leq a_f$. Since $\text{supp } \sigma_f \subset [\Delta_{\text{sc}}, \infty)$,

$$\int t^{-2} d\sigma_f \leq \min\{a_f/\Delta_{\text{sc}}, \|b_f\|^2/\Delta_{\text{sc}}^2\}.$$

Insert the original loop identities $a_f = \kappa\ell(1 - r_C/4)$, $\|b_f\| \leq 4G\kappa\ell$, $d_C \leq 4$, $a_f \leq \kappa\ell$ and (77) to prove (85).

The resolvent identity proves (86). Subtraction of the complete energy integrands gives $t^{-1} - (t+z)^{-1} - z(t+z)^{-2} = z^2/[t(t+z)^2]$. The norm subtraction and the definition of $s_{m,z}$ give the remaining exact differences. On the support of the raw measure, $z/(t+z) \leq \theta$ and $1 - [t/(t+z)]^2 \leq 1 - [\Delta_{\text{sc}}/(\Delta_{\text{sc}} + z)]^2$. Apply these inequalities to the corresponding nonnegative integrands and use the preceding inverse-moment bounds. Squaring (86) first proves its norm estimate. This proves every inequality in (90), with no removal of the measure or its total mass $\|B_m f\|^2$. $\square$

For an elementary coarse face $\ell = 4m$, the last mass bound is $4 + 384m/287$, and its physical side length remains ma . In original parameters the new lower energy scale and range are

$$\Delta_{\text{sc}} = \frac{2g^2}{a} \left(3 - \frac{128}{g^4}\right) \geq \frac{287g^2}{48a}, \quad g^4 \geq 12288.$$

The expectation of H is still $E_0 N_f(z) + e_f(z)$. This is a statement for the interacting finite boxes, uniform as their sizes vary within this explicit coupling range. It takes no spatial continuum limit and provides no small- g continuation. Every induced score covariance and the full resolvent and time memory remain those of Sections 4–6.

11 Original scales and the exact scope of the spectral conclusion

For an elementary coarse face $\ell = 4m$. The actual state has

$$\|u_C\|^2 = r_C - (\overline{W}_C)^2 \in [\alpha, 4], \quad (91)$$

$$\mathfrak{q}_{H-E_0}[u_C] = \frac{8g^2m}{a} \left(1 - \frac{r_C}{4}\right) \in \left[\frac{6\alpha g^2m}{a}, \frac{8g^2m}{a}\right], \quad (92)$$

$$\frac{\mathfrak{q}_{H-E_0}[u_C]}{\|u_C\|^2} \in \left[\frac{3\alpha g^2m}{2a}, \frac{8g^2m}{a\alpha}\right], \quad \Omega_C = \frac{3\alpha g^2m}{4a}. \quad (93)$$

Here $\alpha = 1/(9 \cdot 12^{16/g^4})$, and G, C_*, c_H are the explicit functions (7), (53) of $d = 4/g^4$. The physical length is ma , so $g^2m/a = g^2(ma)/a^2$ as an exact rewriting with no coordinate change. The expectation of the unsubtracted Hamiltonian is

$$\langle u_C, H u_C \rangle = E_0(r_C - (\overline{W}_C)^2) + \frac{8g^2m}{a}(1 - r_C/4).$$

The same scalar $E_0\|u_{z_C}\|^2$ is added to the dressed excitation energy. Also $0 \leq E_0 \leq 2bM$ follows from $H \geq 0$ and the constant unit Haar trial vector, whose plaquette means vanish. The original $2bM$ term has not been omitted from H .

The exact instantaneous kinetic term has the two metric expressions

$$-\kappa m \Delta^T = -(\kappa m/4) \Delta^c, \quad \kappa m = 2g^2 m/a.$$

If its coefficient is written solely as $2g_m^2/(ma)$, algebra forces $g_m^2 = m^2 g^2$. This is only the coefficient dictionary for this marginal kinetic form. It does not make (30) and the energy-dependent memory into the original one-plaquette Hamiltonian at a claimed running coupling.

The map from the spatial coarse observable to the quantum theory is exactly $f \mapsto \psi J_m f$, followed by (51); the retained resolvent is (35), and sequential eliminations obey (45). Theorems 7.2 and Corollary 7.3 are statements about that full quantum spectral measure, with no omission of generated interactions. For fixed $g > 0$, these particular large-loop channels retain a positive fraction of weight at energies proportional to $g^2 m/a$, including the specified fully dressed channels. Their unnormalized spectral mass above the displayed threshold is bounded below by the stated positive multiple of their unchanged squared norm, at every increasing m and fixed a, g .

This assertion does not give a lower bound on the support of the whole measure: some weight can lie below Ω_C . Nor does a lower bound on the quotient of this family bound the infimum over all physical states. At variable $g = g(a)$ the constants α and c_H have the explicit coupling dependence above and can tend to zero. The calculation therefore proves neither a continuum mass gap nor a four-dimensional mass-gap counterexample. It supplies an exact extensive block map and an all-positive-coupling, volume-independent quantitative channel result; Theorem 9.1 additionally relates every physical low-energy state to the zero-shift reconstructed and complementary sectors, with parameter-independent comparison constants and explicit state maps. Section 10 now supplies the proved physical lower bound $\Delta_{sc} = \kappa(3 - 512\xi)$, uniform in box and block size for $0 < \xi \leq 1/49152$, together with the exact skeleton/complement morphism and full-memory zero-shift bounds. The all-positive-coupling finite bound (58) remains valid with its displayed volume dependence. Neither the comparison nor the exhaustive affine calculation in Theorem 8.2 presumes which sector becomes soft or an answer for the joint spatial continuum limit.

12 Primary-source map and reproducibility

The general Feshbach–Schur construction is attributed to its established literature. The locally retained source of Dusson, Sigal and Stamm was read at Theorem 1.2, its reconstruction map, and Eqs. (1.10)–(1.16). Their Corollary 1.3 is the energy-dependent fixed-point formulation. Its full eigenvalue condition tests all retained directions; the scalar affine root (66) tests the fixed retained direction only, as proved after Theorem 8.2. Their operator $H - \lambda$ corresponds here to $\mathcal{L} + z$, with $\lambda = -z$, their complementary inverse to $(C_m + z)^{-1}$, and their reconstruction to (36). Their effective interaction has the negative sign in (34). The infinite-rank form argument and the conditional coefficients are proved in this paper.

Bakry and Ledoux’s reverse Poincaré estimate is their Eq. (1.7), with $d(t) = 2t$ at curvature bound zero, on pp. 686–687. Our single-edge Markov generator is $\kappa \Delta^T$ and its carré du champ is $\kappa \sum_a (X_a f)^2$, giving precisely (8). Its full compact-group proof was included. Simon’s locally retained Theorem 1.1 uses the Euclidean generator $-\Delta/2$ and its Brownian-bridge formula. It is a primary reference for positive potential weights, not a theorem identifying Euclidean space with this compact configuration product; the compact parabolic comparison needed here was proved in (11). Mori’s projection-memory method provides the historical comparison: its projection is Eq. (2.13) and its exact memory equations are Eqs. (3.1)–(3.10), pp. 430–431. Our evolution uses $-\mathcal{L}$ on functions with the vacuum inner product; Mori’s evolution uses a Liouville operator on dynamical

variables. Equation (38) derives the corresponding Euclidean block identity here, including its signs and domains.

The companion exact checker verifies the entire open-box skeleton and path counts on several finite boxes, chain disjointness, nested product paths, every retained loop orientation, exact Pauli rotations and the mixed derivative tensor in explicit noncommuting samples, and exact rational Schur associativity and spectral constants. The zero-shift checker separately verifies physical norm and quotient derivatives, all six interior/boundary and coupled/uncoupled affine examples, the complete energy and norm decomposition, and the harmonic gap comparison in 81 rational interacting block examples. The physical-complement checker additionally verifies noncommuting chain gauge actions, their inverse compositions, the exact original strong-coupling constants, and the full inverse-moment continuity identities and inequalities. These finite checks diagnose algebra; they do not stand in for the all-box proofs above or compute a substitute vacuum. No Lean run or large parallel job is used.

13 Material-tensor transfer into interacting local-energy states

This section retains the original Wilson Hamiltonian, vacuum, gauge action, spacing and coupling from Section 1. It constructs a particular observable map from the full Fabel material tensor. No equality between fluid evolution and quantum Hamiltonian evolution is asserted. The map and its exact spectral measures are calculated below; the remaining infinite-volume spectral question is not replaced by the value of a coordinate.

13.1 Transport conventions and the vacuum-moment correction

For the column-section connection $d + \mathcal{A}$, ordinary parallel transport along an oriented edge $e : [0, a] \rightarrow \mathbb{R}^3$ solves

$$P'_e(s) = -\mathcal{A}(e(s))(\dot{e}(s))P_e(s), \quad P_e(0) = I.$$

Its gauge transform is $P_e^h(s) = h(e(s))^{-1}P_e(s)h(e(0))$. Differentiating this formula verifies the transformed ODE and its initial value; uniqueness gives the assertion. The links in Section 1 have the opposite endpoint convention and are therefore *inverse* transports:

$$U_e = P_e(a)^{-1}, \quad U_e^h = h(e(0))^{-1}U_e h(e(a)), \quad U'(s) = U(s)\mathcal{A}(e(s))(\dot{e}(s)). \quad (94)$$

Inverses and concatenations follow by multiplication of the corresponding transport maps. Thus this is an exact gauge-equivariant map for arbitrary noncommuting connection components. It is not a global inverse from finite links to connections.

The Cartan specialization $\mathcal{A}_i = \lambda u_i T$, $T = -i\sigma_3/2$, has

$$U_e = \exp\left(\lambda T \int_e u \cdot dx\right), \quad U_p = \exp(\vartheta_p T), \quad \vartheta_p = \lambda \oint_{\partial p} u \cdot dx.$$

With positively oriented (i, j) faces, Stokes' theorem gives $\vartheta_p = \lambda \int_p (\partial_i u_j - \partial_j u_i) dx^i dx^j$. For a fixed smooth field on a compact region it equals $a^2 \lambda (\partial_i u_j - \partial_j u_i) + O(a^3)$. In the general non-Abelian case the four edge expansions to second order give $U_p = I + a^2 F_{ij} + O(a^3)$: the differentiated terms are $\partial_i \mathcal{A}_j - \partial_j \mathcal{A}_i$ and the four products give $[\mathcal{A}_i, \mathcal{A}_j]$, with this positive sign. For uniform C^2 bounds the remainder is uniform on that compact region. As $\text{tr } e^{\vartheta T} = 2 \cos(\vartheta/2)$, the exact magnetic value is

$$b(2 - W_p) = \frac{2 - 2 \cos(\vartheta_p/2)}{2g^2 a} = \frac{\vartheta_p^2}{8g^2 a} + O_g(\vartheta_p^4/a). \quad (95)$$

For fixed g, λ and a fixed smooth field, summation on a regular mesh of a bounded region yields $\lambda^2(8g^2)^{-1} \int |\text{curl } u|^2 dx$. This follows by a Riemann sum; each leading cell term is a^3 times the displayed density and each uniform cell error is $O(a^4)$. Pointwise curvature growth alone does not imply growth of this integral: the volume of the concentrating region must also be accounted for.

In Lorentzian coordinates $X^0 = ct, X^i = x^i$, the same Cartan field with $\mathcal{A}_0 = 0$ has the complete conserved current

$$\begin{aligned} j_0 &= 0, \\ j_i &= \frac{\lambda}{g^2} (\Delta u_i - c^{-2} \partial_t^2 u_i) T \\ &= \frac{\lambda}{g^2} \left[\Delta u_i - c^{-2} \left(\nu \Delta w_i - \sum_j w_j \partial_j u_i - \sum_j u_j \partial_j w_i - \partial_i \partial_t p + \partial_t f_i \right) \right] T, \quad w_i = \partial_t u_i. \end{aligned}$$

This is obtained by differentiating the original Navier–Stokes equation, not by changing its viscosity or force. $D^\nu j_\nu = 0$ follows from incompressibility and the common Cartan factor. The field solves that sourced classical equation. It can be used as data for a quantum trial state without claiming it solves source-free Yang–Mills. Proving source-free classical dynamics is not a prerequisite for defining a trial state; its physical Hilbert and spectral properties must instead be proved.

There is a separate important correction concerning Wilson observables. Evaluating W_C at the links of a background gives a number $W_C(U[\mathcal{A}])$. The functional $W_C(U)$ on the full configuration space has no dependence on the chosen background until an additional prescription supplies it. The generic centered state $\psi(W_C - \mu W_C)$ is physical and nonzero, but naming a background does not make it a background-dependent state map. Moreover $r_C = \mu(W_C^2)$ is the actual quantum second moment. It is not $W_C(U[\mathcal{A}])^2$.

Keep the exact bound from Section 7, $\alpha \leq r_C \leq 4 - 3\alpha$, with $\alpha = (9 \cdot 12^{16/g^4})^{-1} > 0$. Then

$$\frac{3\alpha}{4} \kappa \ell \leq a_C = \kappa \ell (1 - r_C/4) \leq (1 - \alpha/4) \kappa \ell. \quad (96)$$

For physical perimeter $P_C = a\ell$ this bounds the *bare* loop numerator between the displayed constants times $2g^2 P_C/a^2$ at fixed g ; it does not give asymptotic ratio one. The reconstructed numerator and norm remain $a_C - M_1(0)$ and $d_C + M_2(0)$. The lower bound Δ_{sc} for these reconstructed states is used only for $\xi \leq 1/49152$. In a spatial continuum path $a \downarrow 0$, $\kappa \rightarrow 0$ would imply $g^2 = (a/2)\kappa \rightarrow 0$ and thus $\xi \rightarrow \infty$. Without $a \downarrow 0$ the latter implication is not valid.

13.2 The full weighted local-energy operator

For arbitrary real weights $f = (f_e)_{e \in E_L}$ define

$$f_p = \frac{1}{4} \sum_{e \in \partial p} f_e, \quad V_p = b(2 - W_p), \quad D_f = \kappa \sum_e f_e E_e + \sum_p f_p V_p, \quad \Xi_f = (D_f - \langle \psi, D_f \psi \rangle) \psi. \quad (97)$$

All sums include the boundary edges and faces. The electric-only operator $\sum f_e E_e$ is a different operator. On a Peter–Weyl product sector with edge spins j_e , the signed electric sum in D_f has eigenvalue $\kappa \sum_e f_e j_e(j_e + 1)$. Its domain consists of square-summable coefficients with the squares of these eigenvalues as weights. Addition of the bounded real magnetic multiplication preserves that self-adjoint domain. Smooth functions lie in the domain, and D_f maps smooth functions to smooth functions. In particular Ξ_f is smooth and belongs to every power domain of H . Gauge

invariance of each link Casimir and each face trace gives gauge invariance of Ξ_f . Its definition proves $\langle \psi, \Xi_f \rangle = 0$ with no rescaling.

Put $\mathcal{E} = E_0$ and $\mathcal{A} = H - \mathcal{E}$. The product rule, including both derivative terms, gives

$$[E_e, V_p]F = - \sum_a [(X_{e,a}^2 V_p)F + 2(X_{e,a} V_p)X_{e,a}F], \quad (98)$$

$$[H, D_f] = \kappa \sum_p \sum_{e \in \partial p} (f_p - f_e)[E_e, V_p]. \quad (99)$$

The first expression vanishes when $e \notin \partial p$. The second identity follows by expanding the commutator: electric terms commute with electric terms, magnetic multiplications commute, and the two remaining electric-magnetic sums have the stated coefficient. It proves

$$\mathcal{A}\Xi_f = [H, D_f]\psi, \quad q_{\mathcal{A}}[\Xi_f] = \frac{1}{2}\langle \psi, [D_f, [H, D_f]]\psi \rangle.$$

For the raw positive measure $\nu_f(B) = \|1_B(\mathcal{A})\Xi_f\|^2$, $B \subset (0, \infty)$, spectral integration yields

$$\|\Xi_f\|^2 = \int d\nu_f, \quad q_{\mathcal{A}}[\Xi_f] = \int E d\nu_f, \quad \|[H, D_f]\psi\|^2 = \int E^2 d\nu_f. \quad (100)$$

All moments are finite by smoothness. For a constant $f_e = c_0$, $D_f = c_0 H$, so its expectation is $c_0 \mathcal{E}$ and $\Xi_f = 0$. This is an exact cancellation, not a zero-energy nonzero state.

13.3 Coordinate-level cubic symmetries and the full covector map

Write $m(e) = n + \mathbf{e}_i/2$ for edge midpoints. A signed permutation matrix R maps the vertex box to itself. If $R\mathbf{e}_i = \epsilon\mathbf{e}_j$, then the positive image edge of $e = (n, i)$ is (Rn, j) for $\epsilon = 1$, and $(Rn - \mathbf{e}_j, j)$ for $\epsilon = -1$. Define $\Gamma_R(U)$ at this image edge to be U_e^ϵ . The inverse is $\Gamma_{R^{-1}}$; composing the signs and endpoints verifies both inverse compositions. Every midpoint is mapped to $Rm(e)$. Each face becomes a face, possibly with its orientation reversed. Its $\text{SU}(2)$ trace is unchanged under reversal. Inversion of a group variable is Haar preserving and takes the left Casimir to the equal right Casimir, since the metric is bi-invariant. Thus the pullback unitary $\mathsf{T}_R F = F \circ \Gamma_R^{-1}$ preserves H , H^1 and H^2 and intertwines the full gauge actions by $h'(Rn) = h(n)$. Uniqueness of the positive unit vacuum gives $\mathsf{T}_R \psi = \psi$. For weights transported by $f_R(e') = f(e)$ it follows that $\mathsf{T}_R D_f \mathsf{T}_R^{-1} = D_{f_R}$ and $\mathsf{T}_R \Xi_f = \Xi_{f_R}$. Spectral projections of $\mathcal{A}$ commute with T_R .

Let $\Xi_r = \Xi_{m_r}$. Reflection in coordinate r changes Ξ_r to $-\Xi_r$ and fixes Ξ_s for $s \neq r$; permutations exchange the three. Consequently, for every real $t \in \mathbb{R}^3$ and $c_0 \in \mathbb{R}$,

$$\Xi_{c_0+t \cdot m} = \sum_r t_r \Xi_r, \quad \langle D_{c_0+t \cdot m} \rangle = c_0 \mathcal{E}, \quad \nu_{c_0+t \cdot m}(B) = |t|^2 \nu_{m_1}(B). \quad (101)$$

Indeed a reflection negates each off-diagonal matrix element $\langle \Xi_r, 1_B(\mathcal{A})\Xi_s \rangle$ while leaving it invariant; it is therefore zero. Permutations equate the diagonal elements. This proof applies at every positive coupling, including when a state or a measure vanishes.

For the retained Fabel time parameter $z = \sqrt{1 - 8\tau} > 0$, $\tau = (1 - z^2)/8$, the full inverse material derivative is computed from the unchanged real polynomial map

$$\begin{aligned} F_1(x, y, w) &= (1 + xy)^3 w + y^2(1 + xy)(4 + 3xy), \\ F_2(x, y, w) &= y + 3x(1 + xy)^2 w + 3xy^2(4 + 3xy), \\ F_3(x, y, w) &= 2x - 3x^2 y - x^3 w. \end{aligned}$$

Here w denotes the original third spatial coordinate, not time. Expanding the three-by-three determinant of its differentiated matrix gives $\det DF = -2$. At the retained points the full matrices are

$$J_0 = \begin{pmatrix} -9/16 & -3/8 & -1/8 \\ 3/8 & 25/4 & 3/4 \\ -17/2 & -3 & -1 \end{pmatrix}, \quad J_\gamma = \begin{pmatrix} -9z^3/16 & -3z/8 & -1/8 \\ 3z^2/8 & 25/4 & 3/(4z) \\ -17/2 & -3/z^2 & -1/z^3 \end{pmatrix}.$$

Direct matrix multiplication gives

$$C_\tau = \begin{pmatrix} (17 - 9z^3)/8 & (3 - 3z^3)/(4z^2) & (1 - z^3)/(4z^3) \\ (51 - 24z^2 - 27z^3)/16 & (9 + 8z^2 - 9z^3)/(8z^2) & (3 - 3z^3)/(8z^3) \\ (-153 + 36z^2 + 117z^3)/8 & (-27 - 12z^2 + 39z^3)/(4z^2) & (-9 + 13z^3)/(4z^3) \end{pmatrix}. \quad (102)$$

It is $J_0^{-1}B_\tau^{-1}J_\gamma$, where $J = DF$ for the original polynomials and

$$q_0 = (1, -3/2, 13/2), \quad \gamma = (z^{-1}, -3z/2, 13z^2/2), \quad B_\tau = I + 6\tau E_{23}.$$

The factorization gives the inverse $A_\tau = J_\gamma^{-1}B_\tau J_0$ and determinant one. The source matrix entries are reproduced and independently rederived by the companion checker from F .

The covector $\zeta_\tau = C_\tau^\top e_3$ has the three entries in the last row of (102). Choose the explicit affine weight $f_\tau(e) = \zeta_\tau \cdot m(e)$. Then

$$\nu_{f_\tau}(B) = Q(z)\nu_{m_1}(B),$$

where the entire, unmodified scalar is

$$Q(z) = \frac{(-153 + 36z^2 + 117z^3)^2}{64} + \frac{(-27 - 12z^2 + 39z^3)^2}{16z^4} + \frac{(-9 + 13z^3)^2}{16z^6},$$

$$\lim_{z \downarrow 0} z^6 Q(z) = 81/16.$$

Thus the full covector, with its changing direction, is realized by this specified parameter-to-state map. Its raw norm, energy, and second spectral moment all have the same factor $Q(z)$; their ratio does not depend on τ whenever the state is nonzero. This is not an identification of the Fabel cotangent space with the whole quantum state space.

13.4 A non-affine map using every entry of the material tensor

Set $K_\tau = C_\tau C_\tau^\top$ and keep all six independent entries. For any real symmetric matrix S define

$$f_S(e) = m(e)^\top S m(e), \quad \mathfrak{M}_{L,a,g}(S) = \Xi_{f_S} \in C^\infty(\mathcal{Q})^g \cap \{\psi\}^\perp. \quad (103)$$

This is a linear, typed map on the six-dimensional real matrix space. It does not change the Hamiltonian metric: S is the spatial weight of the observable D_f , not a replacement kinetic tensor in H . Using physical midpoints a $m(e)$ multiplies f_S and Ξ_{f_S} by a^2 , and every raw spectral measure by a^4 . Those factors remain present if physical midpoint units are chosen.

Use the three unchanged seed functions

$$f_o = m_1^2 + m_2^2 + m_3^2, \quad f_d = m_1^2 - m_2^2, \quad f_o = 2m_1 m_2,$$

and write ν_o, ν_d, ν_o for their actual centered-state spectral measures. These are positive measures of the full interacting Hamiltonian, not free-channel approximations.

Theorem 13.1. *For every fixed $L \geq 2$, $a, g > 0$, every real symmetric S and every Borel $B \subset (0, \infty)$,*

$$\begin{aligned} \nu_{f_S}(B) &= w_0(S)\nu_0(B) + w_d(S)\nu_d(B) + w_o(S)\nu_o(B), \\ w_0(S) &= (\text{tr } S)^2/9, \\ w_d(S) &= \frac{1}{2} \sum_{i=1}^3 (S_{ii} - \text{tr } S/3)^2, \quad w_o(S) = \sum_{i < j} S_{ij}^2. \end{aligned} \tag{104}$$

The same identity holds after integration against $1, E, E^2$ and e^{-sE} for $s \geq 0$, retaining the corresponding raw norms, excitation energies, current norms, and time correlations.

Proof. Put $u_i = \Xi_{m_i^2}$ and $v_{ij} = \Xi_{2m_i m_j}$. A coordinate reflection fixes each u_i and negates v_{ij} if it reflects just one of i, j . Thus every spectral inner product between a u and a v vanishes. For two different off-diagonal pairs there is a reflection negating exactly one, so their cross term also vanishes. Permutations give the same diagonal spectral norm, equal to $\nu_o(B)$, for the three v_{ij} .

For the three u_i , permutations equate diagonal elements, denoted $d(B)$, and equate off-diagonal elements, denoted $h(B)$. A permutation exchanging two indices shows that $h(B)$ equals its complex conjugate; it is real. The diagonal contribution for $s_i = S_{ii}$ is

$$d \sum_i s_i^2 + 2h \sum_{i < j} s_i s_j = (d - h) \sum_i (s_i - \bar{s})^2 + (d + 2h)(\text{tr } S)^2/3, \quad \bar{s} = \text{tr } S/3.$$

But $\nu_0 = 3d + 6h$ and $\nu_d = 2(d - h)$ by the definitions of the unchanged seed states. Substitution gives (104). All identities are identities of finite positive measures. Integration against the listed nonnegative functions is justified by monotone convergence; smoothness makes the moments finite. $\square$

Let $\eta = (1/4, 3/8, -9/4)^\top$. Every entry of (102) yields $z^3 C_\tau \rightarrow \eta e_3^\top$ and hence

$$z^6 K_\tau \longrightarrow S_* = \eta \eta^\top = \frac{1}{64} \begin{pmatrix} 4 & 6 & -36 \\ 6 & 9 & -54 \\ -36 & -54 & 324 \end{pmatrix}.$$

Consequently the full nonlinear spatial profile $m^\top K_\tau m$ has the following exact endpoint:

$$z^{12} \nu_{f_{K_\tau}} \longrightarrow \frac{113569}{36864} \nu_0 + \frac{100825}{12288} \nu_d + \frac{531}{512} \nu_o. \tag{105}$$

This is convergence in total variation at each fixed regulator and convergence of the first two moments. To prove it, note that each w_j is homogeneous of degree two and continuous in S , and apply (104). The total variation of the difference is bounded by the sum of the three coefficient errors times the finite masses of the seed measures; the moment versions use their finite moments. The constants follow by substituting $S_{*,ii} = 4/64, 9/64, 324/64$ and $S_{*,12} = 6/64, S_{*,13} = -36/64, S_{*,23} = -54/64$. Equivalently $z^6 \Xi_{f_{K_\tau}} \rightarrow \Xi_{f_{S_*}}$ in every Sobolev norm at a fixed regulator, by the linear six-state expression. These displayed rescaled limits record the growth of the raw state; the state itself has not been divided by its norm.

All three coefficients in (105) are positive. For sufficiently small $z > 0$, each $w_j(K_\tau)$ is positive as well. Then the positive-energy support of $\nu_{f_{K_\tau}}$ equals the union of the supports of the nonzero seed measures: a Borel neighborhood has zero mass in the sum precisely when it has zero mass in

each seed. Write $d_j = \int d\nu_j$, $n_j = \int E d\nu_j$. If at least one d_j is nonzero, the limiting quotient is exactly

$$\lim_{\tau \uparrow 1/8} \frac{q\mathcal{A}[\Xi_{f_{K\tau}}]}{\|\Xi_{f_{K\tau}}\|^2} = \frac{\frac{113569}{36864}n_0 + \frac{100825}{12288}n_d + \frac{531}{512}n_o}{\frac{113569}{36864}d_0 + \frac{100825}{12288}d_d + \frac{531}{512}d_o}. \quad (106)$$

This is an evaluated relation between actual moments, not an estimate that any unknown moment is small. If all $d_j = 0$, every state in this quadratic family is zero. The next calculation proves that this latter case does not occur for sufficiently small positive ξ on any fixed box.

13.5 The first surviving state and exact open-boundary coefficients

The full local-energy source proof is retained unchanged as `references/LOCAL_ENERGY_CURRENT_INPUT.md`. We reproduce the coefficient argument needed here. Set $H_0 = \sum_e E_e$ and $\mathcal{W} = \sum_p W_p$. On a fixed box, $\|\mathcal{W}\|_\infty = 2M$ and the nonconstant free gap is $3/4$. The circle $|\zeta| = 3/8$ has free resolvent norm at most $8/3$, so the Neumann series for $H_0 - \xi\mathcal{W}$ converges for $|\xi| < 3/(16M)$. The simple ground projection persists there. For real ξ near zero its positive unit vacuum is real analytic in each Sobolev space, by the eigen-equation and elliptic regularity. Keep its unit-vacuum scalar:

$$\begin{aligned} \psi_\xi &= 1 + \xi\mathcal{W}/3 + \xi^2 B + O_{H^k,L}(\xi^3), \\ H_0 B &= (\mathcal{W}^2 - M)/3, \quad \int B = -M/18. \end{aligned}$$

For two faces sharing one edge, decompose $W_p W_q = P_{pq,0} + P_{pq,1}$ by Haar averaging that edge. The first term is one half the six-edge boundary trace, by the fundamental matrix-element identity $\int U_{ij} \bar{U}_{kl} = \delta_{ik} \delta_{jl}/2$. The shared-edge spins are zero and one, and all six outer spins are $1/2$. Hence their free energies are $9/2, 13/2$ and their squared norms are $1/4, 3/4$. Repeated faces give $W_p^2 - 1$ of energy 8; edge-disjoint distinct faces give $W_p W_q$ of energy 6. All unordered pairs are thus retained in

$$B = -M/18 + \sum_p \frac{W_p^2 - 1}{24} + \sum_{\partial p \cap \partial q = \emptyset} \frac{W_p W_q}{9} + \sum_{|\partial p \cap \partial q|=1} \left(\frac{4}{27} P_{pq,0} + \frac{4}{39} P_{pq,1} \right).$$

Inserting this expression in (97), the first-order electric term equals the first-order magnetic term because $E_f(\mathcal{W}/3) = \sum_p f_p W_p$, where $E_f = \sum_e f_e E_e$. Centering subtracts the expectation of the same vacuum. Repeated and edge-disjoint face pairs cancel at order two. For an adjacent pair with shared edge e , the weighted electric eigenvalues are $3(f_p + f_q) - 3f_e/2$ and $3(f_p + f_q) + f_e/2$. The remaining second-order state is therefore

$$\begin{aligned} \Xi_f &= \kappa \xi^2 Z_f + O_{H^k,L,f}(\kappa \xi^3), \\ Z_f &= \sum_{\{p,q\} \text{ adjacent}} \delta_{pq}(f) (P_{pq,0}/9 - P_{pq,1}/39), \quad \delta_{pq}(f) = f_p + f_q - 2f_e. \end{aligned} \quad (107)$$

Different adjacent pairs have different six-edge outer boundaries. Otherwise their symmetric difference would be a nonempty mod-two two-cycle with at most four elementary faces. A face in that cycle needs four distinct other faces to cancel its four edges (two distinct square faces share at most one edge), a contradiction. Central sign reversal of a boundary link present in only one pair therefore kills the cross integral. It also kills the cross integral with H_0 inserted. Both spin channels remain orthogonal. Thus

$$\|Z_f\|^2 = \frac{49}{13689} \mathcal{D}_f, \quad \langle Z_f, H_0 Z_f \rangle = \frac{2}{117} \mathcal{D}_f, \quad \mathcal{D}_f = \sum_{\{p,q\} \text{ adjacent}} \delta_{pq}(f)^2. \quad (108)$$

The constants are respectively $1/(4 \cdot 9^2) + 3/(4 \cdot 39^2)$ and $(9/2)/(4 \cdot 9^2) + 3(13/2)/(4 \cdot 39^2)$. The central transformation $U_i(n) \mapsto (-1)^{\sum_{j < i} n_j} U_i(n)$ negates every W_p and preserves the link Casimirs. It identifies the centered problems at ξ and $-\xi$, including the centered weighted state. Consequently norm and energy are even:

$$\|\Xi_f\|^2 = \frac{49}{13689} \kappa^2 \xi^4 \mathcal{D}_f + O_{L,f}(\kappa^2 \xi^6), \quad q_A[\Xi_f] = \frac{2}{117} \kappa^3 \xi^4 \mathcal{D}_f + O_{L,f}(\kappa^3 \xi^6). \quad (109)$$

Here are complete all-box counts for the new quadratic profiles. For $e = (n, i)$, put $m = n + \mathbf{e}_i/2$. A face in transverse direction j on side ϵ has center $m + \epsilon \mathbf{e}_j/2$. Averaging its four edge-midpoint quadratic values gives

$$f_p = f_S(e) + \epsilon(Sm)_j + (S_{ii} + 3S_{jj})/8.$$

Indeed its midpoint displacements from its center are $\pm \mathbf{e}_i/2, \pm \mathbf{e}_j/2$; the average correction to the center value is $(S_{ii} + S_{jj})/8$. Consequently opposite coplanar faces give $\delta = (S_{ii} + 3S_{jj})/4$. Two hinged faces in directions j, k on sides ϵ, η give

$$\delta = \epsilon(Sm)_j + \eta(Sm)_k + c_i, \quad c_i = (2S_{ii} + 3S_{jj} + 3S_{kk})/8.$$

For each longitudinal coordinate the count and second midpoint moment are $N_i = 2L$ and $I_2 = L(4L^2 - 1)/6$. For each oriented transverse pair (n, ϵ) satisfying $-L \leq n, n + \epsilon \leq L$, the sums are

$$N = 4L, \quad \sum n = \sum \epsilon = 0, \quad N_2 = \sum n^2 = \frac{2L(2L^2 + 1)}{3}, \quad E_n = \sum \epsilon n = -2L.$$

These follow from the sums over $n = -L, \dots, L-1$ at $\epsilon = 1$ and $n = -L+1, \dots, L$ at $\epsilon = -1$. Expansion of every square now yields the exact finite formula

$$\begin{aligned} \mathcal{D}_S^{\text{cop}} &= N_i(2L-1)(2L+1) \sum_{i \neq j} (S_{ii} + 3S_{jj})^2/16, \\ \mathcal{D}_S^{\text{hinge}} &= \sum_i [I_2 N^2 (S_{ij}^2 + S_{ik}^2) + N_i N N_2 (S_{jj}^2 + S_{kk}^2 + 2S_{jk}^2) \\ &\quad + 2N_i E_n^2 (S_{jj} S_{kk} + S_{jk}^2) + 2c_i N_i N E_n (S_{jj} + S_{kk}) + c_i^2 N_i N^2]. \end{aligned}$$

In each summand $\{j, k\} = \{1, 2, 3\} \setminus \{i\}$ and $\mathcal{D}_S = \mathcal{D}_S^{\text{cop}} + \mathcal{D}_S^{\text{hinge}}$. There is no periodic boundary replacement.

Substitution of the three seed matrices gives

$$\begin{aligned} \mathcal{D}_0 &= 4L(2L-1)(2L+1)(4L^2+3), \\ \mathcal{D}_d &= L(256L^4 + 74L^2 - 21)/6, \\ \mathcal{D}_o &= 64L^3(2L^2+1)/3. \end{aligned} \quad (110)$$

All are positive for $L \geq 2$. The same substitution for $S_* = \eta \eta^\top$, or the already proved cubic decomposition of the coefficient Gram matrix, gives

$$\mathcal{D}_{S_*} = \frac{L(14536832L^4 + 5453566L^2 - 1614327)}{24576} > 0. \quad (111)$$

Equation (109) proves nonzero seed states for sufficiently small positive ξ on each fixed box and proves

$$\frac{q_A[\Xi_{f_{S_*}}]}{\|\Xi_{f_{S_*}}\|^2} = \frac{234}{49} \kappa + O_L(\kappa \xi^2).$$

This is an ordered fixed-box coefficient result, not a volume-uniform remainder bound. In original variables $\kappa^2\xi^4 = 1/(64g^{12}a^2)$ and $\kappa^3\xi^4 = 1/(32g^{10}a^3)$. The L^5 growth in (111) belongs to the computed Taylor coefficient of both raw moments. Their common spatial and material growth does not by itself reduce the quotient.

13.6 Full Schur map and the remaining collective spectral question

For a physical state Ξ_f in the original Haar space set $h_f = \Xi_f/\psi$, which is smooth at each finite regulator, and retain

$$F_f = J_m^* h_f, \quad v_f = \mathcal{R}_m F_f, \quad k_f = Q_m h_f + C^{-1} B_m F_f.$$

These are the exact maps and domains of Theorem 9.1. The identity $h_f = v_f + k_f$ has

$$\begin{aligned} q(h_f, h_f) &= s_{m,0}(F_f, F_f) + c_m(k_f, k_f), \\ \|\Xi_f\|^2 &= \|v_f\|^2 + \|k_f\|^2 + 2 \operatorname{Re}\langle v_f, k_f \rangle, \\ \|v_f\|^2 &= \|F_f\|^2 + \int t^{-2} d\sigma_{F_f}(t), \\ s_{m,0}(F_f, F_f) &= a_m(F_f, F_f) - \int t^{-1} d\sigma_{F_f}(t). \end{aligned}$$

No interaction or norm term has been removed. The signed permutations above preserve the skeleton with $m \mid 2L$: the transverse coarse coordinates $-L + m\{0, \dots, 2L/m\}$ are invariant under sign reversal, and chain products reverse by group inversion. Vacuum conditional averaging therefore intertwines these symmetries by change of variables in its defining integral. So do B_m, C and the full memory kernels; for the latter use functional calculus for e^{-sC} or $(C + z)^{-1}$. Their spectral matrix elements have the same symmetry-forced vanishing cross terms. This does not evaluate the remaining same-sector measures by a free approximation.

In the proved regime $\xi \leq 1/49152$, every nonzero constructed physical state obeys $q/\|\Xi\|^2 \geq \kappa(3 - 512\xi) \geq 287\kappa/96$. Outside that regime the exact three-measure formula remains valid but does not compute the unknown seed measures. On any regulator sequence, at least one of the nonzero three seed states has quotient no larger than the quadratic combination, since (104) makes its quotient a weighted mean with weights $w_j d_j \geq 0$. Thus a vanishing quotient in this full-tensor quadratic family would also give a vanishing quotient in a selected seed sector along that sequence. At fixed regulator (106) instead gives the precise endpoint measure and ratio. No exchange of the material endpoint, infinite-volume limit, or spatial continuum limit has been made. The calculation leaves those three interacting seed-sector limits as substantive research, not as assumed estimates. Section 14 now calculates their complete fixed-box weak-coupling pair measures, proves an injective lowest-band tensor map, and constructs an actual positive-coupling projected Fabel sequence with a vanishing energy quotient. It retains the whole-state raw norm and band fraction. The common interacting continuum identification remains unresolved.

14 The exact lowest-energy image of the full Fabel tensor

This section continues the actual tensor map of Section 13 through a spectral projection of the original interacting Hamiltonian. The resulting states are explicitly derived from the full material tensor. Their positive-coupling finite-regulator energy quotients tend to zero. The construction does not identify their changing Hilbert spaces with an interacting four-dimensional continuum theory.

14.1 Full weighted weak-coupling state map

The source proofs retained with this manuscript are *The original finite-box $SU(2)$ Hamiltonian at small positive coupling*, *Actual weak-coupling vacuum*, *observable and spectral maps*, and *Full-vacuum potential moments and the electric-state graph limit*. We retain their full proofs as authored reference files. Here is the complete dictionary and the additional weighted magnetic calculation.

The original vertex gauge law is $U_e \mapsto h_s^{-1} U_e h_t$. Setting $k_v = h_v^{-1}$ gives exactly the same transformation written $U_e \mapsto k_s U_e k_t^{-1}$ in those sources. Thus their invariant function space is identical, with the identity map on link configurations and wavefunctions. Root the specified tree at $(-L, -L, -L)$, using the first coordinate, in order 1, 2, 3, that exceeds $-L$ to select each vertex's parent. If t_v is its ordered root-to-vertex holonomy, put $Z_c = t_{s(c)} U_c t_{t(c)}^{-1}$ for each non-tree chord c . The inverse keeps the tree variables and sets $U_c = t_s^{-1} Z_c t_t$. For each fixed tree configuration these chord changes are independent left/right Haar translations. Fubini gives the exact product measure identity. Based gauge invariance removes the tree variables; the remaining gauge action is simultaneous conjugation of every Z_c . This constructs the same physical Hilbert space, including the residual constraint.

On real cochains with counting inner products let d_0 be the original vertex difference and d_1 the oriented four-edge curl. Let $p_v(x)$ be the additive tree integral and

$$(Tx)_c = p_{s(c)}(x) + x_c - p_{t(c)}(x), \quad G = TT^\top, \quad C = d_1 \iota, \quad R = T^\top G^{-1/2},$$

where ι inserts chords with zero tree entries. Square interchanges of monotone paths prove $\ker T = \ker d_1 = \text{im } d_0$ and $T\iota = I$. Consequently $R^\top R = I$, its range is $\ker d_0^\top$, and $d_1 R = CG^{1/2}$. Choose an orthogonal matrix O with

$$O^\top G^{1/2} C^\top C G^{1/2} O = \Sigma^2, \quad \Sigma = \text{diag}(\sigma_\nu), \quad \sigma_\nu > 0.$$

Thus the columns of RO are an orthonormal transverse edge basis with curl frequencies σ_ν . None of T, G, C, R, O changes the original edge or face inner products.

Write $Z_c = \exp(y_c^\alpha T_\alpha)$, $|y_c| < 2\pi$. Outside a Haar-null set these logarithms are unique. For $\Omega_g = \{x : |x_c| < 2\pi/g\}$ the exact Hilbert map is

$$(\mathcal{B}_g F)(x) = g^{3r/2} \mathcal{J}(gx)^{1/2} F(\exp(gx)), \quad \mathcal{J}(y) = (16\pi^2)^{-r} \prod_c \left(\frac{\sin(|y_c|/2)}{|y_c|/2} \right)^2, \quad (112)$$

extended by zero off Ω_g , where $r = |E_L| - |V_L| + 1$. Change of variables proves that it is an isometry onto the supported subspace; its adjoint includes the reciprocal density and dilation. Both commute with simultaneous colour rotations.

Fix L, a, f before sending g to zero. Put $W_p = \text{tr } U_p$ as before, $w_p = 2 - W_p$, $W = \sum_p w_p$, and $W_f = \sum_p f_p w_p$. The vacuum and spectral source proofs give

$$\mathcal{B}_g \psi_g \longrightarrow \Phi_0, \quad \mathcal{B}_g P_g(I) \mathcal{B}_g^* \longrightarrow P_0(I) \quad \text{in operator norm}$$

for bounded excitation intervals I whose endpoints avoid the comparison spectrum. Here $P_g(I) = \mathbf{1}_I(H_g - \mathcal{E}_g)$ on the physical space and

$$H_{\text{osc}} = \frac{1}{a} \sum_{\nu, \alpha} \left(-2\partial_{z_{\nu, \alpha}}^2 + \frac{\sigma_\nu^2 z_{\nu, \alpha}^2}{8} \right), \quad z^\alpha = O^\top G^{-1/2} x^\alpha.$$

The density and constant Jacobian in this linear coordinate map remain part of the unitary representation. In z coordinates the unit Gaussian has variance $2/\sigma_\nu$ in each real coordinate and ground energy $\mu_0 = 3 \sum_\nu \sigma_\nu / (2a)$.

For specificity, these convergence statements follow from the full operator, as follows. The full potential has a unique zero at all $Z_c = I$, by the contained-square path argument, and its Taylor term is $\frac{1}{4} \sum_\alpha \|Cy^\alpha\|^2$. It bounds $c|y|^2$ from below in a fixed small chart and has positive minimum c_ρ outside that chart. A vector with bounded original energy has chart-tail mass bounded by $2aK/(cR_0^2) + 2aKg^2/c_\rho$ outside $|x| \leq R_0$. Local ellipticity gives a local H^1 bound, so Rellich compactness and that tail estimate give global L^2 compactness. Cut-off invariant oscillator eigenfunctions give the min-max upper bound. The exact partition-of-unity identity for the full kinetic form has error $O_{L,\rho}(g^2)$; the exterior potential and interior quadratic comparison give the lower bound on each finite eigenvalue. The bounded-energy compactness identifies every subsequential eigenvector limit. Equality of finite ranks then proves the stated finite spectral projection convergence. Positivity fixes the vacuum phase. The retained source files contain the complete form estimates, domains, cutoffs and min-max calculation.

The electric graph result for arbitrary real f_e is

$$\mathcal{B}_g \left(g^2 \sum_e f_e E_e \psi_g \right) \longrightarrow - \sum_{\alpha, c, d} (T \operatorname{diag}(f_e) T^\top)_{cd} \partial_{x_c^\alpha} \partial_{x_d^\alpha} \Phi_0.$$

Its derivative control comes from the vacuum equation and all potential moments, not vacuum L^2 convergence alone. Indeed $\sum_{e, \alpha} |X_{e, \alpha} W|^2 \leq 16W$ follows from $\sum_\alpha |X_{e, \alpha} w_p|^2 = w_p - w_p^2/4$ and at most four incident faces. The ground-state transform applied to $W^{k/2}$, with positive regularization, gives

$$b\langle W^{k+1} \rangle \leq \mathcal{E}_g \langle W^k \rangle + 4\kappa k^2 \langle W^{k-1} \rangle.$$

At fixed L, a , $\mathcal{E}_g$ is bounded. Hence every $\langle (W/g^2)^k \rangle$ is bounded. This proves strong convergence of $(W/g^2)\psi_g$ by local Taylor expansion and its third-moment tail bound. The vacuum equation gives the strong graph limit for $g^2 \sum_e E_e \psi_g$. The strongly commuting nonnegative link Casimirs obey $\|\sum_e f_e E_e F\| \leq \|f\|_\infty \|\sum_e E_e F\|$. Apply this inequality to $\psi_g - \mathcal{B}_g^*(\chi_{R_0} \Phi_0)$, use convergence of all differentiated coefficients on the fixed compact support, and then let R_0 grow. Gaussian graph decay proves the displayed arbitrary-weight limit.

The missing magnetic-weight extension is also strong:

$$\mathcal{B}_g((W_f/g^2)\psi_g) \longrightarrow \frac{1}{4} \sum_\alpha (Cx^\alpha)^\top \operatorname{diag}(f_p) Cx^\alpha \Phi_0. \quad (113)$$

To prove it, use the exact inequality $|W_f| \leq \max_p |f_p| W$. The same third-moment tail estimate controls its squared norm outside every chart ball. On a fixed ball every face word has its full Taylor limit; finite summation, strong vacuum convergence and bounded local multipliers give the limit there. The limiting quadratic Gaussian has an integrable squared tail. These statements prove (113) globally, for signed weights as well as positive weights.

Define the full matrices

$$\begin{aligned} A_f &= O^\top R^\top \operatorname{diag}(f_e) R O, & M_f &= O^\top R^\top d_1^\top \operatorname{diag}(f_p) d_1 R O, \\ R_f &= \Sigma^{-1/2} M_f \Sigma^{-1/2} - \Sigma^{1/2} A_f \Sigma^{1/2}. \end{aligned} \quad (114)$$

All magnetic edge averages are the original $f_p = \frac{1}{4} \sum_{\partial p} f_e$. Combining both graph limits with $\kappa = 2g^2/a$ and $b = 1/(2g^2a)$ proves the actual raw centered-vector limit

$$\mathcal{B}_g \Xi_f \longrightarrow V_f = \frac{1}{4a} \sum_{\nu, \eta} (\mathbf{R}_f)_{\nu\eta} \sum_{\alpha=1}^3 a_{\nu, \alpha}^\dagger a_{\eta, \alpha}^\dagger \Phi_0. \quad (115)$$

Here $z_{\nu, \alpha} = \sqrt{2/\sigma_\nu} (a_{\nu, \alpha} + a_{\nu, \alpha}^\dagger)$, with the usual commutator equal to $\delta_{\nu\eta} \delta_{\alpha\beta}$. To check the coefficient, differentiating the Gaussian makes the electric centered creation coefficient $-\Sigma^{1/2} A_f \Sigma^{1/2}/(4a)$. The magnetic quadratic is $\sum_\alpha z_\alpha^\top M_f z_\alpha/(8a)$; its centered creation coefficient is $\Sigma^{-1/2} M_f \Sigma^{-1/2}/(4a)$. The scalar expectations converge by inner products with the converging unit vacuum. This retains both terms and their cancellation; for constant f the matrices give $\mathbf{R}_f = 0$, agreeing with $D_f = fH$.

A diagonal colour contraction has squared norm 6, and an off-diagonal contraction has squared norm 3. Commuting its two annihilators through its two creators proves these values, with different unordered pairs orthogonal. Thus the full limiting raw measure is

$$\nu_{f,0} = \frac{3}{8a^2} \sum_\nu (\mathbf{R}_f)_{\nu\nu}^2 \delta_{2\sigma_\nu/a} + \frac{3}{4a^2} \sum_{\nu < \eta} (\mathbf{R}_f)_{\nu\eta}^2 \delta_{(\sigma_\nu + \sigma_\eta)/a}. \quad (116)$$

Its total mass is $3 \text{tr}(\mathbf{R}_f^2)/(8a^2)$ and its first moment is $3 \text{tr}(\Sigma \mathbf{R}_f^2)/(4a^3)$. Strong vector convergence and finite spectral projection convergence, followed by a finite comparison-energy cutoff, prove weak convergence of the actual raw measures with total-mass convergence. We do not infer convergence of an unbounded energy test from weak convergence. For the bounded spectral intervals used below, the projected energy moments do converge.

14.2 All open-boundary coefficients in the lowest mode block

Put $N = 2L + 1$, $h = \pi/(2N)$, $s = 2 \sin h$, and $\sigma = \sqrt{2}s$. The one-dimensional vertex and edge functions, in the original coordinates, are

$$\begin{aligned} v_0(n) &= N^{-1/2}, & v_1(n) &= -\sqrt{2/N} \sin(\pi n/N), & -L \leq n \leq L, \\ w_1(n) &= -\sqrt{2/N} \cos(\pi(n + \tfrac{1}{2})/N), & & & -L \leq n < L. \end{aligned}$$

Finite geometric sums prove their unit norms, orthogonality of v_0, v_1 , and $dv_1 = sw_1$, $d^*w_1 = sv_1$ with the open endpoints. For each cyclic triple $(i, p, q) = (1, 2, 3), (2, 3, 1), (3, 1, 2)$ define the edge cochain V_i by its only two nonzero directional components:

$$(V_i)_p(n) = \frac{v_0(n_i)w_1(n_p)v_1(n_q)}{\sqrt{2}}, \quad (V_i)_q(n) = -\frac{v_0(n_i)v_1(n_p)w_1(n_q)}{\sqrt{2}}. \quad (117)$$

The exact vertex divergence cancels between the two components. Tensor products give $\langle V_i, V_j \rangle = \delta_{ij}$. Their curls have disjoint face-plane supports and squared norms σ^2 . For example $d_1 V_1$ on plane 23 is $-\sigma v_0(n_1)w_1(n_2)w_1(n_3)$; the other two signs follow from the same original positive face ordering. These are precisely the three lowest transverse modes: the full frequency list is $\sigma_j^2 = \sum_i 4 \sin^2(\pi j_i/(2N))$, $0 \leq j_i \leq N-1$, with multiplicity $k-1$ for $k \geq 2$ positive indices. It follows by tensoring the one-dimensional difference bases; curl on a fixed index triple has squared form $|s|^2|v|^2 - |s \cdot v|^2$. This also proves completeness and all endpoint counts.

Retain $f_S(e) = m(e)^\top S m(e)$ for real symmetric S , $m(e) = n + \mathbf{e}_i/2$. Define

$$\delta_L = \frac{1 - \cot^2 h}{8}, \quad \mathbf{m}_L^2 = \frac{\cos^2 h}{2N^2 \sin^4 h}. \quad (118)$$

Theorem 14.1. *On the three-mode basis (117), the exact block $\mathbf{R}_S^{\min}$ of (114) is*

$$(\mathbf{R}_S^{\min})_{ii} = \sigma \delta_L (\text{tr } S - S_{ii}), \quad (\mathbf{R}_S^{\min})_{ij} = \sigma \mathbf{m}_L^2 S_{ij} \quad (i \neq j). \quad (119)$$

The linear map from real symmetric S to this real symmetric block is injective. Its full raw first-physical-band mass is

$$d_{L,a}(S) = \frac{3\sigma^2}{8a^2} \left[\delta_L^2 \sum_i (\text{tr } S - S_{ii})^2 + 2\mathbf{m}_L^4 \sum_{i < j} S_{ij}^2 \right]. \quad (120)$$

It is strictly positive for every $S \neq 0$.

Proof. Let

$$A = \sum_{n=-L}^L n^2 v_1(n)^2, \quad B = \sum_{n=-L}^{L-1} (n + \tfrac{1}{2})^2 w_1(n)^2, \quad M = \sum_{n=-L}^L n v_1(n) v_0(n).$$

These exact finite sums obey

$$B - A = -\tfrac{1}{4} \cot^2 h, \quad M = -\frac{\sqrt{2} \cos h}{2N \sin^2 h}, \quad M^2 = \mathbf{m}_L^2. \quad (121)$$

Here is a derivation including the endpoint terms. Put

$$D_N(t) = \sum_{n=-L}^L e^{int} = \frac{\sin(Nt/2)}{\sin(t/2)}, \quad D_{N-1}(t) = \sum_{n=-L}^{L-1} e^{i(n+1/2)t}.$$

The second sum equals $D_N(t) \cos(t/2) - \cos(Nt/2)$. Using $\sin^2 u = (1 - \cos 2u)/2$ and $\cos^2 u = (1 + \cos 2u)/2$ gives

$$B - A = \frac{1}{N} \left[-\frac{N(N-1)}{4} - D_N''(2\pi/N) - D_{N-1}''(2\pi/N) \right].$$

Direct differentiation of the displayed quotient and identity yields

$$D_N''(2\pi/N) + D_{N-1}''(2\pi/N) = \frac{N}{4 \sin^2 h} - \frac{N^2}{4}.$$

Substitution gives the first equality in (121). The first derivative at π/N is

$$D_N'(\pi/N) = -\frac{\cos h}{2 \sin^2 h}.$$

This gives the displayed M . Thus neither endpoint lattice has been replaced by an integral.

For a face in plane pq its four edge midpoints have mean $c = n + (\mathbf{e}_p + \mathbf{e}_q)/2$, and averaging the four full quadratics gives

$$f_p = c^\top S c + (S_{pp} + S_{qq})/8.$$

For the mode normal to i , parity removes the mixed-coordinate terms in both diagonal expectations. The electric expectation is $(S_{pp} + S_{qq})(A + B)/2 + S_{ii}C_0$ and the unit-curl face expectation is $(S_{pp} + S_{qq})(B + 1/8) + S_{ii}C_0$, where $C_0 = \sum n^2 v_0(n)^2$ is retained in both and cancels exactly. Their

difference is $(S_{pp} + S_{qq})((B - A)/2 + 1/8)$, which is $\delta_L(\text{tr } S - S_{ii})$. The common frequency factors in (114) give the diagonal part of (119).

For distinct normal indices i, j , the two cochains share one edge direction. The product of their two signed components is minus one half of the squared w_1 factor times one $v_0 v_1$ factor in each of the other directions. Only the term $2S_{ij}n_i n_j$ in the full edge quadratic survives the two odd sums. The electric matrix entry is therefore $-S_{ij}M^2$. Their curls have disjoint face-plane supports, so the magnetic matrix entry is zero even with arbitrary face weights. The negative electric sign in (114) gives the positive off-diagonal entry in (119).

Since $L \geq 2$, $0 < h \leq \pi/10$; hence $\delta_L < 0$ and $\mathbf{m}_L^2 > 0$. A zero block first forces every off-diagonal S_{ij} to vanish and then $S_{ii} = \text{tr } S$ for all i . Summing gives $\text{tr } S = 3 \text{tr } S$, so all entries vanish. For an explicit inverse, if $B = \mathbf{R}_S^{\min}/\sigma$, then $\text{tr } S = \text{tr } B/(2\delta_L)$, $S_{ii} = \text{tr } B/(2\delta_L) - B_{ii}/\delta_L$, and $S_{ij} = B_{ij}/\mathbf{m}_L^2$. Finally the six colour contractions at energy $2\sigma/a$ and their squared norms in (116) give (120). $\square$

The first physical energy is $\Delta = 2\sigma/a$ with multiplicity six. Every one-quantum state transforms as a colour vector and has no invariant component. Among two-quanta states the only energy Δ comes from the three lowest modes, and rotation invariance of a bilinear tensor forces the scalar colour contraction. The next transverse frequency is $\sqrt{3}s$: triples $(1, 1, 1)$ give it, whereas a triple including 2 has squared frequency at least $(1 + 4\cos^2 h)s^2 > 3s^2$. All other two-quanta energies are at least $(\sqrt{2} + \sqrt{3})s/a > (11/10)\Delta$. Every state with three or more quanta has energy at least $3\sigma/a > (11/10)\Delta$. Consequently

$$I_{L,a} = (\Delta/2, 11\Delta/10)$$

isolates precisely the first six-dimensional physical comparison space. For $S \neq 0$, (115) and spectral projection convergence prove

$$\|P_g(I_{L,a})\Xi_{f_S}\|^2 \longrightarrow d_{L,a}(S) > 0. \quad (122)$$

This is a conclusion for the actual positive-coupling vector, not an assignment of the comparison Gaussian as its vacuum.

14.3 The entire raw state and its lowest-band fraction

Set

$$\mathcal{N}_{L,a}(S) = \frac{3}{8a^2} \text{tr}(\mathbf{R}_{f_S}^2).$$

This is the limiting squared norm of the entire unchanged tensor state, including all higher mode pairs in (116). It is positive for $S \neq 0$, since its lowest block is nonzero. The exact limiting fraction in the first physical band is therefore

$$\rho_{L,a}(S) = \frac{d_{L,a}(S)}{\mathcal{N}_{L,a}(S)} = \frac{\text{tr}((\mathbf{R}_S^{\min})^2)}{\text{tr}(\mathbf{R}_{f_S}^2)}. \quad (123)$$

This ratio records two separately retained raw masses; it does not change the vector. Both strong vector and projection convergence prove that the corresponding fraction for Ξ_{f_S} at positive g tends to (123).

There is an explicit lower bound retaining all original modes:

$$1 \geq \rho_{L,a}(S) \geq \frac{\sigma^2 \delta_L^2}{432rL^4} > 0, \quad r = 4L^2(4L + 3). \quad (124)$$

To prove it, let $V = RO$ and $Y = d_1 V \Sigma^{-1}$. Both are isometries into the original edge and face spaces, respectively. Equation (114) becomes

$$\mathbf{R}_{f_S} = \Sigma^{1/2} [Y^\top \text{diag}(f_p)Y - V^\top \text{diag}(f_e)V] \Sigma^{1/2}.$$

Every original midpoint has squared length at most $B_L = 2L^2 + (L - \frac{1}{2})^2 < 3L^2$. Thus $|f_e|, |f_p| \leq B_L \|S\|_{\text{op}}$. Since $\sigma_{\max}^2 \leq 12$, compression and finite-rank Hilbert–Schmidt estimates give

$$\text{tr}(\mathbf{R}_{f_S}^2) \leq 4\sigma_{\max}^2 r B_L^2 \|S\|_{\text{op}}^2 \leq 432r L^4 \|S\|_{\text{op}}^2.$$

On the lowest block, $\sum_i (\text{tr } S - S_{ii})^2 = \sum_i S_{ii}^2 + (\text{tr } S)^2$. Moreover

$$\frac{\mathbf{m}_L^2}{|\delta_L|} = \frac{4 \cos^2 h}{N^2 \sin^2 h \cos(2h)} > \frac{16}{\pi^2} > 1.$$

Here $\cos(2h) > 0$, $\cos^2 h > \cos(2h)$ and $\sin h < h$ give the strict inequalities. Consequently $\text{tr}((\mathbf{R}_S^{\min})^2) \geq \sigma^2 \delta_L^2 \|S\|_{\text{F}}^2$. Use $\|S\|_{\text{F}} \geq \|S\|_{\text{op}}$ to obtain (124). The upper bound follows because the full Hilbert–Schmidt sum contains the lowest block. The displayed lower bound tends to zero at large L ; it is not an asserted positive volume-uniform spectral fraction.

14.4 The original material endpoint and an actual diagonal

For integers $j \geq 2$ retain the original full matrices from Section 13 and set

$$L_j = j^2, \quad a_j = \frac{1}{100j}, \quad z_j = \frac{1}{j}, \quad \tau_j = \frac{1 - j^{-2}}{8}, \quad S_j = K_{\tau_j} = C_{\tau_j} C_{\tau_j}^\top. \quad (125)$$

These times stay on the regular material orbit; no singular matrix is inserted at its endpoint. Each S_j is positive definite because C_{τ_j} is invertible. In particular the preceding theorem applies. Section 13 proves $j^{-6} S_j \rightarrow S_* = \eta \eta^\top$, $\eta = (1/4, 3/8, -9/4)^\top$, with every matrix entry retained. Define $d_j = d_{L_j, a_j}(S_j)$, $\Delta_j = 2\sigma_j/a_j$, and

$$\mathcal{C}_* = \frac{204976875}{512\pi^2} + \frac{3982500}{\pi^6} > 0. \quad (126)$$

Theorem 14.2. *There is a definite sequence of strictly positive dyadic couplings $g_j < 1/j$ and actual smooth physical vectors*

$$w_j = \mathbf{1}_{(\Delta_j/2, 11\Delta_j/10)} (H_{g_j, L_j, a_j} - \mathcal{E}_{g_j, L_j, a_j}) \Xi_{f_{S_j}}, \quad (127)$$

orthogonal to the actual vacuum, for which

$$j^{-18} \|w_j\|^2 \longrightarrow \mathcal{C}_*, \quad (128)$$

$$j^{-17} \langle w_j, (H_{g_j, L_j, a_j} - \mathcal{E}_{g_j, L_j, a_j}) w_j \rangle \longrightarrow 100\sqrt{2}\pi \mathcal{C}_*, \quad (129)$$

$$j \frac{\langle w_j, (H_{g_j, L_j, a_j} - \mathcal{E}_{g_j, L_j, a_j}) w_j \rangle}{\|w_j\|^2} \longrightarrow 100\sqrt{2}\pi. \quad (130)$$

No vector is divided by its norm in this construction.

Proof. For each fixed j , fixed-box eigenvalue convergence and (122) imply that every sufficiently small $g > 0$ satisfies the following finite set of strict inequalities. The first six physical excitation energies lie within $\Delta_j/(10j)$ of Δ_j , the seventh exceeds $11\Delta_j/10$, and

$$\left| \frac{\|P_g(I_{L_j, a_j})\Xi_{f_{S_j}}\|^2}{d_j} - 1 \right| < \frac{1}{j}.$$

Also require $|\|\Xi_{f_{S_j}}\|^2/\mathcal{N}_{L_j, a_j}(S_j) - 1| < 1/j$, which follows from the proved full-vector norm convergence. Let n_j be the least positive integer for which $g_j = 2^{-n_j} < 1/j$ and all these inequalities hold. Their established fixed-box limits prove existence of this integer. This definition asserts no unproved volume-uniform convergence rate or prescribed renormalized coupling law.

The projection in (127) has rank six. Its range consists of smooth physical eigenvectors; it excludes the vacuum, since its energy interval is strictly positive. The norm inequality gives $\|w_j\|^2 > (1 - 1/j)d_j > 0$. On that range every excitation eigenvalue differs from Δ_j by at most $\Delta_j/(10j)$. Expansion in its actual orthonormal eigenbasis therefore gives the exact two-sided bound

$$\left| \frac{\langle w_j, (H - \mathcal{E})w_j \rangle}{\Delta_j \|w_j\|^2} - 1 \right| < \frac{1}{10j}.$$

The basis is used to evaluate the original vector, not to rescale it.

It remains to compute the unchanged raw scale d_j . For $N_j = 2j^2 + 1$ and $h_j = \pi/(2N_j)$, the limit $\sin h_j/h_j \rightarrow 1$ gives

$$\frac{\sigma_j^2}{2\pi^2/N_j^2} \rightarrow 1, \quad \frac{\delta_{L_j}}{N_j^2} \rightarrow -\frac{1}{2\pi^2}, \quad \frac{\mathbf{m}_{L_j}^2}{N_j^2} \rightarrow \frac{8}{\pi^4}, \quad \frac{N_j^2}{a_j^2 j^6} \rightarrow 40000.$$

The exact tensor entries give

$$\sum_i (\text{tr } S_* - S_{*,ii})^2 = \frac{333^2 + 328^2 + 13^2}{64^2} = \frac{109321}{2048}, \quad \sum_{i < j} S_{*,ij}^2 = \frac{531}{512}.$$

Insert these into the full expression (120), using $j^{-6}S_j \rightarrow S_*$, to obtain

$$j^{-18}d_j \rightarrow \frac{7500}{\pi^2} \frac{109321}{2048} + \frac{3840000}{\pi^6} \frac{531}{512} = \mathcal{C}_*.$$

Moreover $j\Delta_j = 400\sqrt{2}j^2 \sin(\pi/(4j^2+2)) \rightarrow 100\sqrt{2}\pi$. The norm and energy inequalities above now prove all three limits. The additionally retained whole-state norm gives the actual positive-coupling fraction bound

$$\frac{\|w_j\|^2}{\|\Xi_{f_{S_j}}\|^2} \geq \frac{j-1}{j+1} \frac{\sigma_j^2 \delta_{L_j}^2}{432r_j L_j^4}.$$

Its right side is asymptotic to $j^{-10}/(3456\pi^2)$. No limit of the full fraction is inferred from this lower estimate. Every finite Hamiltonian still has $\kappa_j = 200jg_j^2$, $b_j = 50j/g_j^2$ and $\xi_j = 1/(4g_j^4)$, including the full scalar $2b_j|\mathcal{P}_{L_j}|$. No strong-coupling inverse estimate was used. $\square$

If the chosen observable uses physical midpoints $am(e)$ instead, its exact state is $a_j^2 w_j$ by linearity of D_f , centering and spectral projection. Its raw norm and energy are multiplied by a_j^4 . Thus their powers become j^{14} and j^{13} with constants $\mathcal{C}_*/10^8$ and $100\sqrt{2}\pi\mathcal{C}_*/10^8$. The quotient is identical. This is the full coordinate/unit dictionary, not a silent change of weight.

The construction proves the full morphism from each material tensor through the energy-current state and the actual physical spectral projection. The six-dimensional coefficient map has the explicit inverse in Theorem 14.1. The spectral projection itself is global in space and depends on the full Hamiltonian. It has not been proved to be a local observable map between the changing regulators, nor a dynamical intertwiner of the forced Navier–Stokes solution and source-free Yang–Mills. The compactly supported Navier–Stokes profile is not replaced here by the stationary Fabel material orbit. The next unresolved part of the original objective is a common interacting continuum state/observable construction preserving this low-energy image. The finite-regulator sequence alone does not supply that identification. Section 15 constructs the next common vacuum-plus-band frame with its exact Gram tensor, calculates compact local-energy and Navier–Stokes curvature-state matrix elements, and retains the explicit higher-mode leakage in products. The common full local interacting continuum representation remains unresolved.

15 Local energy, curvature profiles, and the common lowest-band map

The global spectral projection of Section 14 now receives local weighted-energy states and observables. We calculate its entire vacuum-plus-first-band operator, its raw matrix elements against the full Fabel state, and its two physical time scales. We also retain the exact terms that leave this band. A compressed product is not substituted for a product of the original local operators.

15.1 Graph convergence on the actual excited space

All definitions and coefficients of Sections 1, 13 and 14 remain in force. We first extend the vacuum graph calculation to the actual finite physical eigenspaces. At fixed L, a , let u_g be an original physical eigenvector, with eigenvalue e_g in a bounded interval and bounded squared norm. Positivity of u_g is not required. For a real smooth multiplier F , expand the full kinetic product and use the eigenvalue equation, taking its real part. The exact identity is

$$q_H[Fu_g] - e_g \|Fu_g\|^2 = \kappa \sum_{e,\alpha} \int |u_g|^2 |X_{e,\alpha} F|^2 dU. \quad (131)$$

The terms $F^2 |Xu_g|^2$ and $2F \operatorname{Re}(\overline{u_g} Xu_g) XF$ occur on both sides of the integration-by-parts eigenvalue identity; their cancellation proves the formula also for complex u_g .

Put $M_k(u_g) = \int W^k |u_g|^2 dU$. The same regularized powers and the exact bound $\sum |XW|^2 \leq 16W$ used in Section 14 give

$$bM_{k+1}(u_g) \leq e_g M_k(u_g) + 4\kappa k^2 M_{k-1}(u_g), \quad bM_1(u_g) \leq e_g M_0(u_g).$$

Thus for every fixed k , $M_k(u_g)/g^{2k}$ is bounded at fixed L, a . The constants are uniform over a bounded-energy unit eigenspace. If $\mathcal{B}_g u_g \rightarrow u_0$ strongly and $e_g \rightarrow e_0$, the third moment gives the same chart-tail estimate for $(W/g^2)u_g$ as for the vacuum. Local Taylor convergence proves

$$\mathcal{B}_g((W/g^2)u_g) \rightarrow V_2 u_0, \quad \mathcal{B}_g(g^2 H_0 u_g) \rightarrow (ae_0/2 - V_2/4)u_0 = Du_0.$$

The second identity is the full eigenvalue equation $g^2 H_0 u_g = (ae_g/2)u_g - (W/(4g^2))u_g$. For every fixed real edge weight f , the strongly commuting Casimir inequality $\|\sum_e f_e E_e F\| \leq \|f\|_\infty \|H_0 F\|$

and compactly supported Gaussian-polynomial graph approximations prove the arbitrary electric-weight graph limit, exactly as in Section 14. The weighted magnetic limit follows from $|W_f| \leq \max_p |f_p|W$. These arguments prove

$$\mathcal{B}_g D_f u_g \longrightarrow D_f^{\text{osc}} u_0 \quad (132)$$

in full L^2 , with the original factors $2/a$ and $1/(8a)$, for every eigenvector limit in any fixed finite physical spectral range.

To obtain this result for a finite spectral frame, choose an eigenbasis of that range along an arbitrary sequence $g \downarrow 0$. The retained compactness and projection theorem gives a subsequence on which every basis vector converges. Apply the preceding moment and graph estimates to each vector; bounded finite linear combinations preserve them. The limit identifies the same operator matrix independently of the chosen subsequence and eigenbasis. Therefore the graph conclusion holds for the original finite spectral projections, including degenerate bands, not merely a chosen eigenvector subsequence.

15.2 Exact common coordinates and the whole compressed operator

Let

$$\mathcal{K} = \mathbb{C} \oplus \text{Sym}_3(\mathbb{C}), \quad \langle (c, B), (d, E) \rangle_{\mathcal{K}} = \bar{c}d + 6 \text{tr}(B^*E).$$

The factor 6 is the unchanged creation-vector norm:

$$\Phi(B) = \sum_{i,k=1}^3 B_{ik} \sum_{\alpha=1}^3 a_{i,\alpha}^\dagger a_{k,\alpha}^\dagger \Phi_0, \quad \langle \Phi(B), \Phi(E) \rangle = 6 \text{tr}(B^*E).$$

Commuting two annihilators through two creators proves this formula for complex symmetric matrices; off-diagonal entries occur twice in the displayed sum and twice in the matrix trace. Define the comparison isometry $I_{L,a}(c, B) = c\Phi_0 + \Phi(B)$ using the three exact lowest modes of Section 14.

Write P_g for the first-band projection on $(\Delta/2, 11\Delta/10)$ and $\Pi_g = |\psi_g\rangle\langle\psi_g| + P_g$. For sufficiently small positive g define

$$M_g(c, B) = c\psi_g + P_g \mathcal{B}_g^* \Phi(B), \quad G_g = M_g^* M_g. \quad (133)$$

Here every adjoint on $\mathcal{K}$ uses its displayed factor-6 inner product. Projection convergence proves $G_g \rightarrow I$ in operator norm. In particular G_g is positive definite for small g , M_g is a bijection onto the actual seven-dimensional range of Π_g , and its exact inverse there is

$$M_g^{-1} = G_g^{-1} M_g^*, \quad M_g^{-1} M_g = I, \quad M_g M_g^{-1} = \Pi_g.$$

The last equality on the full Hilbert space means the indicated orthogonal projection. These identities follow by multiplication and the equality of the range dimensions. No original vector or frame column is divided by its norm. The complete finite- g Gram tensor G_g is retained.

Put $C_{g,f} = D_f - \langle \psi_g, D_f \psi_g \rangle$. Let $J_f = \Sigma^{1/2} A_f \Sigma^{1/2}$ and $K_f = \Sigma^{-1/2} M_f \Sigma^{-1/2}$ in the notation of (114); here the face matrix M_f is not the Hilbert frame M_g . Set

$$R_f = K_f - J_f, \quad Q_f = K_f + J_f.$$

The full centered oscillator operator is

$$C_{0,f} = \frac{1}{4a} \sum_{\nu, \eta, \alpha} \left[(R_f)_{\nu\eta} (a_{\nu,\alpha} a_{\eta,\alpha} + a_{\nu,\alpha}^\dagger a_{\eta,\alpha}^\dagger) + 2(Q_f)_{\nu\eta} a_{\nu,\alpha}^\dagger a_{\eta,\alpha} \right]. \quad (134)$$

Expand $\partial_z = \sqrt{\sigma}/(2\sqrt{2})(a - a^\dagger)$ and $z = \sqrt{2/\sigma}(a + a^\dagger)$ in the full electric and magnetic quadratic forms to obtain this identity. Their vacuum scalar is $3 \operatorname{tr}(\mathbf{Q}_f)/(4a)$; subtracting that scalar gives exactly (134). No magnetic or number-preserving term is discarded.

Use superscript min for the three-lowest-mode blocks. The resulting exact compressed operator on $\mathcal{K}$ is

$$\mathcal{D}_{0,f}(c, B) = \left(\frac{3}{2a} \operatorname{tr}(\mathbf{R}_f^{\min} B), \frac{c}{4a} \mathbf{R}_f^{\min} + \frac{1}{2a} (\mathbf{Q}_f^{\min} B + B \mathbf{Q}_f^{\min}) \right). \quad (135)$$

Indeed the pair annihilation gives $6 \operatorname{tr}(\mathbf{R}_f^{\min} B)/(4a)$ in the vacuum, pair creation from the vacuum gives the second component, and the commutator of $\sum Q_{\nu\eta} a_\nu^\dagger a_\eta$ with $\Phi(B)$ is $\Phi(QB + BQ^\top)$. Higher-mode terms are outside the compressed range; they will be retained below. The graph result (132) proves

$$\mathcal{D}_{g,f} := G_g^{-1} M_g^* \Pi_g C_{g,f} \Pi_g M_g \longrightarrow \mathcal{D}_{0,f} \quad (136)$$

in the finite-dimensional operator norm. The original compressed operator is self-adjoint for the Gram inner product G_g : $G_g \mathcal{D}_{g,f} = \mathcal{D}_{g,f}^* G_g$.

15.3 Compact physical supports and all leading spatial factors

For real $h \in C_c(\mathbb{R}^3)$ use the exact local weights

$$f_{h,j}(e) = h(a_j m(e)), \quad f_{h,j}(p) = \frac{1}{4} \sum_{e \in \partial p} h(a_j m(e)), \quad L_j = j^2, \quad a_j = \frac{1}{100j}.$$

These define a finite sum of the original local electric and magnetic operators. A face adjacent to the sampled support is retained whenever its four-edge average is nonzero. Put

$$N_j = 2L_j + 1, \quad \ell_j = a_j N_j, \quad H_h = \int_{\mathbb{R}^3} h(x) dx, \quad H_{h,ik} = \int_{\mathbb{R}^3} x_i x_k h(x) dx.$$

The vertex endpoints remain at $\pm L_j a_j$; ℓ_j is the specified difference-basis length $a_j(2L_j + 1)$, not a replacement for the endpoint distance.

Theorem 15.1. *For the full comparison matrices of these local weights,*

$$\frac{\ell_j^4}{a_j} \mathbf{R}_{h,j}^{\min} \longrightarrow 4\sqrt{2}\pi H_h I_3, \quad \frac{\ell_j^4}{a_j} \mathbf{Q}_{h,j}^{\min} \longrightarrow 4\sqrt{2}\pi H_h I_3, \quad (137)$$

$$\frac{\ell_j^6}{a_j} (\mathbf{R}_{h,j}^{\min})_{ik} \longrightarrow 2\sqrt{2}\pi^3 H_{h,ik} \quad (i \neq k). \quad (138)$$

In particular the complete raw first-band mass obeys

$$\ell_j^8 d_{L_j, a_j}(f_{h,j}) \longrightarrow 36\pi^2 H_h^2. \quad (139)$$

Here $d(f)$ denotes $3 \operatorname{tr}((\mathbf{R}_f^{\min})^2)/(8a^2)$ for arbitrary weights, extending the quadratic-weight notation of (120).

Proof. Let V_i be the original edge functions (117) and $Y_i = d_1 V_i / \sigma_j$. Put

$$A_{ik}^h = \sum_e f_{h,j}(e) V_i(e) V_k(e), \quad F_{ik}^h = \sum_p f_{h,j}(p) Y_i(p) Y_k(p).$$

Then, exactly, $R_h^{\min} = \sigma_j(F^h - A^h)$ and $Q_h^{\min} = \sigma_j(F^h + A^h)$. Different Y_i have disjoint face-plane supports, so $F_{ik}^h = 0$ for $i \neq k$ with any face weight. For the plane (p, q) normal to i ,

$$Y_i(n)^2 = \frac{4}{N_j^3} \cos^2 \left(\frac{\pi a_j(n_p + 1/2)}{\ell_j} \right) \cos^2 \left(\frac{\pi a_j(n_q + 1/2)}{\ell_j} \right).$$

The sign of the ordered curl disappears here by squaring; it was not changed. On every fixed physical compact set the two cosine factors tend uniformly to one. The quarter-face sample differs uniformly from h at the face center by at most its modulus of continuity at distance $a_j/2$. The support increases by at most that distance. The face-center Riemann sums with volume factor a_j^3 consequently give

$$\ell_j^3 F_{ii}^h \longrightarrow 4H_h.$$

For clarity, uniform continuity bounds each sample error, the number of nonzero samples times a_j^3 stays bounded in a fixed enlarged compact cube, and the ordinary shifted cubic partition proves convergence of the remaining Riemann sum. Thus the four edges and the open endpoints contribute no unrecorded limit term.

For V_i on direction p its squared factor is

$$\frac{2}{N_j^3} \cos^2 \left(\frac{\pi a_j(n_p + 1/2)}{\ell_j} \right) \sin^2 \left(\frac{\pi a_j n_q}{\ell_j} \right),$$

and there is the other directional term with p, q interchanged. On a fixed support, $|\sin(\pi x/\ell_j)| \leq \pi|x|/\ell_j$. The same sample-count estimate gives $A_{ii}^h = O_h(\ell_j^{-5})$. Cauchy-Schwarz with $|h|$ also bounds every $A_{ik}^h = O_h(\ell_j^{-5})$.

More precisely, distinct normal indices i, k share only the remaining edge direction d . On that edge the exact product is

$$V_i(n, d)V_k(n, d) = -\frac{2}{N_j^3} \sin \left(\frac{\pi a_j n_i}{\ell_j} \right) \sin \left(\frac{\pi a_j n_k}{\ell_j} \right) \cos^2 \left(\frac{\pi a_j(n_d + 1/2)}{\ell_j} \right).$$

Keeping the two original signs and using uniform convergence $\ell_j \sin(\pi x/\ell_j) \rightarrow \pi x$ on the fixed support proves

$$\ell_j^5 A_{ik}^h \longrightarrow -2\pi^2 H_{h,ik}.$$

Finally $\ell_j \sigma_j/a_j = N_j \sqrt{8} \sin(\pi/(2N_j)) \rightarrow \sqrt{2}\pi$. These identities prove both matrix limits. Insert the first into the exact raw mass formula to get $3 \cdot 3(4\sqrt{2}\pi)^2 H_h^2/8 = 36\pi^2 H_h^2$. $\square$

This calculation retains spatial information beyond the leading integral: (138) is an explicit map to three mixed spatial moments. A vanishing integral does not imply that the original weighted state or these next coefficients vanish.

15.4 One positive-coupling diagonal and its raw common representation

The finite-regulator convergence can be enforced for every bounded edge weight at once at each fixed j . There are finitely many original edges. Let $b^{(e)}$ be the weight one at e and zero elsewhere, with its prescribed face averages. Both the actual centered operator and its comparison matrix are linear in this weight. Choose

$$\epsilon_j = j^{-40}(1 + \ell_j)^{-10}.$$

At each fixed j , take the least positive integer n_j such that $g_j = 2^{-n_j} < 1/j$ satisfies the finite inequalities of Theorem 14.2, $\|G_{g_j} - I\| < \epsilon_j$, and

$$\|\mathcal{D}_{g_j, b(e)} - \mathcal{D}_{0, b(e)}\| < \epsilon_j / |\mathbf{E}_{L_j}| \quad (e \in \mathbf{E}_{L_j}).$$

Also require the exact frame energy matrix to differ from $0 \oplus \Delta_j I$ by less than ϵ_j , and the frame coefficient of the full projected Fabel state to differ from $(0, \mathbf{R}_{S_j}^{\min} / (4a_j))$ by less than ϵ_j . Projection, graph and eigenvalue convergence prove existence for this finite list. This is a defined stricter coupling sequence; it does not silently reuse a previously chosen g_j for new inequalities. All conclusions of Section 14 persist.

For every real edge weight f , finite linearity now gives the uniform bound

$$\|\mathcal{D}_{g_j, f} - \mathcal{D}_{0, f}\| \leq \epsilon_j \|f\|_\infty.$$

The energy and Fabel coordinate requirements also follow from the fixed-box frame and vector convergence already proved. All actual states and operators retain their original coefficients $\kappa_j = 200jg_j^2$, $b_j = 50j/g_j^2$, and full Wilson scalar.

Theorem 15.2. *For every real $h \in C_c(\mathbb{R}^3)$, the actual Gram and compressed operator coordinates satisfy*

$$G_{g_j} \rightarrow I, \quad \ell_j^4 \mathcal{D}_{g_j, f_{h,j}} \rightarrow \mathcal{T}_h$$

in operator norm on the fixed space $\mathcal{K}$, where

$$\mathcal{T}_h(c, B) = \sqrt{2}\pi H_h (6 \operatorname{tr} B, cI_3 + 4B). \quad (140)$$

The actual excitation matrix $\mathcal{A}_j$ satisfies

$$\mathcal{A}_j \rightarrow 0, \quad \ell_j \mathcal{A}_j \rightarrow \mathcal{A}_* := 0 \oplus 2\sqrt{2}\pi I. \quad (141)$$

These limits include all finite products of the displayed compressed operators and their compressed time evolution.

Proof. Apply (137) to each term of (135), and use $\ell_j^4 \epsilon_j \|h\|_\infty \rightarrow 0$. The first component becomes $6\sqrt{2}\pi H_h \operatorname{tr} B$; the other two become $\sqrt{2}\pi H_h cI$ and $4\sqrt{2}\pi H_h B$. This proves (140), including its adjoint relation in the original factor-6 metric. The exact energy frame is block diagonal because the vacuum and excited range are invariant and orthogonal. Its specified error and $\ell_j \Delta_j \rightarrow 2\sqrt{2}\pi$ give (141). Operator-norm convergence in a fixed finite-dimensional space preserves finite products. The exponential power series, uniformly bounded on each fixed compact time interval, gives convergence of the displayed time evolutions. $\square$

The map $h \mapsto \mathcal{T}_h$ is explicitly the integral functional times the single displayed operator. On the traceless matrices its eigenvalue is $4\sqrt{2}\pi H_h$; on the vacuum/trace coordinates its matrix is $\sqrt{2}\pi H_h \begin{pmatrix} 0 & 18 \\ 1 & 4 \end{pmatrix}$ in the basis $(1, 0), (0, I)$, whose squared norms are 1, 18. Its characteristic roots are $\sqrt{2}\pi H_h (2 + \sqrt{22})$ and $\sqrt{2}\pi H_h (2 - \sqrt{22})$. This is a finite-dimensional self-adjoint compressed observable with all raw metric factors, not an assertion that the full local-energy algebra has collapsed.

Let $v_{h,j} = P_{g_j} \Xi_{f_{h,j}}$. The exact vacuum component of Ξ vanishes. Equations (133)–(140) therefore give its coordinate limit

$$\ell_j^4 M_{g_j}^{-1} v_{h,j} \longrightarrow (0, \sqrt{2}\pi H_h I).$$

This equation records the raw amplitude ℓ_j^{-4} . It defines no unit-vector rescaling. In particular, for real compact h, k and fixed physical Euclidean time $t \geq 0$,

$$\ell_j^8 \langle v_{h,j}, e^{-t(H_j - \mathcal{E}_j)} v_{k,j} \rangle \longrightarrow 36\pi^2 H_h H_k, \quad (142)$$

$$\ell_j^8 \langle v_{h,j}, e^{-\ell_j \theta (H_j - \mathcal{E}_j)} v_{k,j} \rangle \longrightarrow 36\pi^2 H_h H_k e^{-2\sqrt{2}\pi\theta} \quad (\theta \geq 0). \quad (143)$$

These follow by inserting the exact Gram G_{g_j} and the exact energy matrix in the inner product. The second equation records the explicit time change $t = \ell_j \theta$; it is not used to replace the original Hamiltonian or its physical energy quotient. At fixed physical time this particular finite band becomes a zero-energy sector. At times proportional to the box length its nontrivial dynamics are the explicitly displayed finite matrix. The same actual energy matrix gives the raw first-moment limit

$$\ell_j^9 \langle v_{h,j}, (H_j - \mathcal{E}_j) v_{h,j} \rangle \longrightarrow 72\sqrt{2}\pi^3 H_h^2.$$

When $H_h \neq 0$, the raw norm limit is strictly positive after its displayed ℓ_j^8 factor, so $v_{h,j} \neq 0$ eventually and

$$\ell_j \frac{\langle v_{h,j}, (H_j - \mathcal{E}_j) v_{h,j} \rangle}{\|v_{h,j}\|^2} \longrightarrow 2\sqrt{2}\pi.$$

The vectors themselves retain their vanishing raw norms.

For the unchanged Fabel states w_j of Section 14 let $S_* = \eta\eta^\top$ as there, and define the full fixed matrix

$$(B_*)_{ii} = -\frac{25\sqrt{2}}{\pi} (\text{tr } S_* - S_{*,ii}), \quad (B_*)_{ik} = \frac{400\sqrt{2}}{\pi^3} S_{*,ik} \quad (i \neq k).$$

The exact lowest-mode coefficients give $M_{g_j}^{-1} w_j = j^9(0, B_* + o(1))$ and $6 \text{tr}(B_*^2) = \mathcal{C}_*$. Consequently the original raw matrix elements obey

$$\frac{\ell_j^4}{j^{18}} \langle w_j, C_{g_j, f_{h,j}} w_j \rangle \longrightarrow 4\sqrt{2}\pi H_h \mathcal{C}_*, \quad (144)$$

$$\frac{\ell_j^4}{j^9} \langle \psi_j, C_{g_j, f_{h,j}} w_j \rangle \longrightarrow -\frac{25275}{8} H_h. \quad (145)$$

The first follows from the band component $4\sqrt{2}\pi H_h B$ of (140). For the sign and constant in the second, $\text{tr } S_* = 337/64$ and $\text{tr } B_* = -8425\sqrt{2}/(32\pi)$; multiplication by $6\sqrt{2}\pi H_h$ yields the result. Thus the quotient of the first raw expectation by $\|w_j\|^2$ decays as $4\sqrt{2}\pi H_h / \ell_j^4$ when $H_h \neq 0$, even though the unscaled raw expectation itself grows. Neither conclusion is inferred by dropping the Fabel amplitude.

15.5 The map from a compact Navier–Stokes curvature profile

Take any actual smooth compactly supported incompressible velocity profile $u(s, \cdot)$ at a regular time s in the Navier–Stokes construction under study. Its viscosity, force, pressure and time interval remain the ones of that construction. Set

$$\mathcal{A}_0 = 0, \quad \mathcal{A}_i = \lambda u_i T, \quad T = -i\sigma_3/2, \quad h_s(x) = \sum_{i < k} -2 \text{tr}(F_{ik}(s, x)^2) = \lambda^2 |\text{curl } u(s, x)|^2.$$

Here $F_{ik} = \lambda(\partial_i u_k - \partial_k u_i)T$ and $-2 \text{tr}(T^2) = 1$ prove the equality. The weighted local-energy operator uses the full profile: $f_{s,j}(e) = h_s(a_j m(e))$, with all prescribed face averages. The composition

$$u(s, \cdot) \longmapsto h_s \longmapsto D_{f_{s,j}} \longmapsto \Xi_{f_{s,j}} \longmapsto P_{g_j} \Xi_{f_{s,j}}$$

is now a specified map into the actual physical Hilbert space. Its raw first-band masses and mixed-time spectral matrix elements are (139), (142) and (143), with

$$H_{h_s} = \lambda^2 \|\operatorname{curl} u(s, \cdot)\|_2^2, \quad H_{h_s, ik} = \lambda^2 \int x_i x_k |\operatorname{curl} u(s, x)|^2 dx.$$

For $\lambda \neq 0$ and a nonzero compact smooth incompressible profile, $H_{h_s} > 0$. Indeed integration by parts gives $\int |\nabla u|^2 = \int |\operatorname{curl} u|^2 + \int |\operatorname{div} u|^2$; if its curl vanishes, all derivatives vanish and compact support forces $u = 0$. Thus this particular local-curvature state has nonzero first-band mass at sufficiently late regulators, with all norm powers explicit.

This is a regular-time profile map. It does not assert that H_{h_s} diverges as a blowup time is approached: pointwise concentration and its spatial integral remain distinct quantities in the displayed exact map. The full sourced current $j_i = (\lambda/g^2)(\Delta u_i - c^{-2}\partial_s^2 u_i)T$ from Section 13 is unchanged. No equality between evolution in the fluid parameter s and the quantum time t or θ is used or claimed. The limits above apply to every fixed regular profile; a simultaneous singular-fluid-time limit is not exchanged with the regulator limits. Only the nonzero compact incompressible profile, rather than its singular endpoint, was used to prove positive band mass. The gap-closing projected sequence therefore does not establish an implication specific to Navier–Stokes breakdown.

15.6 All terms leaving the seven-dimensional range

Let $\hat{Q}_g = I - \Pi_g$. On the original smooth spectral range, the exact identity is

$$\Pi_g C_{g,h} C_{g,k} \Pi_g = \Pi_g C_{g,h} \Pi_g C_{g,k} \Pi_g + (\hat{Q}_g C_{g,h} \Pi_g)^* (\hat{Q}_g C_{g,k} \Pi_g). \quad (146)$$

Each finite-range eigenvector is smooth, each weighted differential operator preserves smoothness, and its formal adjoint is its self-adjoint realization there. Inserting $I = \Pi_g + \hat{Q}_g$ therefore proves the identity with no domain omission. For $h = k$ the last term is positive. Products of the compressed limits alone do not determine this term.

The fixed-box graph limit proves the exact comparison for every leakage matrix element as well. Here are its full coordinates. Extend a lowest-mode symmetric B by zero to all modes and put $\mathcal{S}_B = \sum_{\nu, \eta, \alpha} B_{\nu\eta} a_{\nu, \alpha}^\dagger a_{\eta, \alpha}^\dagger$. Let $P_\perp$ remove the lowest 3-by-3 block of a symmetric mode matrix. From (134),

$$\hat{Q}_0 C_{0,f} \Phi_0 = \frac{1}{4a} \Phi(P_\perp R_f), \quad (147)$$

$$\hat{Q}_0 C_{0,f} \Phi(B) = \frac{1}{4a} \mathcal{S}_{R_f} \mathcal{S}_B \Phi_0 + \frac{1}{2a} \Phi(P_\perp (Q_f B + B Q_f)). \quad (148)$$

In these equations Φ on a full mode matrix uses the same colour-contracted formula over every spatial mode. The two terms on the right in (148) have respectively four and two creation operators, hence are orthogonal. The annihilation term has landed in the vacuum and is included in Π_0 , so it is absent for that exact reason. In particular

$$\|\hat{Q}_0 C_{0,f} \Phi_0\|^2 = \frac{3}{8a^2} [\operatorname{tr}(R_f^2) - \operatorname{tr}((R_f^{\min})^2)].$$

The four-creation contraction also has a completely explicit finite sum. With combined indices $I = (\nu, \alpha)$ define $T_{I_1 I_2 I_3 I_4} = (R_f)_{\nu_1 \nu_2} B_{\nu_3 \nu_4} \delta_{\alpha_1 \alpha_2} \delta_{\alpha_3 \alpha_4}$. For another pair of matrices producing coefficients U ,

$$\langle \mathcal{S}_{R_f} \mathcal{S}_B \Phi_0, \mathcal{S}_{R_k} \mathcal{S}_E \Phi_0 \rangle = \sum_{I_1, \dots, I_4} \sum_{J_1, \dots, J_4} \overline{T_{I_1 I_2 I_3 I_4}} U_{J_1 J_2 J_3 J_4} \sum_{\pi \in S_4} \prod_{r=1}^4 \delta_{I_r, J_{\pi(r)}}.$$

Repeatedly commute the four annihilators through the four creators; every surviving vacuum term is one of these 24 pairings, proving the formula with its multiplicities and colour factors. The two-creation term has the already-proved factor-6 trace inner product. These equations give every omitted-intermediate covariance in (146) at fixed box and coupling limit.

We have constructed a common finite-dimensional representation of the exact compressed local-energy maps, including their raw coupling to Fabel and compact Navier–Stokes profiles. Its fixed physical-time energy is zero and its leading spatial functional is the integral, with explicitly retained next mixed moments. Section 16 evaluates the full unprojected vacuum two-creation spectral measure in the joint infinite-volume continuum limit, using the exact mode maps above. Products on excited states still retain the four-creation terms displayed here. Their continuum products and the survival of nonabelian interactions require further calculation before this sequence can identify a local interacting four-dimensional Yang–Mills theory.

16 The full local-energy spectral measure in the expanding-box limit

We now retain every transverse mode of the original open box. The observable remains the original weighted electric and magnetic energy, with the original face average. Positive Euclidean time acts by the original excitation semigroup. It is an explicitly recorded operation on the vector; no vector is divided by its norm. The result below constructs a common two-creation spectral representation, including its continuous low-energy tail. The representation is identified explicitly with three colour copies of a transverse free field. This identification is part of the result and does not identify that representation with an interacting four-dimensional Yang–Mills theory.

16.1 All-mode coordinates and the observable to be taken to the limit

Take real $h \in C_c^\infty(\mathbb{R}^3)$, and put

$$h_e = h(am(e)), \quad h_p = \frac{1}{4} \sum_{e \in \partial p} h_e, \quad \Xi_{h,g} = (D_h - \langle \psi_g, D_h \psi_g \rangle) \psi_g, \quad A_g = H_g - \mathcal{E}_g.$$

All original edges and faces occur in D_h , even where the weight is zero. In particular, faces adjacent to the support are retained. For $t > 0$ define the unchanged, time-filtered state

$$y_{h,g}(t) = e^{-tA_g/2} \Xi_{h,g}, \quad C_{g;h,k}(t) = \langle \Xi_{h,g}, e^{-tA_g} \Xi_{k,g} \rangle. \quad (149)$$

The inner product is conjugate-linear in the first argument. These vectors are physical and orthogonal to the vacuum, since each operation commutes with the original gauge action and with the vacuum projection.

Write $V_\nu = ROe_\nu$, as in Section 14, for the real transverse edge eigenvectors with counting norm one and $d_1^\top d_1$ eigenvalue σ_ν^2 . Put $\omega_\nu = \sigma_\nu/a$. The full pair matrix divided by the retained spacing is exactly

$$B_{h;\nu\eta}^{L,a} := \frac{R_{h;\nu\eta}}{a} = \frac{\sum_p h_p (d_1 V_\nu)_p (d_1 V_\eta)_p}{a^2 \sqrt{\omega_\nu \omega_\eta}} - \sqrt{\omega_\nu \omega_\eta} \sum_e h_e V_\nu(e) V_\eta(e). \quad (150)$$

Both terms have been carried from the full original operator; neither is dropped in the limit. Section 14’s graph theorem and its exact Fock norm give, at fixed L, a , the finite-measure limit as

$g \downarrow 0$,

$$d\nu_{h,0}^{L,a}(\omega) = \frac{3}{8} \sum_{\nu,\eta} |B_{h;\nu\eta}^{L,a}|^2 \delta_{\omega_\nu + \omega_\eta}(d\omega), \quad C_{0;h,h}^{L,a}(t) = \int e^{-t\omega} d\nu_{h,0}^{L,a}(\omega). \quad (151)$$

The sum is ordered, so an off-diagonal pair occurs twice. Three colours and the two bosonic contractions give $3 \cdot 2/4^2 = 3/8$. Total mass converges at fixed box by strong convergence of the full weighted vacuum vector. Finite spectral projections converge as proved in Section 14. To obtain convergence for a bounded continuous spectral function, first truncate to a finite union of spectral intervals and then use the convergent vector norms to make the remaining mass small. This proves in particular convergence of every expression (149) at fixed L, a . No unbounded first moment is inferred from weak convergence alone.

16.2 Exact open-box modes, parity transform, and convergence of the full sum

Here $N = 2L + 1$ and $\ell = aN$, retaining the distinction between this basis length and the distance between the extreme vertices. For $j = 0, \dots, N - 1$ define on vertices

$$v_j(n) = \sqrt{\frac{2 - \delta_{j0}}{N}} \cos\left(\frac{\pi j(n + L + 1/2)}{N}\right), \quad s_j = 2 \sin \frac{\pi j}{2N}.$$

For $j > 0$, the edge function is

$$w_j(n) = \frac{v_j(n+1) - v_j(n)}{s_j} = -\sqrt{\frac{2}{N}} \sin\left(\frac{\pi j(n + L + 1)}{N}\right).$$

There is no w_0 . Finite geometric sums give $\sum v_j v_k = \delta_{jk}$ and $\sum w_j w_k = \delta_{jk}$ on their respective original vertex and edge ranges. The displayed difference identity gives $d_0 v_j = s_j w_j$ and, by transposition, $d_0^\top w_j = s_j v_j$ for $j > 0$.

For a triple $\mathbf{j}$, let $I = \{i : j_i > 0\}$ and $s = (s_{j_1}, s_{j_2}, s_{j_3})$, with missing components zero. Each unit vector $\epsilon \in \mathbb{R}^I$ satisfying $s \cdot \epsilon = 0$ gives the exact transverse edge cochain

$$V_{j,\epsilon}(n, i) = \epsilon_i w_{j_i}(n_i) \prod_{d \neq i} v_{j_d}(n_d), \quad \sigma = |s|. \quad (152)$$

Its curl on the (i, k) face is $(s_{j_i} \epsilon_k - s_{j_k} \epsilon_i) w_{j_i}(n_i) w_{j_k}(n_k) \prod_{d \neq i, k} v_{j_d}(n_d)$. The preceding orthogonality and difference identities prove its norm, transversality, and curl eigenvalue. For each triple there are $|I| - 1$ such polarizations if $|I| \geq 2$, and none otherwise. Decomposing the tensor product complex into these finite frequency blocks proves completeness: each block of one-forms is $\mathbb{R}^I$, its gradient line is $\mathbb{R}s$, and its transverse complement is precisely the one just constructed. This reproduces all r columns of RO up to orthogonal changes inside eigenspaces, which leave (151) unchanged.

Theorem 16.1. *For any $a \downarrow 0$, $\ell = a(2L + 1) \rightarrow \infty$, any real $h \in C_c^\infty(\mathbb{R}^3)$, and every $t > 0$, the complete sum in (151) converges to*

$$C_h(t) = \frac{3}{4(2\pi)^6} \int_{\mathbb{R}^3} \int_{\mathbb{R}^3} |p||q|(1 + \widehat{p} \cdot \widehat{q})^2 |\widehat{h}(p+q)|^2 e^{-t(|p|+|q|)} dp dq, \quad \widehat{h}(k) = \int e^{-ik \cdot x} h(x) dx. \quad (153)$$

The values assigned to unit directions at $p = 0$ or $q = 0$ do not affect the integral. All positive-time spectral moments converge as well. Real mixed covariances follow by polarization.

Proof. We give bounds for the complete sum before passing to any bounded frequency region. Each component of V has modulus at most $2\sqrt{2}N^{-3/2}$, and each component of d_1V has modulus at most $4N^{-3/2}\sigma$. Only $O_h(a^{-3})$ original edges or faces have nonzero weight once $a \leq 1$ and the box contains a fixed unit neighbourhood of the support. Consequently

$$|B_{h;\nu\eta}^{L,a}| \leq K_h \ell^{-3} \sqrt{\omega_\nu \omega_\eta}, \quad (154)$$

where K_h is finite and independent of a, L . For clarity, one may take a cube $[-R_h, R_h]^3$ containing that neighbourhood, bound its edge and face counts by $6(2R_h/a + 3)^3$, use $|h_p| \leq \|h\|_\infty$, and insert the component bounds in (150); this gives such a K_h directly. Because $2x/\pi \leq \sin x \leq x$ on $[0, \pi/2]$,

$$\frac{2|\mathbf{j}|}{\ell} \leq \omega_{\mathbf{j}} \leq \frac{\pi|\mathbf{j}|}{\ell}.$$

Grouping integer triples in shells $m \leq |\mathbf{j}| < m+1$, whose counts are at most $K(m+1)^2$, proves for $\ell \geq 1$ and fixed $u > 0$ that $\ell^{-3} \sum_\nu \omega_\nu e^{-u\omega_\nu} \leq K_u$. Indeed the bound is a constant times $\ell^{-4} \sum_{m \geq 1} (m+1)^3 e^{-2um/\ell}$; comparison with the integral on each interval $[m, m+1]$ bounds it uniformly for $\ell \geq 1$. It follows from (154) that the part with $\omega_\nu + \omega_\eta > R$ is bounded by $K_{h,t} e^{-tR/2}$, uniformly in the regulators. Multiplying by any fixed power of the pair frequency preserves a tail tending to zero, by applying this estimate with a smaller positive time.

The part with $\omega_\nu < \varepsilon$ tends to zero uniformly in the limiting sense needed here: the same shell count gives

$$\limsup_{\ell \rightarrow \infty} \ell^{-3} \sum_{\omega_\nu < \varepsilon} \omega_\nu \leq K\varepsilon^4.$$

The analogous estimate holds in the other index. In a bounded frequency region, triples having any $j_i = 0$ contribute at most $K_R \ell^2$ modes, so their contribution to the paired sum is $O_{h,R}(\ell^{-1})$. Strips of width δ adjacent to coordinate planes have limiting contribution $O_{h,R}(\delta)$ by counting the allowed integers in that coordinate. These bounds permit us to work first where every momentum coordinate is positive and bounded away from zero, and both total frequencies are bounded above and below.

Set $k_i = \pi j_i / \ell$ and $\beta_i = j_i \bmod 2$. At the physical location of its component, (152) is exactly a product of $\cos(k_i x_i + \pi j_i / 2)$ and $-\sin(k_i x_i + \pi j_i / 2)$, times $\sqrt{8} N^{-3/2} \epsilon_i$. For fixed bounded frequencies, the lattice frequency vector s/a converges uniformly to k . Choose the two polarizations by Gram-Schmidt applied to a fixed coordinate vector and the direction, on a region where these are not parallel. Finitely many such regions cover the retained compact frequency set; alternatively the polarization sum uses directly the matrix $I - kk^\top / |k|^2$. Thus all formulas below are independent of the choice across overlaps.

The elementary exponential transform of the product is

$$\ell^{3/2} a^{-3/2} V_{\mathbf{j},\epsilon}(x) = \chi_{\mathbf{j}} \sum_{\rho \in \{-1,1\}^3} U_{\beta\rho} \epsilon^\rho e^{i(\rho k) \cdot x}, \quad U_{\beta\rho} = \frac{i}{\sqrt{8}} e^{i\pi\rho \cdot \beta/2}, \quad \epsilon_i^\rho = \rho_i \epsilon_i, \quad \chi_{\mathbf{j}} = (-1)^{\sum_i [j_i/2]}. \quad (155)$$

The left side means the vector of component expressions evaluated at a common physical x , whose restrictions are the original edge values; no relocation of the original samples is made in the sums. For each pair of signs, summing $e^{i\pi(\rho_i - \rho'_i)\beta_i/2}$ over $\beta_i = 0, 1$ gives two when the signs agree and zero otherwise. Hence $U^*U = UU^* = I_8$. The sign χ remains in every individual mode and drops out of the squared modulus of its pair coefficient.

Ordinary Riemann sums in physical space now apply uniformly on the retained bounded frequency set. Derivatives of each trigonometric factor and of h are bounded there, so replacing the

integral on each spacing- a cell by its original sample has total error at most $K_{h,R}a$. For the magnetic term, the quarter-edge face weight differs from the value at the face centre by at most $a\|\nabla h\|_\infty$, and the exact discrete curl coefficients converge to $ik \times \epsilon$. This proves convergence of both terms in (150), with uniform error $o(1)\ell^{-3}$ per mode pair. There are $O_R(\ell^6)$ pairs, so the squared-sum error tends to zero by the uniform coefficient bound. No derivative or magnetic coefficient is discarded.

For each parity, the momentum grid has step $2\pi/\ell$ in every coordinate. The eight grids can be aligned with one such grid: their coordinate differences are at most π/ℓ , and the preceding compact-set uniform continuity makes the resulting error tend to zero. Thereafter the sum over the two parities of a mode pair is the Hilbert–Schmidt norm under $U \otimes U$ in (155). Its unitarity replaces the two standing-wave parity sums by the two sums over momentum signs. The factor ℓ^{-6} in the squared pair coefficient and the two grid densities $(\ell/(2\pi))^3$ leave exactly $(2\pi)^{-6}$. The sign sums cover the whole momentum space, except null coordinate planes already controlled above.

For real unit transverse polarizations $e_r(p), e_s(q)$, the remaining plane-wave pair coefficient, without its Fourier measure factor, is

$$-\widehat{h}(-p-q) \left[\frac{(p \times e_r(p)) \cdot (q \times e_s(q))}{\sqrt{|p||q|}} + \sqrt{|p||q|} e_r(p) \cdot e_s(q) \right]. \quad (156)$$

The two factors of i in the two curls produce the first minus sign; the electric term supplies the second term with the same minus sign. In particular the bracket vanishes for $q = -p$ with matched polarizations, as required by the exact identity $D_1 = H$. For the polarization sum rotate p to the third axis and q to the first–third plane, and put $c = \widehat{p} \cdot \widehat{q}$. Take $e_1(p) = (1, 0, 0)$, $e_2(p) = (0, 1, 0)$, $e_1(q) = (c, 0, -\sqrt{1-c^2})$, $e_2(q) = (0, 1, 0)$. The bracket divided by $\sqrt{|p||q|}$ is the diagonal matrix $(1+c)I_2$. Thus its squared polarization sum is $2|p||q|(1+c)^2$. Rotational invariance extends this calculation to every direction; its endpoints follow by continuity. Multiplication by $3/8$ proves (153). The previously proved tails justify removal of all cutoffs and also prove every positive-time moment assertion. $\square$

16.3 Evaluation of the measure and a common physical state map

Theorem 16.2. *The locally finite raw measure associated to (153) is absolutely continuous, with*

$$d\nu_h(\omega) = \rho_h(\omega) d\omega, \quad \rho_h(\omega) = \frac{1}{320\pi^5} \int_{|k| \leq \omega} |k|^4 |\widehat{h}(k)|^2 dk, \quad \omega \geq 0, \quad (157)$$

$$C_h(t) = \frac{1}{320\pi^5 t} \int_{\mathbb{R}^3} |k|^4 e^{-t|k|} |\widehat{h}(k)|^2 dk. \quad (158)$$

For nonzero compact smooth h , its support meets every interval $(0, \varepsilon)$ and it has no atom at zero. Its total mass is infinite, but $e^{-t\omega} d\nu_h$ has finite mass and all moments for every $t > 0$.

Proof. In (153) set $k = p + q$, $r = |p|$, $s = |q|$, $K = |k|$ and $\omega = r + s$. For $K > 0$, align k with the third axis. The change of the polar angle of p into s has absolute Jacobian $s/(Kr)$. Integration over the azimuth and over $\delta(\omega - r - s)$ therefore gives the measure $(2\pi/K)rs dr$. Set $u = r - s = 2r - \omega$. The admissible range is $-K \leq u \leq K$ precisely when $\omega \geq K$; its boundary has zero measure. The cosine of the angle between p and q satisfies $2rs(1+c) = K^2 - u^2$. The complete inner integral is

$$\frac{2\pi}{K} \int \frac{(K^2 - u^2)^2}{4} dr = \frac{\pi}{4K} \int_{-K}^K (K^2 - u^2)^2 du = \frac{4\pi K^4}{15}.$$

The set $k = 0$ is null in the outer integral, and the expression extends there by zero. Tonelli's theorem applies to the nonnegative integrand. Since $[3/(4(2\pi)^6)](4\pi/15) = 1/(320\pi^5)$, this proves (157). Integrating $e^{-t\omega}$ from K to infinity proves (158).

For completeness, compact support gives an everywhere convergent Taylor series for $\hat{h}$, by integrating the exponential series on the fixed support. If all Taylor coefficients vanished, this series would give $\hat{h} = 0$ everywhere and hence $h = 0$. One direct proof of the last implication is to convolve h with a Gaussian: the elementary Gaussian Fourier integral and Fubini give a zero convolution for every positive Gaussian width; these Gaussians are an approximate identity, so the convolution tends to h . For $h \neq 0$ let m be the smallest degree with nonzero homogeneous Taylor polynomial P_m . Uniformly on the unit sphere, $\hat{h}(r\theta) = r^m P_m(\theta) + O(r^{m+1})$. Consequently

$$\rho_h(\omega) = \frac{\int_{S^2} |P_m(\theta)|^2 d\theta}{320\pi^5(7+2m)} \omega^{7+2m} + O(\omega^{8+2m}). \quad (159)$$

Its leading coefficient is positive because a nonzero homogeneous polynomial cannot vanish on the whole sphere. This proves the support assertion. Repeated integration by parts in h makes $\hat{h}$ decay faster than every inverse power. Thus $\rho_h(\omega)$ tends as $\omega \rightarrow \infty$ to the finite positive constant $(320\pi^5)^{-1} \int |k|^4 |\hat{h}(k)|^2 dk$. The raw total mass is therefore infinite. Boundedness of that density proves all the positive-time integrability claims. $\square$

We specify the common state space and map, so that convergence of numbers is not substituted for a spectral representation. Let $\mathfrak{h}$ be the complex Hilbert space of square-integrable transverse vector functions on $\mathbb{R}^3$, with three colour coordinates and Lebesgue momentum measure. Let $\mathcal{F}$ be its symmetric Fock space, with vacuum Ω , and let $A_{\text{fr}} = d\Gamma(|p|)$. This definition means that on the n -particle space the energy multiplies by $|p_1| + \dots + |p_n|$; the domain consists of vectors for which the corresponding squared weighted norm is summable over n . The physical subspace used here consists of the vectors fixed by every simultaneous adjoint $\text{SU}(2)$ rotation of the three colours. Orthogonality of those rotations follows from the original trace metric $-2\text{tr}(T_\alpha T_\beta) = \delta_{\alpha\beta}$.

Choose any measurable real transverse polarization basis. Define

$$\begin{aligned} R_h^{rs}(p, q) &= -\frac{\hat{h}(-p-q)}{(2\pi)^3} \left[\frac{(p \times e_r(p)) \cdot (q \times e_s(q))}{\sqrt{|p||q|}} + \sqrt{|p||q|} e_r(p) \cdot e_s(q) \right], \\ J_t h &= \frac{1}{4} \sum_{\alpha=1}^3 \sum_{r,s=1}^2 \int e^{-t(|p|+|q|)/2} R_h^{rs}(p, q) a_{r\alpha}^\dagger(p) a_{s\alpha}^\dagger(q) \Omega dp dq. \end{aligned} \quad (160)$$

The integral is the symmetric two-particle Hilbert-space integral: the convention with creation symbols has norm twice the squared norm of its symmetric kernel. Equivalently its two-particle wave function is $\sqrt{2}/4$ times that kernel with equal colours. This gives an explicit definition without distributional creation operators. The preceding integral proves that it is well-defined, that it is a colour singlet, and that

$$\|J_t h\|^2 = C_h(t), \quad \langle J_t h, A_{\text{fr}}^n J_t h \rangle = \int \omega^n e^{-t\omega} d\nu_h(\omega), \quad e^{-sA_{\text{fr}}/2} J_t h = J_{t+s} h.$$

The positive density proves injectivity on the real test functions; the same kernel defines a complex-linear injective map on complex test functions, with the Hermitian Fourier modulus in its norm. It intertwines spatial translations: replacing $h(x)$ by $h(x-b)$ multiplies its creation kernel by $e^{i(p+q) \cdot b}$, precisely the specified second-quantized translation on the two momenta. Polarization changes are pointwise orthogonal changes of the one-particle coordinates and induce unitary maps on their symmetric tensor squares, preserving the state and its raw norm.

16.4 One actual positive-coupling sequence for every local profile

Return to the original Hamiltonian, taking $L_j = j^2$, $a_j = 1/(100j)$ and $\ell_j = a_j(2L_j + 1)$, for $j \geq 2$. The following construction strengthens, rather than silently reuses, the coupling selection of Section 15. Enumerate the positive rational times as $t_1, t_2, \dots$. Keep all the finite inequalities already used there. For each of the finitely many original single-edge weights $\mathbf{1}_e$ and each $n \leq j$, require additionally

$$\left| C_{g; \mathbf{1}_e, \mathbf{1}_{e'}}(t_n) - C_{0; \mathbf{1}_e, \mathbf{1}_{e'}}^{L_j, a_j}(t_n) \right| < \frac{\varepsilon_j}{|\mathbf{E}_{L_j}|^2} \quad (e, e' \in \mathbf{E}_{L_j}), \quad \varepsilon_j = j^{-40}(1 + \ell_j)^{-10}. \quad (161)$$

Each centered single-edge vector includes its adjacent quarter-weight faces. At fixed j its convergence was established above; mixed terms follow by the real polarization identity. Hence the finite list holds for every sufficiently small positive g . Define n_j as the smallest positive integer for which $g_j = 2^{-n_j} < 1/j$ satisfies this list and all the earlier finite lists. The well-ordering of the positive integers and fixed-box convergence prove existence. This specifies actual positive couplings in the original model; it does not replace H_{g_j} by its quadratic part.

Linearity in the weights bounds the covariance error at each listed time by $\varepsilon_j \|h\|_\infty \|k\|_\infty$. Theorem 16.1 therefore gives convergence for every real compact smooth profile and every positive rational time on this one sequence. For an arbitrary positive real time, squeeze the diagonal covariance between two rational times: its spectral measure is nonnegative, so it decreases with time. The limiting function (158) is continuous on $(0, \infty)$ by dominated convergence. Taking rational bounds to the specified time proves

$$C_{g_j; h, h}(t) \longrightarrow C_h(t), \quad t > 0. \quad (162)$$

Real polarization proves mixed convergence for all times.

For precision this also proves weak convergence of the actual damped raw spectral measures, not just of a few moments. Fix $t > 0$ and let $\mu_j(d\omega) = e^{-t\omega} d\nu_{h, g_j}(\omega)$. Their masses converge to $C_h(t)$, and their Laplace transforms at $s > 0$ converge to $C_h(t + s)$. The inequality

$$(1 - e^{-sR})\mu_j([R, \infty)) \leq C_{g_j; h, h}(t) - C_{g_j; h, h}(t + s)$$

first with small s and then large R proves tightness, by continuity of C_h ; finitely many initial measures can be covered separately. On a compact interval, the algebra generated by $e^{-s\omega}$ separates points and contains constants, so its polynomials uniformly approximate continuous functions there. An explicit version follows by setting $z = e^{-\omega}$ and using Bernstein polynomials for the resulting continuous function on $[e^{-R}, 1]$. Thus the transforms and tightness identify the weak limit uniquely as $e^{-t\omega} d\nu_h$. Applying this assertion at time $t/2$ to the bounded continuous function $\omega^n e^{-t\omega/2}$ proves convergence of every damped moment. In particular the original vectors satisfy

$$\|y_{h, g_j}(t)\|^2 \longrightarrow C_h(t), \quad \langle y_{h, g_j}(t), A_{g_j} y_{h, g_j}(t) \rangle \longrightarrow -C'_h(t). \quad (163)$$

They lie in every excitation-power domain since the semigroup multiplier damps all powers. For any finite list of profiles and times, their Gram matrices and their semigroup matrix elements converge to those of the vectors $J_t h$ defined in (160). Indeed the mixed time argument is $(t + s)/2$ plus the intervening positive evolution time, and the mixed covariance follows by polarization. This constructs the common spectral correspondence at the level actually proved. No norm-preserving identification of the entire changing Hilbert spaces, or convergence of products of several local operators, has been asserted.

16.5 Low-energy tail, original raw amplitudes, and curvature concentration

Put $H_h = \int h$. For $H_h \neq 0$, equation (159) and the substitution $v = t\omega$ give

$$\rho_h(\omega) \sim \frac{H_h^2}{560\pi^4} \omega^7, \quad C_h(t) \sim \frac{9H_h^2}{\pi^4 t^8}, \quad -C'_h(t) \sim \frac{72H_h^2}{\pi^4 t^9}, \quad \frac{\langle J_t h, A_{\text{fr}} J_t h \rangle}{\|J_t h\|^2} \sim \frac{8}{t}. \quad (164)$$

To justify the substitution over the entire integration range, split at a fixed small frequency. On its complement the positive time exponential is exponentially small, while on the first piece the uniform Taylor remainder above is bounded by the next power; the integrals are the elementary gamma integrals $\int_0^\infty v^n e^{-v} dv = n!$, obtained by repeated integration by parts. This also proves that for general nonzero h with first degree m , the quotient is $(8 + 2m)/t + o(t^{-1})$. These are raw amplitudes: the norm tends to zero as time grows. They are nevertheless nonzero at every finite time, with a ratio tending to zero. The positive-coupling convergence in (163) transfers both raw quantities at each fixed time, and a further explicitly selected subsequence transfers this time sequence: take $t = n$ and the least increasing j_n for which both errors are less than $1/n$ times their positive limiting quantities. Existence follows from that equation. Thus actual vectors $y_{h,g_{j_n}}(n)$ also have the quotient $8/n + o(n^{-1})$ when $H_h \neq 0$, with their unaltered norms and energies asymptotic to the displayed limits.

The map has a useful exact continuity estimate for concentration. For $h \in C_c^\infty$, $|\hat{h}| \leq \|h\|_1$ in (158), so

$$\|J_t h\|^2 \leq \frac{9}{\pi^4 t^8} \|h\|_1^2. \quad (165)$$

The constant is the integral $4\pi \int_0^\infty K^6 e^{-tK} dK / (320\pi^5 t)$, with every Fourier factor retained. The same estimate for differences extends J_t uniquely to $L^1(\mathbb{R}^3)$, by approximation with compact smooth functions. More generally, a finite signed measure defines the kernel (160) by its bounded Fourier transform; the identical estimate with total variation proves existence. Point measures have the exact raw norm given by equality in (165).

For an explicit concentration family let ρ be compact smooth and real, x_* fixed, and $h_\epsilon(x) = A\epsilon^{-\beta}\rho((x - x_*)/\epsilon)$, where $A, \beta \in \mathbb{R}$, $\epsilon > 0$, and all these parameters are retained. Direct change of variables gives

$$\begin{aligned} \hat{h}_\epsilon(k) &= A\epsilon^{3-\beta} e^{-ik \cdot x_*} \hat{\rho}(\epsilon k), \\ \|J_t h_\epsilon\|^2 &= \frac{A^2 \epsilon^{6-2\beta}}{320\pi^5 t} \int |k|^4 e^{-t|k|} |\hat{\rho}(\epsilon k)|^2 dk, \\ \epsilon^{2\beta-6} \|J_t h_\epsilon\|^2 &\longrightarrow \frac{9A^2 (\int \rho)^2}{\pi^4 t^8}. \end{aligned} \quad (166)$$

Dominated convergence uses $|\hat{\rho}| \leq \|\rho\|_1$. It proves convergence of the correspondingly scaled kernels in the same Fock space, including their energy forms, to the point measure kernel. This is a displayed coordinate and amplitude map, not a deletion of A or a division by the vector norm. For $\int \rho \neq 0$, the limiting energy quotient at fixed t is $8/t$ as $\epsilon \downarrow 0$, while the raw norm vanishes, stays finite, or diverges according to the exponent $6 - 2\beta$.

Finally insert an actual regular smooth compact incompressible Navier–Stokes profile from the preceding sections:

$$u(s) \longmapsto h_s = \lambda^2 |\nabla \times u(s)|^2 \longmapsto D_{h_s} \longmapsto \Xi_{h_s, g_j} \longmapsto e^{-tA_{g_j}/2} \Xi_{h_s, g_j} \longleftrightarrow J_t h_s. \quad (167)$$

Every map on the left is an exact coordinate operation on the original objects; the last correspondence is the proved common Gram and spectral limit. For $\lambda \neq 0$ and a nonzero profile,

$H_{h_s} = \lambda^2 \|\nabla \times u(s)\|_2^2 > 0$, using the compact-support integration identity proved in Section 15. Hence (164) applies with its exact squared coefficient $\lambda^4 \|\nabla \times u(s)\|_2^4$, and (165) bounds its raw norm with the same coefficient. To record the spatial scaling explicitly, for any fixed nonzero compact smooth divergence-free U , put $u_\epsilon(x) = \epsilon^{-\alpha} U((x - x_*)/\epsilon)$. Its curvature weight has $A = \lambda^2$, $\beta = 2\alpha + 2$, $\rho = |\nabla \times U|^2$, so

$$\|u_\epsilon\|_2^2 = \epsilon^{3-2\alpha} \|U\|_2^2, \quad \|\nabla \times u_\epsilon\|_2^2 = \epsilon^{1-2\alpha} \|\nabla \times U\|_2^2, \quad \|J_t h_\epsilon\|^2 \sim \frac{9\lambda^4 \|\nabla \times U\|_2^4}{\pi^4 t^8} \epsilon^{2-4\alpha}.$$

This is an explicit family of regular profiles; it is not asserted to be the singular profile of the supplied flow. The original spacetime connection and current remain $\mathcal{A}_i = \lambda u_i T$, $\mathcal{A}_0 = 0$ and $j_i = (\lambda/g^2)(\Delta u_i - c^{-2} \partial_s^2 u_i)T$. No identification of the fluid time s with the quantum time t has been made. A singular endpoint can be passed through this map only by evaluating its actual Fourier-weighted curvature measures; a pointwise infinity does not specify those measures.

The full spectral correspondence proved here is stronger than the lowest-band correspondence: its limiting measure has nonzero weight at every positive energy scale and a continuous tail at zero. Its Hamiltonian is exactly the explicitly defined $d\Gamma(|p|)$, and its states are the equal-colour transverse two-creation kernels above. Convergence of higher local products and survival of a nonabelian interaction on a continuum scaling trajectory have not been established by this calculation. Those are further calculations in the original counterexample programme; the free spectral correspondence itself is not a disproof of the interacting Yang–Mills mass-gap assertion.

17 A nonzero nonabelian vertex and the actual vacuum response

The free spectral correspondence in Section 16 does not remove the nonabelian terms of the original operator. We calculate those terms in the same exact tree and Haar coordinates, including the kinetic contribution, and exhibit a nonzero physical three-creation channel. Every statement in this section fixes the original finite box and $a > 0$ before the small-positive-coupling limit. Bounds depending on that box are not asserted to be uniform in the continuum limit.

17.1 Exact logarithmic fields and their retained density terms

Use the tree and chords of Section 14. To avoid confusing a coefficient matrix with the chart $\mathcal{B}_g$, write t_{ce} for the original additive tree matrix and b_{ce} for the following second integer matrix. If e is a chord, set $l_{ce} = \delta_{ce}$ and $r_{ce} = 0$. If e is a tree edge, set $l_{ce} = \epsilon(s(c), e)$ and $r_{ce} = -\epsilon(t(c), e)$, where $\epsilon(v, e)$ records whether the root-to- v tree path uses e . Then

$$t_{ce} = l_{ce} + r_{ce}, \quad b_{ce} = r_{ce} - l_{ce}, \quad \mathcal{X}_{e,\alpha} = \sum_c (l_{ce} L_{c,\alpha} + r_{ce} R_{c,\alpha}). \quad (168)$$

This is exactly the original field obtained by varying that edge, recomputing the rooted chord products and differentiating. In particular each downstream tree holonomy has derivative T_α , and the inverse target holonomy contributes the displayed minus sign. Product Haar measure makes each left and right field divergence-free.

Identify $y^\alpha T_\alpha$ with the real colour vector y . The Pauli multiplication law is

$$(u \cdot T)(v \cdot T) = -\frac{1}{4}(u \cdot v)I + \frac{1}{2}(u \times v) \cdot T, \quad [u \cdot T, v \cdot T] = (u \times v) \cdot T. \quad (169)$$

Write $\text{ad}_y v = y \times v$. Differentiation of the exponential power series, grouping the positions of the differentiated factor, gives

$$(d\exp)_Y(V)e^{-Y} = \int_0^1 e^{s\text{ad}_Y} V ds.$$

Thus the exact logarithmic coordinate derivatives of left and right multiplication are respectively

$$F_-(\text{ad}_y)T_\alpha, \quad F_+(\text{ad}_y)T_\alpha, \quad F_-(z) = \frac{z}{e^z - 1}, \quad F_+(z) = F_-(z) + z.$$

The values at zero are their removable continuations. On a sufficiently small original chart these are analytic matrix functions. Multiplying their series by the exponential series proves $F_-(z) = 1 - z/2 + z^2/12 + O(z^4)$ and $F_+(z) = 1 + z/2 + z^2/12 + O(z^4)$, including the zero cubic coefficient. The exact functions, rather than a truncated replacement, define all remainders below.

Put $d = 3r$ and keep the exact chart (112), with $Z_c = \exp(gx_c \cdot T)$. On compactly supported smooth functions of x , whose support lies inside that chart for small g , define

$$Y_{e,\alpha}(g) = g\mathcal{B}_g\mathcal{X}_{e,\alpha}\mathcal{B}_g^*.$$

Let $D_{e,\alpha}(g)$ be the vector field whose c -component is $l_{ce}F_-(g\text{ad}_{x_c})e_\alpha + r_{ce}F_+(g\text{ad}_{x_c})e_\alpha$. Density conjugation gives the exact equality

$$Y_{e,\alpha}(g) = D_{e,\alpha}(g) - \frac{1}{2}D_{e,\alpha}(g) \log \mathcal{J}(gx). \quad (170)$$

Indeed apply the derivative to $\mathcal{J}(gx)^{-1/2}f(x)$ before multiplying by the chart density. Direct expansion of the retained sine density gives

$$\log \mathcal{J}(gx) = \log(16\pi^2)^{-r} - \frac{g^2}{12} \sum_c |x_c|^2 - \frac{g^4}{1440} \sum_c |x_c|^4 + O(g^6|x|^6).$$

For each c , all positive powers of ad_{x_c} are orthogonal to x_c . Consequently the radial contraction in (170) retains exactly the t_{ce} term. The coefficients through order two are therefore

$$\begin{aligned} D_{e,\alpha}^0 &= \sum_c t_{ce} \partial_{c,\alpha}, \\ D_{e,\alpha}^1 &= \frac{1}{2} \sum_c b_{ce}(x_c \times e_\alpha) \cdot \nabla_c, \\ D_{e,\alpha}^2 &= \frac{1}{12} \sum_c t_{ce}[x_c \times (x_c \times e_\alpha)] \cdot \nabla_c, \quad M_{e,\alpha}^2 = \frac{1}{12} \sum_c t_{ce} x_c^\alpha, \\ Y_{e,\alpha}(g) &= D_{e,\alpha}^0 + gD_{e,\alpha}^1 + g^2(D_{e,\alpha}^2 + M_{e,\alpha}^2) + O(g^4). \end{aligned} \quad (171)$$

Here M^2 is multiplication. Its sign can also be verified without the density series: the divergences of D^0, D^1 are zero and $\text{div } D^2 = \sum_c t_{ce} x_c^\alpha / 6$. Thus $D^2 + M^2$ is skew-adjoint on flat compact test functions, as required by the exact Haar isometry. This check retains the term that would be missed by treating the chart density as constant.

17.2 Every face coefficient and the full first two vertices

In the original ordered four-link word of a face, let $\zeta_1, \dots, \zeta_4$ be the signed chord vectors, with zero for each tree link. Reversed traversal supplies the minus sign in that factor, without reversing the order of the four factors. For $k \geq 1$ the exact Taylor coefficient of the full potential is

$$W_{p,k}(x) = - \sum_{n_1 + \dots + n_4 = k} \text{tr} \left(\prod_{m=1}^4 \frac{(\zeta_m \cdot T)^{n_m}}{n_m!} \right), \quad 2 - \text{tr} \prod_{m=1}^4 e^{g\zeta_m \cdot T} = \sum_{k \geq 1} g^k W_{p,k}. \quad (172)$$

The constant term is zero because the original scalar 2 cancels the trace of the identity at this configuration, not because the scalar Wilson term was removed from H .

The first coefficients can be written as the following complete recursion. Start with $A_0 = B_0 = C_0 = 0$ and set

$$\begin{aligned} A_m &= A_{m-1} + \zeta_m, \\ B_m &= B_{m-1} + \frac{1}{2}A_{m-1} \times \zeta_m, \\ C_m &= C_{m-1} + \frac{1}{2}B_{m-1} \times \zeta_m + \frac{1}{12}\{A_{m-1} \times (A_{m-1} \times \zeta_m) + \zeta_m \times (\zeta_m \times A_{m-1})\}. \end{aligned} \quad (173)$$

To verify the recursion, multiply the two exponential series through degree three using (169), and then use $\log(I + Q) = Q - Q^2/2 + Q^3/3 + O(Q^4)$. The linear coefficient is the sum, the quadratic cross coefficient is half the commutator, and the cubic coefficients are respectively one half and one twelfth as displayed. This finite multiplication proves that the logarithm of the face word is $(gA_4 + g^2B_4 + g^3C_4) \cdot T + O(g^4)$. Since $2 - 2\cos(|z|/2) = |z|^2/4 - |z|^4/192 + O(|z|^6)$, substitution proves, with $A = A_4, B = B_4, C = C_4$,

$$W_{p,1} = 0, \quad W_{p,2} = \frac{1}{4}|A|^2, \quad W_{p,3} = \frac{1}{2}A \cdot B, \quad W_{p,4} = \frac{1}{4}|B|^2 + \frac{1}{2}A \cdot C - \frac{1}{192}|A|^4. \quad (174)$$

In particular the nested-commutator and negative fourth-order trace contributions remain alongside the positive square.

Write $\{S, T\} = ST + TS$. The original operator in the exact chart is

$$\tilde{H}_g = -\frac{2}{a} \sum_{e,\alpha} Y_{e,\alpha}(g)^2 + \frac{1}{2g^2a} \sum_p \left(2 - \text{tr} \prod_{m=1}^4 e^{g\zeta_m \cdot T} \right), \quad (175)$$

$$\begin{aligned} H^{(1)} &= -\frac{2}{a} \sum_{e,\alpha} \{D_{e,\alpha}^0, D_{e,\alpha}^1\} + \frac{1}{2a} \sum_p W_{p,3}, \\ H^{(2)} &= -\frac{2}{a} \sum_{e,\alpha} ((D_{e,\alpha}^1)^2 + \{D_{e,\alpha}^0, D_{e,\alpha}^2 + M_{e,\alpha}^2\}) + \frac{1}{2a} \sum_p W_{p,4}. \end{aligned} \quad (176)$$

Its zeroth coefficient is exactly H_{osc} of Section 14. Define the remainder to be the difference of (175) and $H_{\text{osc}} + gH^{(1)} + g^2H^{(2)}$; this definition retains every higher term.

We record the remainder control used below. Fix a small closed original ball $|y| \leq \rho$ inside the logarithmic chart, with a slightly larger ball still inside it. Analyticity bounds all derivatives through any fixed finite order on that larger ball. Taylor's integral remainder for each coefficient of Y , and its first x derivative, is bounded by a constant times $g^3(1+|x|)^m$ on $|gx| \leq \rho$, for some fixed integer m ; the zero cubic coefficient of Y gives a stronger bound where needed. For the potential, fifth-order Taylor remainder before division by g^2 is bounded by $Cg^5|x|^5$, so its operator remainder is bounded by $Cg^3|x|^5/a$. These constants are finite sums of suprema of derivatives of the explicit functions above and depend on the fixed box, a, ρ . They are not uniform-limit claims.

Let χ be a smooth colour-invariant cutoff equal to one on $|y| \leq \rho/2$ and zero before $|y| = \rho$. For any fixed polynomial times the comparison Gaussian F , the exact chart vector $\chi(gx)F(x)$, extended by zero, is smooth on the original compact manifold after applying $\mathcal{B}_g^*$. The preceding coefficient bounds and all Gaussian polynomial moments give

$$\|[\tilde{H}_g - H_{\text{osc}} - gH^{(1)} - g^2H^{(2)}]\chi(gx)F\| \leq C_F g^3. \quad (177)$$

Derivatives of the cutoff are supported at $|x| \geq \rho/(2g)$. Their polynomial factors are bounded by powers of g^{-1} and their Gaussian integrals by $C_N g^N$ for every fixed N : bound the smallest Gaussian quadratic eigenvalue below and integrate $r^m e^{-cr^2}$ beyond that radius. This proves the same estimate when the cutoff is commuted past any displayed coefficient operator.

17.3 A physical three-creation image of the complete cubic term

Let V denote the matrix of all original transverse edge eigenvectors and $\Sigma = \text{diag } \sigma_\nu$. Let ι insert chord coordinates with zero tree entries. The comparison Gaussian in x has the matrix

$$K = \iota^\top V \Sigma V^\top \iota = G^{-1/2} O \Sigma O^\top G^{-1/2}, \quad \Phi_0(x) = N_0 \exp \left(-\frac{1}{8} \sum_\alpha (x^\alpha)^\top K x^\alpha \right). \quad (178)$$

Here N_0 is the already specified unit-vacuum constant in this coordinate measure. No trial state is changed by it. To prove the matrix identity, use $T\iota = I$ and $V = T^\top G^{-1/2} O$. Thus $V^\top \iota = O^\top G^{-1/2}$, which proves (178), positivity of K , and $T^\top K = V \Sigma V^\top$.

For each colour put $p_c = (Kx)_c$, as a three-component vector, and set

$$v_e = \sum_c t_{ce} p_c, \quad \theta_e = \sum_c b_{ce} x_c \times p_c.$$

Direct differentiation of the full Gaussian gives $D_{e,\alpha}^0 \Phi_0 = -v_e^\alpha \Phi_0/4$ and $D_{e,\alpha}^1 \Phi_0 = \theta_e^\alpha \Phi_0/8$. In $\sum_\alpha D_{e,\alpha}^0 \theta_e^\alpha$ every derivative contracts two equal indices of the alternating colour tensor, so the sum is zero. In $\sum_\alpha D_{e,\alpha}^1 v_e^\alpha$ the derivative forces the component $(x_c \times e_\alpha)_\alpha$, which is zero. The full first vertex therefore satisfies

$$H^{(1)} \Phi_0 = P_3(x) \Phi_0, \quad P_3(x) = \frac{1}{8a} \sum_e v_e \cdot \theta_e + \frac{1}{4a} \sum_p A_p \cdot B_p. \quad (179)$$

This calculation includes the kinetic part of the original nonlinear operator, rather than identifying the Wilson cubic term alone with the quantum vertex. Every term is invariant under simultaneous adjoint colour rotations and is odd under $x \mapsto -x$.

Expand this exact alternating polynomial as

$$P_3(x) = \sum_{i < j < k} c_{ijk} x_i \cdot (x_j \times x_k).$$

Repeated chord indices give zero, and collecting distinct indices with their permutation signs determines each coefficient uniquely. Put

$$D = \sqrt{2} G^{1/2} O \Sigma^{-1/2}, \quad d_I = \sum_J c_J \det D_{J,I}, \quad |I| = |J| = 3, \quad E_I = \frac{\sigma_{i_1} + \sigma_{i_2} + \sigma_{i_3}}{a}.$$

The coordinate identity $x_c^\alpha = \sum_\nu D_{c\nu} (a_{\nu\alpha} + a_{\nu\alpha}^\dagger)$ then gives exactly

$$P_3 \Phi_0 = \sum_{i < j < k} d_{ijk} \sum_{\alpha\beta\gamma} \epsilon_{\alpha\beta\gamma} a_{i\alpha}^\dagger a_{j\beta}^\dagger a_{k\gamma}^\dagger \Phi_0. \quad (180)$$

Each attempted one-creation contraction identifies two colours in $\epsilon_{\alpha\beta\gamma}$ and vanishes. The remaining coefficient is the alternating determinant of the three coordinate columns, proving the formula. Distinct ordered triples have disjoint mode occupations; the six nonzero colour assignments are orthogonal and each has norm one. Consequently

$$\begin{aligned} \|P_3 \Phi_0\|^2 &= 6 \sum_I |d_I|^2 = 6c^* \bigwedge^3 (2K^{-1}) c, \\ \langle P_3 \Phi_0, A_0 P_3 \Phi_0 \rangle &= 6 \sum_I E_I |d_I|^2, \quad A_0 = H_{\text{osc}} - \mu_0. \end{aligned} \quad (181)$$

The second equality in the first line follows also by direct Gaussian contraction: two ordered colour determinants have inner product six times the determinant of their cross-covariance submatrix. Their covariance is exactly $2K^{-1}$ by (178). Thus the entire map preserves the original raw Gram factors and the physical three-colour singlet constraint.

17.4 A finite rational certificate that the full vertex is nonzero

Theorem 17.1. *On the original $L = 2$ box, $H^{(1)}\Phi_0$ is a nonzero physical three-creation vector. For every original $a > 0$,*

$$\|H^{(1)}\Phi_0\|^2 > \frac{14641}{125000\sqrt{3}a^2}, \quad \|A_0^{-1}H^{(1)}\Phi_0\|^2 > \frac{14641}{13500000\sqrt{3}}. \quad (182)$$

Proof. Use the following three positively oriented original chords, in the displayed order:

$$c_1 = ((-1, 1, -2), 2), \quad c_2 = ((-1, 2, -2), 3), \quad c_3 = ((-1, 1, -1), 2).$$

Directions here are 1, 2, 3 as in the original manuscript. Assign $x_{c_1} = e_1, x_{c_2} = e_2, x_{c_3} = e_3$ and all other chords zero. The face $((-1, 1, -2); 2, 3)$ has first two factors with those positive vectors, third factor with $-e_3$, and fourth vector zero. All other faces involve fewer than these three nonzero colour vectors. Equation (174) gives the magnetic coefficient $-1/(8a)$.

Let K_I be the three-by-three submatrix of K in this ordered list and put $Q_I = (bV\Sigma V^\top)_I$. The complete kinetic coefficient multiplied by a is

$$\frac{1}{8} \sum_{j=1}^3 (Q_I)_{j,\cdot} \cdot (e_j \times (K_I)_{j,\cdot}). \quad (183)$$

This follows from $T^\top K = V\Sigma V^\top$ in (179); all original tree paths in b are retained.

Here are short rational enclosures for every entry. Each entry lies strictly between the corresponding integer below divided by 10^6 and that integer plus one divided by 10^6 :

$$10^6 K_I : \begin{pmatrix} 1273974 & 432279 & -317748 \\ 432279 & 1273974 & -386480 \\ -317748 & -386480 & 1563504 \end{pmatrix}, \quad 10^6 Q_I : \begin{pmatrix} -845265 & -977768 & 417911 \\ 473487 & -1342407 & 1208367 \\ 421454 & 840903 & -1200597 \end{pmatrix}. \quad (184)$$

We specify the exact finite calculation producing these bounds, so they are rational certificates rather than rounded numerical evidence. For $N = 5$, use every frequency triple in $\{0, 1, 2, 3, 4\}^3$ with at least two positive entries. There are $125 - 1 - 12 = 112$ such blocks. Set $s_j = 2 \sin(\pi j/10)$ and $\sigma = \sqrt{\sum_i s_{j_i}^2}$. For an original edge $e = (n, i)$ put

$$\phi_e(j) = w_{j_i}(n_i) \prod_{d \neq i} v_{j_d}(n_d),$$

with zero when $j_i = 0$, and with the exact v, w of Section 16. The contribution of that block to the full edge spectral matrix is

$$S_{e,e'}(j) = \sigma \phi_e(j) \phi_{e'}(j) \left(\delta_{ii'} - \frac{s_{j_i} s_{j_{i'}}}{\sigma^2} \right). \quad (185)$$

The two matrices in (184) are respectively the indicated entries of $\sum_j S$ and $b \sum_j S$. This is the complete transverse projection in each exact frequency block, so no choice or rounding of polarization vectors occurs.

Every trigonometric value here is an integer multiple of $\pi/10$. Start with $\sin(\pi/10) = (\sqrt{5}-1)/4$ and $\cos(\pi/10) = \sqrt{(5+\sqrt{5})}/8$ and use the angle-addition recurrence up to multiples $0, \dots, 19$. These radical identities follow, for example, by $\cos 5\theta = 0$ and the larger positive root $\cos^2 \theta =$

$(5 + \sqrt{5})/8$; the bound $0 < \theta < \pi/6$ selects $\theta = \pi/10$. Also retain the factors $\sqrt{1/5}$ and $\sqrt{2/5}$ in the original vertex and edge functions.

The included program `certify_nonabelian_vertex.py` evaluates precisely this finite sum using endpoints that are integers divided by 10^{40} . Addition and negation are exact. For multiplication take the minimum and maximum of the four endpoint products and round outward by integer floor and ceiling division. For a nonzero-sign divisor apply the same rule to the four rational quotients. To enclose a positive square root with denominator $M = 10^{40}$, take the integer floor of $\sqrt{lM}$ for the lower numerator and the integer ceiling of $\sqrt{uM}$ for the upper numerator, for input numerators l, u . The integer-square inequalities are checked directly. Monotonicity of each operation proves inductively that every resulting interval contains the exact radical value. Integer tree paths and the complete 112-block loops are specified in that source. The full interval endpoints, short bounds above, and final inequality are recorded in `NONABELIAN_VERTEX_CERTIFICATE.json`.

Inserting even the wider short intervals (184) into (183), and adding the exact $-1/8$ magnetic term, proves by rational arithmetic that the full coefficient c obeys

$$-\frac{243}{1000} < ac < -\frac{242}{1000}. \quad (186)$$

Sorting the three chord indices changes only its sign and not its absolute value in the alternating coefficient vector of (181).

Finally all original frequencies are at most $\sqrt{12}$, since each $s_j \leq 2$. Hence $0 < K \leq \sqrt{12}I$ by (178), and $2K^{-1} \geq I/\sqrt{3}$. Diagonalizing this positive matrix proves $\bigwedge^3(2K^{-1}) \geq I/(3\sqrt{3})$: its eigenvalues are the products of three of the original eigenvalues. Thus (181) is at least $2 \sum_I |c_I|^2/\sqrt{3}$, which with (186) gives the first bound in (182). Each $E_I \leq 6\sqrt{3}/a$, so the inverse excitation multiplies the squared coefficient by at least $a^2/108$. This proves the second bound, with every physical spacing factor retained. $\square$

17.5 The actual vacuum, raw quasimode Gram factor, and an observable response

The finite-box comparison spectrum has an isolated simple physical ground state and the actual low eigenvalues and projections converge, as proved in Section 14. We now use those facts with the controlled remainder, rather than assuming a formal perturbation series for the actual vacuum. Define

$$\begin{aligned} \phi_1 &= -A_0^{-1}P_3\Phi_0, \\ E^{(2)} &= \langle \Phi_0, H^{(2)}\Phi_0 \rangle - 6 \sum_I \frac{|d_I|^2}{E_I}, \\ \phi_2 &= -A_0^{-1}(I - |\Phi_0\rangle\langle\Phi_0|)(H^{(1)}\phi_1 + H^{(2)}\Phi_0). \end{aligned} \quad (187)$$

All inverses in this formula act on the physical vacuum complement of the fixed oscillator. These are finite explicit Fock calculations: the inputs are polynomial Gaussians, the coefficient operators are polynomial differential operators, and dividing each nonvacuum Fock coefficient by its sum of positive frequencies defines the inverse. They retain all intermediate physical modes produced by those polynomials. Oddness of P_3 gives the exact first energy coefficient $\langle \Phi_0, H^{(1)}\Phi_0 \rangle = 0$.

Set $F_g = \chi(gx)(\Phi_0 + g\phi_1 + g^2\phi_2)$ without dividing by its norm. Substitution of (187) into the exact operator, using (177), gives

$$\|[\tilde{H}_g - \mu_0 - g^2E^{(2)}]F_g\| = O(g^3), \quad \|F_g\|^2 = 1 + g^2\|\phi_1\|^2 + O(g^3). \quad (188)$$

For the residual, orders zero, one and two cancel by the displayed equations; their remaining polynomial-Gaussian terms have finite norm, and cutoff commutators have the Gaussian tail bound already proved. For the Gram factor, both ϕ_1, ϕ_2 are orthogonal to the vacuum by definition. Expanding the unaltered square and using the same tail bound proves the second equation.

The spectral theorem applied to this nonzero quasimode places an actual eigenvalue within $O(g^3)$ of $\mu_0 + g^2 E^{(2)}$: otherwise the residual norm would exceed that spectral distance times $\|F_g\|$. The fixed-box spectral separation identifies it as the unique actual vacuum eigenvalue. The orthogonal remainder of F_g from its actual vacuum line has norm $O(g^3)$, by the same spectral theorem and the fixed positive separation from that line. Positivity and the already proved vacuum convergence fix the positive overlap. Its squared magnitude equals $\|F_g\|^2$ minus that orthogonal remainder's squared norm. Therefore the overlap is $1 + O(g^2)$, and we obtain the actual statements

$$\mathcal{E}_g = \mu_0 + g^2 E^{(2)} + O(g^3), \quad \mathcal{B}_g \psi_g = \Phi_0 + g\phi_1 + O(g^2). \quad (189)$$

The only unit vector here is the originally specified actual vacuum. The raw trial vector and its nontrivial Gram factor in (188) have been left unchanged. In particular the second lower bound of (182) proves a nonzero first derivative of the actual vacuum in this exact coordinate frame.

There is also an explicit bounded physical observable that detects it. Choose even radial cutoffs $\chi_R(x)$, equal to one on $|x| \leq R$ and zero for $|x| \geq 2R$, with $0 \leq \chi_R \leq 1$, and set $O_R(x) = \chi_R(x)P_3(x)$. Its multiplication norm is finite at each fixed R . Lift it by the original logarithmic coordinate $x = y/g$ and set it zero outside the chart; for $2Rg < \rho$ this defines a globally smooth gauge-invariant multiplier $O_{R,g}$. Its prescribed dependence on g is part of this exact observable map. Equation (189) gives

$$\langle \psi_g, O_{R,g} \psi_g \rangle = -2g \langle \chi_R P_3 \Phi_0, A_0^{-1} P_3 \Phi_0 \rangle + O_R(g^2). \quad (190)$$

The constant term is zero by parity. Multiplication is bounded, so the norm error in (189) controls every remaining term of this expectation. Gaussian domination gives $\chi_R P_3 \Phi_0 \rightarrow P_3 \Phi_0$ in norm as $R \rightarrow \infty$. The limiting inner product is $6 \sum_I |d_I|^2 / E_I > 0$ by the proved nonzero vertex. Fix the least integer R for which the inner product exceeds half that positive quantity. Its existence follows from that norm convergence. This produces a fixed bounded observable prescription with a strictly negative nonzero first response for the actual positive-coupling vacuum. Neither a Gaussian replacement of that vacuum nor an assumption of interaction survival in the continuum is used.

17.6 Exact relation to the full nonlinear coupling derivative

The retained source *Local-energy spacetime kernel and the exact nonlinear coupling response* proves differentiation on the original fixed Hilbert space. With $K_{\text{el}} = \sum_e E_e$ and $W = \sum_p (2 - W_p)$ its derivative is

$$V_g = \partial_g H_g = \frac{4g}{a} K_{\text{el}} - \frac{1}{g^3 a} W, \quad \psi'_g = -R_g (V_g - \mathcal{E}'_g) \psi_g, \quad R_g = (A_g|_{\psi_g^\perp})^{-1} P_{\psi_g^\perp}.$$

At each fixed $g > 0$, smoothness of the simple vacuum follows directly from the resolvent identity on the common elliptic H^2 domain and a contour around that eigenvalue. Differentiating the eigen-equation and its original unit-norm constraint gives the displayed formula; the reduced inverse is bounded at that fixed regulator. The complete source, including all unbounded-operator domains and Duhamel terms, is retained with this manuscript.

There is an exact morphism from this derivative to the vertices calculated above. Differentiate the full density-bearing chart, at fixed original F , on any compact subset inside its logarithmic

domain. The density terms in the two derivatives agree, giving

$$(\partial_g \mathcal{B}_g)F = \frac{1}{g} \mathcal{D} \mathcal{B}_g F, \quad \mathcal{D} = x \cdot \nabla + \frac{d}{2}, \quad \mathcal{D}^* = -\mathcal{D} \quad (191)$$

on compact test functions. Explicitly, differentiating $g^{d/2}$ gives $d/(2g)$, differentiating $\mathcal{J}(gx)^{1/2}$ gives $x \cdot \nabla_y \log \mathcal{J}(gx)/2$, and differentiating $F(gx)$ gives $x \cdot \nabla_y F(gx)$. Expanding $g^{-1}x \cdot \nabla_x (\mathcal{B}_g F)$ reproduces the latter two terms exactly. Integration by parts gives the stated skew-adjoint identity. It follows that

$$\begin{aligned} \partial_g \tilde{H}_g &= \mathcal{B}_g V_g \mathcal{B}_g^* + \frac{1}{g} [\mathcal{D}, \tilde{H}_g], \\ \partial_g (\mathcal{B}_g \psi_g) &= \frac{1}{g} \mathcal{D} (\mathcal{B}_g \psi_g) - \mathcal{B}_g R_g (V_g - \mathcal{E}'_g) \psi_g. \end{aligned} \quad (192)$$

These equalities hold locally on the stated smooth coordinate domains; no derivative of a discontinuous boundary extension is taken. Products with the fixed interior cutoffs used above give the corresponding global test-vector identities. Thus $H^{(1)}$ is the regular first coefficient of the full coordinate derivative in (192). Omitting its commutator would confuse a changing coordinate frame with the fixed-Hilbert-space derivative and leave false singular terms. This explicit relation retains both derivatives and shows how they enter the same original operator.

17.7 The first nonlinear coefficient of the original local covariance

The odd vacuum response above does not by itself give a nonzero first-order response for an even energy covariance. We calculate that covariance, retaining its full quadratic coefficient. For the fixed real original edge weights f , let $D_f^{(n)}$ be the coefficients of $\mathcal{B}_g D_{f,g} \mathcal{B}_g^*$. Their exact formulas are (175)–(176), with each electric edge term multiplied by f_e and each face term by its original quarter-boundary average f_p . Write $D_f^{(0)} = D_0$, $D_f^{(1)} = D_1$, $D_f^{(2)} = D_2$ only in this subsection; these symbols denote the weighted operators, not the individual fields $D_{e,\alpha}^n$ above.

We first establish the order of the expansion on which this calculation depends. The same exact coefficient functions give

$$Y_{e,\alpha}^{(3)} = 0, \quad Y_{e,\alpha}^{(4)} = -\frac{1}{720} \sum_c t_{ce} (\text{ad}_{x_c})^4 e_\alpha \cdot \nabla_c + \frac{1}{720} \sum_c t_{ce} |x_c|^2 x_c^\alpha.$$

The fourth density coefficient follows by differentiating the retained $-g^4 |x_c|^4 / 1440$ term; its radial contraction with every positive adjoint power is zero. Set $Y^{(2)} = D^2 + M^2$. Then

$$\begin{aligned} H^{(3)} &= -\frac{2}{a} \sum_{e,\alpha} \{Y_{e,\alpha}^{(1)}, Y_{e,\alpha}^{(2)}\} + \frac{1}{2a} \sum_p W_{p,5}, \\ H^{(4)} &= -\frac{2}{a} \sum_{e,\alpha} \{(Y_{e,\alpha}^{(2)})^2 + \{Y_{e,\alpha}^{(0)}, Y_{e,\alpha}^{(4)}\}\} + \frac{1}{2a} \sum_p W_{p,6}. \end{aligned}$$

The coefficients $W_{p,5}, W_{p,6}$ are the full finite sums (172); no fifth or sixth power in an oriented factor is omitted. These formulas specify all coefficients needed for the following fourth-order quasimode.

Put $\phi_0 = \Phi_0$. For $1 \leq n \leq 4$, recursively set

$$\begin{aligned} E^{(n)} &= \left\langle \Phi_0, \sum_{k=1}^n H^{(k)} \phi_{n-k} \right\rangle, \\ \phi_n &= -A_0^{-1} P_{\Phi_0^\perp} \sum_{k=1}^n (H^{(k)} - E^{(k)}) \phi_{n-k}. \end{aligned} \quad (193)$$

These agree with the preceding coefficients at $n = 1, 2$. Induction proves that all inputs are finite physical polynomial Gaussians. It also proves their parity: conjugation by $\mathcal{P}F(x) = F(-x)$ multiplies $H^{(k)}$ by $(-1)^k$, as follows directly from the homogeneous coefficient degrees in the fields and potential. Hence ϕ_n has parity $(-1)^n$ and $E^{(1)} = E^{(3)} = 0$. Taylor's integral remainder through the next order, with the same Gaussian cutoff estimates as (177), proves that $F_g^{[4]} = \chi(gx) \sum_{n=0}^4 g^n \phi_n$ has residual $O(g^5)$ for its displayed eigenvalue polynomial. Its component off the actual vacuum line has both L^2 norm and $\tilde{H}_g$ graph norm $O(g^5)$ by the fixed positive spectral separation and its residual.

This stronger graph estimate justifies applying the unbounded local operator to the actual vacuum expansion. Indeed on the original common smooth domain the commuting edge Casimirs give

$$\|D_{f,g}u\| \leq \|f\|_\infty \left(\|H_g u\| + \frac{4|\mathbf{P}_L|}{g^2 a} \|u\| \right). \quad (194)$$

To verify it, the joint edge-Casimir expansion gives $\|\sum f_e E_e u\| \leq \|f\|_\infty \|\sum E_e u\|$. Use $\kappa \sum E_e = H_g - bW$, $\|W\| \leq 4|\mathbf{P}_L|$, and $\|W_f\| \leq \|f\|_\infty 4|\mathbf{P}_L|$ to obtain the displayed sum, including both potential contributions. Consequently the $O(g^5)$ vacuum-complement remainder becomes only $O_f(g^3)$ after applying $D_{f,g}$. All polynomial terms themselves have the stronger local coefficient estimates already proved.

The raw overlap squared of $F_g^{[4]}$ with the actual vacuum is $1 + g^2 \|\phi_1\|^2 + O(g^3)$. Comparing coefficients with the original unit-vacuum constraint therefore fixes

$$\chi_2 = \phi_2 - \frac{1}{2} \|\phi_1\|^2 \Phi_0, \quad \mathcal{B}_g \psi_g = \Phi_0 + g\phi_1 + g^2 \chi_2 + O(g^3).$$

The expansion also holds after applying the weighted operator, with the corresponding coefficients, by (194). The term along Φ_0 is thus explicitly retained; the raw quasimode was not normalized.

Define the following finite Fock vectors and scalars:

$$\begin{aligned} m_0 &= \langle \Phi_0, D_0 \Phi_0 \rangle, \\ m_2 &= \langle \Phi_0, D_2 \Phi_0 \rangle + 2 \operatorname{Re} \langle \phi_1, D_1 \Phi_0 \rangle + 2 \operatorname{Re} \langle \chi_2, D_0 \Phi_0 \rangle + \langle \phi_1, D_0 \phi_1 \rangle, \\ \xi_0 &= (D_0 - m_0) \Phi_0, \quad \xi_1 = D_1 \Phi_0 + (D_0 - m_0) \phi_1, \\ \xi_2 &= (D_2 - m_2) \Phi_0 + D_1 \phi_1 + (D_0 - m_0) \chi_2. \end{aligned} \quad (195)$$

Parity gives $m_1 = 0$, ξ_0, ξ_2 even and ξ_1 odd. The preceding graph argument proves the actual expansions $m_{f,g} = m_0 + g^2 m_2 + O_f(g^3)$ and $\mathcal{B}_g \Xi_{f,g} = \xi_0 + g \xi_1 + g^2 \xi_2 + O_f(g^3)$ in the required norms. This includes the motion of the vacuum and its scalar subtraction.

Put $S_0(t) = e^{-tA_0}$ and $A_2 = H^{(2)} - E^{(2)}$. Define

$$\begin{aligned} S_1(t) &= - \int_0^t S_0(t-s) H^{(1)} S_0(s) ds, \\ S_2(t) &= - \int_0^t S_0(t-s) A_2 S_0(s) ds \\ &\quad + \int_{0 \leq r \leq s \leq t} S_0(t-s) H^{(1)} S_0(s-r) H^{(1)} S_0(r) dr ds. \end{aligned} \quad (196)$$

These are finite-dimensional integrals on each polynomial input and the finite Fock spaces its indicated insertions generate. To prove the expansion for the actual semigroup, let $U_0 = S_0(t)F$, $U_1 = S_1(t)F$, $U_2 = S_2(t)F$ for a fixed physical polynomial Gaussian F . Differentiation of these finite integrals gives $(\partial_t + A_0)U_0 = 0$, $(\partial_t + A_0)U_1 = -H^{(1)}U_0$ and $(\partial_t + A_0)U_2 = -H^{(1)}U_1 - A_2U_0$.

Their coefficients stay bounded on each fixed compact time interval. The cutoff sum $\chi(gx)(U_0 + gU_1 + g^2U_2)$ therefore has residual $O(g^3)$ for the actual equation $(\partial_t + \tilde{H}_g - \mathcal{E}_g)U = 0$, uniformly on that interval. Variation of constants and the actual excitation semigroup's contraction bound make its difference from the exact evolution at most the time integral of this residual. The cutoff tail is smaller than every power of g . This proves the actual semigroup expansion on these vectors without a bounded-operator perturbation assumption.

It follows that, at every fixed $L, a, f, t \geq 0$, the original nonlinear covariance has the expansion

$$\begin{aligned} C_{g,f}(t) &= \langle \Xi_{f,g}, e^{-tA_g} \Xi_{f,g} \rangle = C_{0,f}(t) + g^2 C_{2,f}(t) + O_{L,a,f,t}(g^3), \\ C_{2,f}(t) &= 2 \operatorname{Re} \langle \xi_2, S_0(t) \xi_0 \rangle + \langle \xi_1, S_0(t) \xi_1 \rangle \\ &\quad + 2 \operatorname{Re} \langle \xi_1, S_1(t) \xi_0 \rangle + \langle \xi_0, S_2(t) \xi_0 \rangle. \end{aligned} \quad (197)$$

There is no linear term: its two external-vector terms pair an odd vector with an even one, and its semigroup insertion is odd between two even vectors. The displayed second coefficient retains the quartic operator, both cubic insertions, every intermediate mode, the actual second vacuum coefficient and the centering correction.

For an entirely algebraic evaluation, take the original occupation basis with energies $\lambda_n = \sum_{\nu,\alpha} n_{\nu\alpha} \sigma_\nu / a$. Its creation and annihilation matrix entries are $\sqrt{n_{\nu\alpha} + 1}$ and $\sqrt{n_{\nu\alpha}}$ with the corresponding occupation changed by one. The polynomial formulas above determine every matrix entry from those rules. The time integrals are exactly

$$\begin{aligned} I_t(\alpha, \beta) &= \begin{cases} (e^{-t\alpha} - e^{-t\beta})/(\beta - \alpha), & \alpha \neq \beta, \\ te^{-t\alpha}, & \alpha = \beta, \end{cases} \\ (S_1)_{nm} &= -H_{nm}^{(1)} I_t(\lambda_n, \lambda_m), \\ (S_2)_{nm} &= -(A_2)_{nm} I_t(\lambda_n, \lambda_m) + \sum_k H_{nk}^{(1)} H_{km}^{(1)} J_t(\lambda_n, \lambda_k, \lambda_m), \end{aligned} \quad (198)$$

where for three distinct arguments $J_t(\alpha, \beta, \gamma) = \sum_{z \in \{\alpha, \beta, \gamma\}} e^{-tz} / \prod_{w \neq z} (w - z)$. For exactly two equal arguments, symmetry gives

$$J_t(\alpha, \alpha, \gamma) = \frac{[t(\gamma - \alpha) - 1]e^{-t\alpha} + e^{-t\gamma}}{(\gamma - \alpha)^2}, \quad J_t(\alpha, \alpha, \alpha) = \frac{1}{2} t^2 e^{-t\alpha}.$$

These identities follow by integrating the two exponentials on the simplex in (196); the repeated cases follow by direct integration of the resulting linear factor. Only finitely many occupations enter (197), because all of its input vectors and insertions are polynomials of bounded degree. The formula does not discard other physical intermediate states that are reached by those insertions.

As an exact check on every subtraction, take $f_e = 1$ on the full finite box. Then $D_n = H^{(n)}$, $D_0 = H_{\text{osc}}$, $m_0 = \mu_0$ and $m_2 = E^{(2)}$. The equations defining ϕ_1, ϕ_2 give $\xi_0 = \xi_1 = \xi_2 = 0$ individually. Thus both sides of (197) vanish, as the exact identity $\Xi_{1,g} = 0$ requires. A formula omitting the vacuum response or its raw Gram coefficient would not pass these coefficient identities.

We have proved that the original nonlinear Hamiltonian has a nonzero colour-singlet cubic vertex, with an actual vacuum and observable response, even though Section 16's selected covariance limit admits the displayed free representation. The exact finite-box response and the full intermediate-state terms are now available for the joint regulator calculation. Their survival at fixed physical scales, higher local products, and an interacting continuum mass-gap violation are not established by the fixed-box bounds of this section. The original counterexample programme remains unfinished.

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